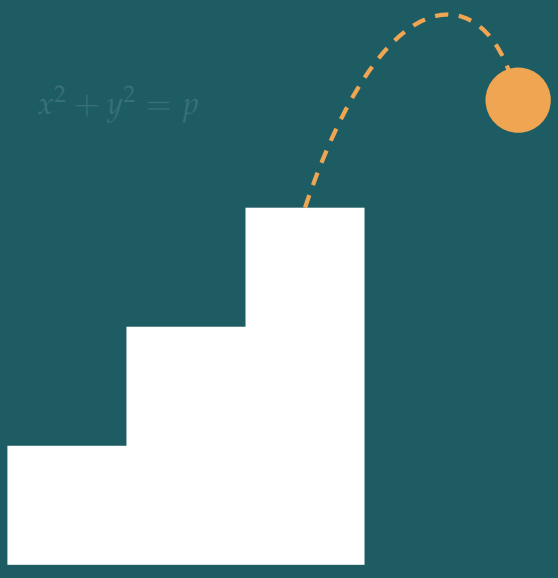
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1987 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1987 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1987 Worked Solutions

Worked solutions to all 48 questions of the 1987 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

Contents

STEP I

Q1 Curve Sketching Pure . . . 3
Q2 Areas of Circle Packings Pure . . . 7
Q3 Separable Differential Equations Pure . . . 11
Q4 Logarithms and Geometric Series Pure . . . 14
Q5 Integration by Substitution Pure . . . 16
Q6 Areas and Inverse Functions Pure . . . 19
Q7 Trigonometric Series Pure . . . 22
Q8 Integration by Substitution Pure . . . 25
Q9 Barycentric Coordinates Pure . . . 28
Q10 Elastic Strings Mechanics . . . 32
Q11 Circular Motion and Energy Mechanics . . . 36
Q12 Projectiles with Restitution Mechanics . . . 39
Q13 Variable Forces Mechanics . . . 42
Q14 Geometric Series of Probabilities Probability and Statistics . . . 45
Q15 Continuous Random Variables Probability and Statistics . . . 49
Q16 Bayes and Normal Approximation Probability and Statistics . . . 52

STEP II

Q1 Algebraic Identities Pure . . . 55
Q2 Trigonometric Identities Pure . . . 58
Q3 Conjugate Surds Pure . . . 61
Q4 Complex Numbers and Geometry Pure . . . 64
Q5 Lagrange Inversion Series Pure . . . 68
Q6 Definite Integrals by Symmetry Pure . . . 72
Q7 Trapezium Rule and Factorials Pure . . . 76
Q8 Vector Loci in 3D Pure . . . 80
Q9 Orthogonal Matrices Pure . . . 84
Q10 Group Axioms Pure . . . 87
Q11 Friction and Moments Mechanics . . . 90

Q12 Equilibrium and Energy Mechanics . . . 93
Q13 Oblique Impacts with Walls Mechanics . . . 97
Q14 Elastic Band on Cone Mechanics . . . 101
Q15 Geometric Probability Probability and Statistics . . . 104
Q16 Bayes' Theorem Probability and Statistics . . . 108

STEP III

Q1 Divisibility of Factorials Pure . . . 111
Q2 Trigonometric Reduction Formulae Pure . . . 114
Q3 Arguments and Machin's Formula Pure . . . 117
Q4 Volumes of Revolution Pure . . . 120
Q5 Modular Arithmetic Ciphers Pure . . . 123
Q6 Coupled Differential Equations Pure . . . 126
Q7 Series for Arctangent Pure . . . 130
Q8 Parallelogram Area Inequalities Pure . . . 133
Q9 Direct Products of Groups Pure . . . 137
Q10 Bernoulli Polynomials Pure . . . 141
Q11 Relative Velocity Optimisation Mechanics . . . 144
Q12 Variable Mass Rotation Mechanics . . . 148
Q13 Rotating Rod with Friction Mechanics . . . 152
Q14 Gravitational Collapse Mechanics . . . 155
Q15 Expected Cost Optimisation Probability and Statistics . . . 159
Q16 Order Statistics and Poisson Probability and Statistics . . . 162

WORKED SOLUTION

STEP I 1987, Question 1

Pure • Curve Sketching

> **STEP I 1987, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/87-s1-q1

The key idea

Differentiating with the product rule leaves the factor e^{ax}, which is never zero, times a combination of $\cos bx$ and $\sin bx$. So the stationary points come from a single equation $\tan bx =$ constant, whose solutions are equally spaced, a distance π/b apart. Moving from one to the next multiplies the exponential by a fixed factor and flips the sign of the cosine, and that is exactly a geometric progression.

Solution

Stationary points

By the product rule,

$$f'(x) = ae^{ax}\cos bx - be^{ax}\sin bx = e^{ax}(a\cos bx - b\sin bx).$$

Since $e^{ax} > 0$ for every x, we have $f'(x) = 0$ exactly when $a\cos bx = b\sin bx$.

> **Ask yourself** Can we safely divide by $\cos bx$ here?

Yes. If $\cos bx = 0$ then $\sin bx = \pm 1$, and the equation would say $0 = \pm b$, which is false because $b > 0$. So at any stationary point $\cos bx \neq 0$, and dividing gives

$$\tan bx = \frac{a}{b}.$$

Let $\alpha = \arctan(a/b)$. Because $a/b > 0$, we have $0 < \alpha < \frac{\pi}{2}$. As tan has period π, the solutions are $bx = \alpha + n\pi$, so the stationary points are at

$$x_n = \frac{\alpha + n\pi}{b}, \qquad n \in \mathbb{Z}, \qquad \text{where } \tan\alpha = \frac{a}{b}.$$

From a right-angled triangle with legs a and b, $\cos\alpha = \dfrac{b}{\sqrt{a^2+b^2}}$. Since $\cos(\theta + n\pi) =$

$(-1)^n \cos\theta$, the value of f there is

$$\begin{aligned} \mathrm{f}(x_n) &= \mathrm{e}^{a(\alpha+n\pi)/b}\cos(\alpha+n\pi) \\ &= (-1)^n \frac{b}{\sqrt{a^2+b^2}}\,\mathrm{e}^{a\alpha/b}\,\mathrm{e}^{na\pi/b}. \end{aligned}$$

Maximum or minimum? Differentiating again,

$$\mathrm{f}''(x) = \mathrm{e}^{ax}((a^2-b^2)\cos bx - 2ab\sin bx).$$

At a stationary point $b\sin bx = a\cos bx$, so $2ab\sin bx = 2a^2\cos bx$ and

$$\mathrm{f}''(x_n) = \mathrm{e}^{ax_n}\cos bx_n\,(a^2-b^2-2a^2) = -(a^2+b^2)\,\mathrm{f}(x_n).$$

So x_n is a maximum when $\mathrm{f}(x_n) > 0$ (n even) and a minimum when $\mathrm{f}(x_n) < 0$ (n odd). The stationary points alternate: maximum, minimum, maximum, ...

The geometric progression

Since $0 < \alpha < \pi$, we have $x_n > 0$ exactly when $n \geq 0$. So the stationary points with $x > 0$ are $x_0 < x_1 < x_2 < \cdots$, and for each $n \geq 0$

$$\frac{\mathrm{f}(x_{n+1})}{\mathrm{f}(x_n)} = \frac{\mathrm{e}^{a(\alpha+(n+1)\pi)/b}\cos(\alpha+(n+1)\pi)}{\mathrm{e}^{a(\alpha+n\pi)/b}\cos(\alpha+n\pi)} = \mathrm{e}^{a\pi/b}\cdot(-1) = -\mathrm{e}^{a\pi/b},$$

using $\cos(\theta+\pi) = -\cos\theta$ and $\cos(\alpha+n\pi) \neq 0$. The ratio doesn't depend on n, so the values $\mathrm{f}(x_0), \mathrm{f}(x_1), \ldots$ form a geometric progression with first term $\dfrac{b\,\mathrm{e}^{a\alpha/b}}{\sqrt{a^2+b^2}}$ and common ratio $-\mathrm{e}^{a\pi/b}$, as required.

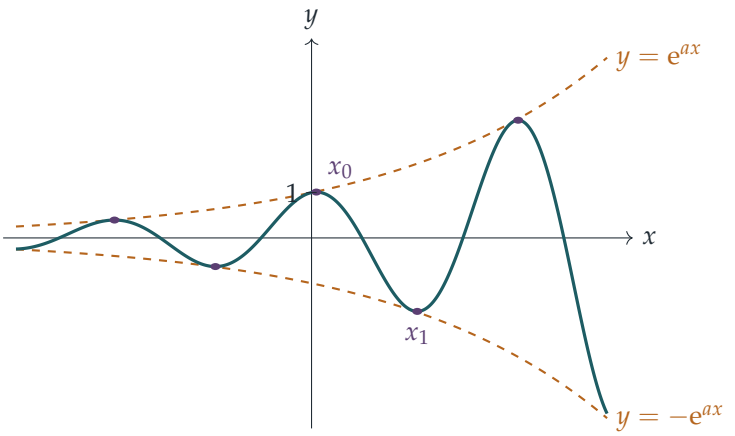
(The same calculation works for negative n too, so in fact all the stationary values form a single two-way geometric progression. The restriction to $x > 0$ just gives it a natural first term.)

Sketch

The facts we need:

- Since $-1 \leq \cos bx \leq 1$, the graph lies between the curves $y = \mathrm{e}^{ax}$ and $y = -\mathrm{e}^{ax}$. It touches $y = \mathrm{e}^{ax}$ where $\cos bx = 1$ ($bx = 2m\pi$) and $y = -\mathrm{e}^{ax}$ where $\cos bx = -1$ ($bx = (2m+1)\pi$). At these points $\sin bx = 0$, so $\mathrm{f}'(x) = \pm a\mathrm{e}^{ax}$, the same gradient as the envelope: the curve is tangent to it there, not turning.
- The zeros are at $bx = \frac{\pi}{2} + m\pi$, spaced π/b apart, and $\mathrm{f}(0) = 1$.
- Each turning point $bx = \alpha + n\pi$ lies just *after* the corresponding touching point $bx = n\pi$, because $0 < \alpha < \frac{\pi}{2}$: the rising envelope drags each peak slightly to the right.
- As $x \to \infty$ the oscillations grow without bound; as $x \to -\infty$, $\mathrm{e}^{ax} \to 0$, so $\mathrm{f}(x) \to 0$ while still oscillating.

The sketch shows $a = 0.3$, $b = 2$, with the envelopes $y = \pm e^{ax}$ dashed.



Checks

- Take $a = b = 1$: then $\alpha = \frac{\pi}{4}$, $x_0 = \frac{\pi}{4}$ and $f(x_0) = e^{\pi/4}/\sqrt{2} \approx 1.551$, while $f(x_1) = e^{5\pi/4}\cos\frac{5\pi}{4} \approx -35.89$. The ratio is $\approx -23.14 = -e^{\pi}$.
- As $a \to 0$ the function becomes $\cos bx$: $\alpha \to 0$, the turning points move to $bx = n\pi$, and the ratio tends to -1, giving the familiar $1, -1, 1, \ldots$
- Numerically, for 500 random pairs (a, b) we confirmed that $f'(x_n) = 0$, that $f(x_n)$ matches the formula above, that $f''(x_n) = -(a^2 + b^2)f(x_n)$, that consecutive ratios equal $-e^{a\pi/b}$, and that f' has no other zeros.

Common slips

- Putting the turning points where the curve touches the envelope, at $bx = n\pi$. Those points have gradient $\pm ae^{ax} \neq 0$; the turning points are a little to the right.
- Taking only $bx = \alpha$ and forgetting the general solution $bx = \alpha + n\pi$ of $\tan bx = a/b$.
- Missing the minus sign in the ratio: the cosine changes sign between consecutive stationary points, so the values alternate maximum, minimum.

Beyond STEP

The function $e^{ax}\cos bx$ is the real part of $e^{(a+ib)x}$, and functions like this are exactly the solutions of linear differential equations $y'' + py' + qy = 0$ whose auxiliary equation has complex roots. With $a < 0$ you get damped oscillation (a swinging pendulum with friction); with $a > 0$, as here, an unstable system whose oscillations grow. Successive peaks are in constant ratio, and the logarithm of that ratio is what engineers call the

logarithmic decrement. Boyce and DiPrima, *Elementary Differential Equations*, develops this in its chapter on second-order linear equations, and Needham, *Visual Complex Analysis*, shows why multiplying by $e^{(a+ib)x}$ traces a spiral.

WORKED SOLUTION

STEP I 1987, Question 2

Pure • Areas of Circle Packings

STEP I 1987, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q2

The key idea

Join the centres of the small circles. They all lie on the circle of radius R, midway between the two big circles, and they form a regular n-gon whose sides pass through the points where neighbouring small circles touch. That polygon is the dividing line between the pieces called Y (inside it) and X (outside it). So every area comes from three simple shapes: the polygon, a disc, and sectors of the small circles cut off by the polygon's corners.

Solution

The relationship between n, R and r

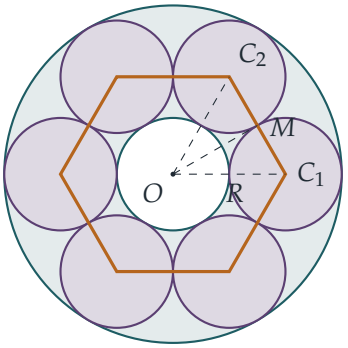
Each small circle touches both big circles, so its centre is at distance $R - r + r = R$ from the common centre O: all n centres lie on the circle of radius R. By symmetry they are equally spaced, so neighbouring centres C_1, C_2 subtend an angle $2\pi/n$ at O.

Two neighbouring circles touch, so $C_1C_2 = 2r$. In the isosceles triangle OC_1C_2, with $OC_1 = OC_2 = R$, dropping the perpendicular from O to the midpoint M of C_1C_2 gives

$$r = C_1M = R\sin\frac{\pi}{n}.$$

This is the required relationship. (We need $n \geq 3$; for $n = 2$ the inner circle would have radius $R - r = 0$.) The same triangle gives

$$OM = R\cos\frac{\pi}{n} = \sqrt{R^2 - r^2}.$$



The figure (drawn for $n = 6$) shows the polygon of centres in amber. Its sides pass through the points of contact of neighbouring circles, such as M.

The area Y

The polygon is made of n triangles congruent to OC_1C_2, each of area $\frac{1}{2} \cdot 2r \cdot \sqrt{R^2 - r^2}$. So

$$\text{area of polygon} = nr\sqrt{R^2 - r^2}.$$

What lies inside the polygon? Its inradius is $OM = \sqrt{R^2 - r^2}$, and $R^2 - r^2 \geq (R - r)^2$ because this rearranges to $2r(R - r) \geq 0$. So the inner disc of radius $R - r$ lies inside the polygon. At each vertex there is a sector of a small circle. The interior angle of a regular n-gon is $\pi - \frac{2\pi}{n}$, so each such sector has area

$$\tfrac{1}{2}r^2\left(\pi - \frac{2\pi}{n}\right) = \pi r^2\left(\frac{1}{2} - \frac{1}{n}\right).$$

These n sectors and the inner disc don't overlap one another, since the small circles only touch each other and the inner circle. What is left of the polygon is exactly the n curved triangles next to the inner circle, and together these make up Y:

$$Y = nr\sqrt{R^2 - r^2} - \pi(R - r)^2 - n\pi r^2\left(\frac{1}{2} - \frac{1}{n}\right),$$

as required.

Ask yourself Why is the side C_1C_2 the right place to split X from Y?

The side C_1C_2 runs from centre to centre through the point of contact M, so it never crosses the gaps between the circles. Every gap next to the inner circle lies on the O side of it, and every gap next to the outer circle lies on the far side. Cutting along the polygon therefore separates the two kinds of gap cleanly, and it cuts each small circle along two radii, which is why we get sectors.

The areas X and Z

Outside the polygon but inside the outer circle, we remove the rest of each small circle. That rest is the reflex sector of angle $2\pi - (\pi - \frac{2\pi}{n}) = \pi + \frac{2\pi}{n}$, of area $\pi r^2\left(\frac{1}{2} + \frac{1}{n}\right)$.

Hence

$$X = \pi(R+r)^2 - nr\sqrt{R^2-r^2} - n\pi r^2\left(\frac{1}{2}+\frac{1}{n}\right).$$

The n discs of radius r give

$$Z = n\pi r^2.$$

The limit as $n \to \infty$

Adding, the polygon terms cancel and the sector terms combine to $n\pi r^2$:

$$\begin{aligned} X+Y &= \pi(R+r)^2 - \pi(R-r)^2 - n\pi r^2 \\ &= 4\pi Rr - n\pi r^2. \end{aligned}$$

This is just the area of the annulus minus the area of the circles, as it should be. Therefore

$$\frac{X+Y}{Z} = \frac{4\pi Rr - n\pi r^2}{n\pi r^2} = \frac{4R}{nr} - 1 = \frac{4}{n\sin(\pi/n)} - 1,$$

using $r = R\sin(\pi/n)$. As $n \to \infty$, $\pi/n \to 0$ and $\dfrac{\sin(\pi/n)}{\pi/n} \to 1$, so $n\sin(\pi/n) \to \pi$ and

$$\frac{X+Y}{Z} \to \frac{4}{\pi} - 1 \approx 0.273.$$

Checks

- For $n = 3$, $R = 1$ we get $r = \frac{\sqrt{3}}{2}$, $Y \approx 0.0646$ and $X \approx 3.7497$. A numerical integration over a fine polar grid, classifying each point as inside a circle, inside the polygon or outside it, agreed to four significant figures for $n = 3, 5, 8, 12$.
- The ratio $\frac{4}{n\sin(\pi/n)} - 1$ is 0.2944 at $n = 10$, 0.27345 at $n = 100$ and 0.273240 at $n = 1000$. It is closing in on $4/\pi - 1 = 0.273240\ldots$.
- Heuristically, for large n the circles look like a row of discs of diameter $2r$ in a strip of width $2r$. A disc fills πr^2 of each $2r \times 2r$ square, and the gaps fill $4r^2 - \pi r^2$, giving the ratio $(4-\pi)/\pi$.

Common slips

- Using the exterior angle $2\pi/n$ of the polygon for the sectors instead of the interior angle $\pi - 2\pi/n$.
- Forgetting that the inner disc must be removed in Y, or not checking that it really lies inside the polygon.

Beyond STEP

The final limit is $\lim_{x\to 0} \frac{\sin x}{x} = 1$ in disguise. It is the same fact that lets Archimedes squeeze π between perimeters of inscribed and circumscribed polygons. The question is also a toy version of a circle-packing density problem, here asking what fraction of a thin annulus the circles can fill. To make limits like this rigorous, try Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1987, Question 3

Pure • Separable Differential Equations

STEP I 1987, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q3

The key idea

The equation looks unpleasant in v, but the left-hand side is begging to be recognised: $x^3v' + x^2v$ is x^2 times the derivative of xv. So the suggested substitution turns the left-hand side into x^2y', and the right-hand side, rewritten in terms of y, separates neatly. After that it is partial fractions and a careful choice of sign.

Solution

Changing variable

With $y = xv$ the product rule gives $y' = v + xv'$, so

$$x^3\frac{\mathrm{d}v}{\mathrm{d}x} + x^2v = x^2\left(x\frac{\mathrm{d}v}{\mathrm{d}x} + v\right) = x^2\frac{\mathrm{d}y}{\mathrm{d}x}.$$

On the right, $x^2v^2 = y^2$ and $v = y/x$, so the right-hand side is

$$\frac{1+y^2}{(1+x^2)\,y/x} = \frac{x(1+y^2)}{(1+x^2)\,y}.$$

The equation becomes

$$x^2\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x(1+y^2)}{(1+x^2)\,y}, \qquad \text{that is,} \qquad \frac{y}{1+y^2}\,\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x(1+x^2)},$$

dividing by x^3 (we work near $x = 1$, so $x > 0$) and multiplying by $y/(1+y^2)$. The variables have separated.

Integrating

The left-hand side integrates to $\frac{1}{2}\ln(1+y^2)$. For the right, partial fractions give

$$\frac{1}{x(1+x^2)} = \frac{1}{x} - \frac{x}{1+x^2},$$

which is easily checked by putting the right-hand side over a common denominator. Hence, for $x > 0$,

$$\tfrac{1}{2}\ln(1+y^2) = \ln x - \tfrac{1}{2}\ln(1+x^2) + C, \qquad \text{so} \qquad 1+y^2 = \frac{Ax^2}{1+x^2}$$

for a positive constant $A = \mathrm{e}^{2C}$.

When $x = 1$ we have $v = 1$, so $y = 1$ too, giving $2 = A/2$ and $A = 4$. Therefore

$$y^2 = \frac{4x^2}{1+x^2} - 1 = \frac{3x^2-1}{1+x^2}, \qquad v^2 = \frac{y^2}{x^2} = \frac{3x^2-1}{x^2(1+x^2)}.$$

? **Ask yourself** Which square root do we take, and where is the solution valid?

The original equation has v in a denominator, so a solution can never pass through $v = 0$; being continuous, it keeps the sign it has at $x = 1$, which is positive. So we take the positive root:

$$\boxed{v(x) = \frac{\sqrt{3x^2-1}}{x\sqrt{1+x^2}}}$$

valid for $x > 1/\sqrt{3}$. As x decreases to $1/\sqrt{3}$, v falls to 0 and the equation breaks down, so the solution cannot be continued below that point.

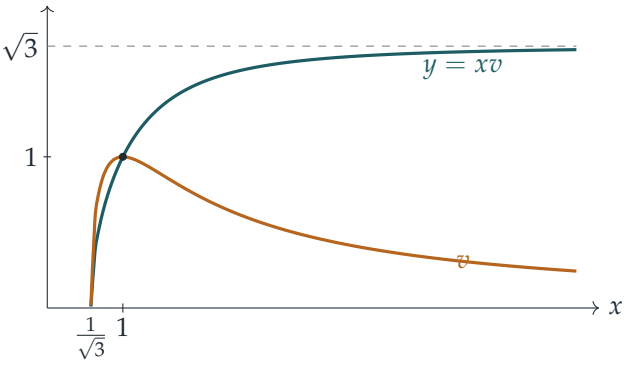
Large x

Dividing the top and bottom by x^2,

$$v(x) = \frac{1}{x}\cdot\frac{\sqrt{3-1/x^2}}{\sqrt{1+1/x^2}} \approx \frac{\sqrt{3}}{x},$$

so $v(x) \to 0$ as $x \to \infty$. Equivalently, $y = xv \to \sqrt{3}$: the graph of y has the horizontal asymptote $y = \sqrt{3}$, and $v = y/x$ decays like $\sqrt{3}/x$.

In fact the starting point is the peak: writing $u = x^2$, the derivative of $(3u-1)/(u^2+u)$ has numerator $-(3u+1)(u-1)$, so v rises to its maximum value 1 at $x = 1$ and then decreases towards 0.



Checks

- At $x = 1$: $v = \sqrt{2}/(1 \cdot \sqrt{2}) = 1$, as required.
- Substituting back: from $1 + y^2 = 4x^2/(1 + x^2)$, implicit differentiation gives $2yy' = 8x/(1 + x^2)^2$, and $(1 + y^2)/(x(1 + x^2)) = 4x/(1 + x^2)^2$, so $yy'/(1 + y^2) = 1/(x(1 + x^2))$ as it should.
- Numerically, stepping the original equation from $(1, 1)$ with a Runge–Kutta method gives $v(10) \approx 0.172058$, matching the formula, and at $x = 10^4$ the formula gives $xv \approx 1.73205 \approx \sqrt{3}$.

Common slips

- Answering $\sqrt{3}$ for the limit: that is the limit of y, but the question asks about v, which tends to 0.
- Exponentiating the constant wrongly: $\ln(1 + y^2) = \ln \frac{x^2}{1+x^2} + 2C$ gives $1 + y^2 = Ax^2/(1 + x^2)$, a multiplicative constant, not $x^2/(1 + x^2) + 2C$.
- Taking $\pm$ in the square root without deciding: the condition $v(1) = 1$ and the ban on $v = 0$ fix the positive sign.

Beyond STEP

The substitution here works because the left-hand side is an exact derivative after multiplying through, the same idea that lies behind integrating factors and homogeneous equations. The care needed over the sign and the domain is a first glimpse of existence and uniqueness theory: a solution through a given point exists until it runs into a place where the equation stops making sense, here $v = 0$. Boyce and DiPrima, *Elementary Differential Equations*, treats separable equations and the question of how far a solution extends.

WORKED SOLUTION

STEP I 1987, Question 4

Pure • Logarithms and Geometric Series

STEP I 1987, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q4

The key idea

Logarithms to awkward bases are best turned into natural logarithms, because then the base shows up as a simple factor. Every base here is a power of 2, and the exponents 2^n come down as multipliers. Once each term is written in terms of $\ln 2$, the series is just a geometric series in disguise.

Solution

Rewriting a general term

Fix $n \geq 0$ and let $c = \log_{2^{2^n}} \mathrm{e}$. By the definition in the hint, this means

$$\left(2^{2^n}\right)^c = \mathrm{e}.$$

Taking natural logarithms of both sides (both are positive),

$$c \cdot 2^n \ln 2 = \ln \mathrm{e} = 1, \qquad \text{so} \qquad \log_{2^{2^n}} \mathrm{e} = \frac{1}{2^n \ln 2}.$$

As a check on the indexing, $n = 0, 1, 2$ give the bases 2, 4, 16, matching the first three terms, with values $\frac{1}{\ln 2}$, $\frac{1}{2\ln 2}$, $\frac{1}{4\ln 2}$.

Ask yourself Where did the 2^n come from?

From the rule $\ln(a^k) = k \ln a$, used with $a = 2$ and $k = 2^n$: the base 2^{2^n} has natural logarithm $2^n \ln 2$. The "tower" of exponents is what makes each base the *square* of the one before, and squaring the base halves the logarithm.

Summing the series

The series is therefore

$$\sum_{n=0}^{\infty} (-1)^n \log_{2^{2^n}} \mathrm{e} = \frac{1}{\ln 2} \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n.$$

This is a geometric series with first term 1 and common ratio $r = -\frac{1}{2}$. Since $|r| < 1$ it converges, to $\dfrac{1}{1-r} = \dfrac{1}{1+\frac{1}{2}} = \dfrac{2}{3}$. Hence

$$\sum_{n=0}^{\infty} (-1)^n \log_{2^{2^n}} \mathrm{e} = \frac{2}{3\ln 2} = \frac{1}{\frac{3}{2}\ln 2} = \frac{1}{\ln\left(2^{3/2}\right)} = \frac{1}{\ln(2\sqrt{2})},$$

using $\ln(a^k) = k \ln a$ again, and $2^{3/2} = 2 \cdot 2^{1/2} = 2\sqrt{2}$. This is the required sum.

Checks

Numerically, $\frac{1}{\ln(2\sqrt{2})} \approx 0.96180$. The partial sums of the series are 1.4427, 0.7213, 1.0820, 0.9017, 0.9919, 0.9468, …, swinging either side of this value with the gap halving each time, as an alternating geometric series should. Sixty terms agree with 0.961797 to machine precision.

Common slips

- Reading $\log_{2^{2^n}}$ as $\log_{2^{2n}}$: that would give bases 2, 16, 256, … and the wrong ratio $-\frac{1}{4}$. Check the indexing against the first few terms.
- Writing $\log_a \mathrm{e} = \ln a$ instead of $\dfrac{1}{\ln a}$. The change-of-base rule gives $\log_a b = \dfrac{\ln b}{\ln a}$, and here $\ln b = 1$.

Beyond STEP

Whether an infinite sum "has a value" at all is the central question of a first course in analysis. Geometric series are the basic benchmark: many convergence tests, such as the ratio test, work by comparing a series with a geometric one. Abbott's *Understanding Analysis* builds up this theory carefully and readably.

WORKED SOLUTION

STEP I 1987, Question 5

Pure • Integration by Substitution

STEP I 1987, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q5

The key idea

The substitution is built so that both factors under the square root become perfect squares: $x-\alpha$ turns into a multiple of $\sin^2\theta$ and $\beta-x$ into the same multiple of $\cos^2\theta$. The square root then cancels exactly against $\mathrm{d}x/\mathrm{d}\theta$, and the first integral collapses to a constant integrand. The second integral keeps one extra factor $1/x$, which becomes a standard $\tan\theta$ integral once we divide top and bottom by $\cos^2\theta$.

Solution

The first integral, with $\alpha < \beta$

Put $x = \alpha\cos^2\theta + \beta\sin^2\theta$. Using $\cos^2\theta + \sin^2\theta = 1$,

$$x-\alpha = \alpha\cos^2\theta + \beta\sin^2\theta - \alpha(\cos^2\theta+\sin^2\theta) = (\beta-\alpha)\sin^2\theta,$$
$$\beta-x = \beta(\cos^2\theta+\sin^2\theta) - \alpha\cos^2\theta - \beta\sin^2\theta = (\beta-\alpha)\cos^2\theta,$$

and

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -2\alpha\cos\theta\sin\theta + 2\beta\sin\theta\cos\theta = 2(\beta-\alpha)\sin\theta\cos\theta.$$

For the limits: $\theta = 0$ gives $x = \alpha$ and $\theta = \frac{\pi}{2}$ gives $x = \beta$. Since $\mathrm{d}x/\mathrm{d}\theta > 0$ on $(0, \frac{\pi}{2})$, as θ runs over $[0, \frac{\pi}{2}]$, x runs once, increasingly, over $[\alpha, \beta]$.

On this range $\sin\theta \geq 0$ and $\cos\theta \geq 0$, and $\beta - \alpha > 0$, so

$$\sqrt{(x-\alpha)(\beta-x)} = \sqrt{(\beta-\alpha)^2\sin^2\theta\cos^2\theta} = (\beta-\alpha)\sin\theta\cos\theta.$$

Therefore

$$\int_\alpha^\beta \frac{\mathrm{d}x}{\sqrt{(x-\alpha)(\beta-x)}} = \int_0^{\pi/2} \frac{2(\beta-\alpha)\sin\theta\cos\theta}{(\beta-\alpha)\sin\theta\cos\theta}\,\mathrm{d}\theta = \int_0^{\pi/2} 2\,\mathrm{d}\theta = \pi,$$

as required. (The integrand blows up at both ends, so strictly this is an improper integral. The substitution shows it converges, because the transformed integrand is the bounded constant 2.)

Ask yourself Where exactly did we use $\alpha < \beta$?

Only in taking the square root: $\sqrt{(\beta-\alpha)^2}$ equals $\beta-\alpha$ precisely because $\beta-\alpha>0$. In general it is $|\beta-\alpha|$, and that is what drives the next part.

The first integral, with $\alpha > \beta$

Now the integral runs from the larger number α down to the smaller number β. On the interval between them, $x-\alpha<0$ and $\beta-x<0$, so the product is still positive and the integrand still makes sense.

The substitution goes through word for word, except that now $\sqrt{(\beta-\alpha)^2}=\alpha-\beta$, so

$$\frac{2(\beta-\alpha)\sin\theta\cos\theta}{(\alpha-\beta)\sin\theta\cos\theta}=-2,$$

and the integral is

$$\int_0^{\pi/2}(-2)\,\mathrm{d}\theta=-\pi.$$

Here is a quicker way to see it. The expression $(x-\alpha)(\beta-x)$ is unchanged when we swap α and β, so the integrand is the same function as in the first case with the roles of the two letters exchanged. Reversing the limits changes the sign, so

$$\int_\alpha^\beta = -\int_\beta^\alpha = -\pi,$$

using the first result with β as the smaller endpoint.

The second integral, with $0 < \alpha < \beta$

Exactly the same substitution, with the same cancellation, leaves only the factor $1/x$:

$$\int_\alpha^\beta \frac{\mathrm{d}x}{x\sqrt{(x-\alpha)(\beta-x)}}=\int_0^{\pi/2}\frac{2\,\mathrm{d}\theta}{\alpha\cos^2\theta+\beta\sin^2\theta}.$$

Because $\alpha,\beta>0$, the denominator is positive throughout $[0,\frac{\pi}{2}]$, so this is a proper integral. To evaluate it, divide numerator and denominator by $\cos^2\theta$ (valid for $0\le\theta<\frac{\pi}{2}$):

$$\int_0^{\pi/2}\frac{2\sec^2\theta\,\mathrm{d}\theta}{\alpha+\beta\tan^2\theta}.$$

Now put $t=\tan\theta$, so $\mathrm{d}t=\sec^2\theta\,\mathrm{d}\theta$, and t runs from 0 to ∞ as θ runs from 0 up to $\frac{\pi}{2}$:

$$\begin{aligned}\int_0^\infty\frac{2\,\mathrm{d}t}{\alpha+\beta t^2}&=\frac{2}{\beta}\int_0^\infty\frac{\mathrm{d}t}{t^2+\alpha/\beta}=\frac{2}{\beta}\sqrt{\frac{\beta}{\alpha}}\left[\arctan\left(t\sqrt{\frac{\beta}{\alpha}}\right)\right]_0^\infty\\&=\frac{2}{\sqrt{\alpha\beta}}\cdot\frac{\pi}{2}=\frac{\pi}{\sqrt{\alpha\beta}},\end{aligned}$$

as required. We needed $\alpha>0$ twice: so that $1/x$ has no singularity inside $[\alpha,\beta]$, and so that $\sqrt{\beta/\alpha}$ is real.

Ask yourself The upper limit became ∞. Is that a problem?

No. The θ-integral is perfectly ordinary; the infinite limit only appears because $\tan\theta \to \infty$ as $\theta \to \frac{\pi}{2}$. The t-integral converges, since arctan tends to $\frac{\pi}{2}$, and it equals the θ-integral as a limit of the integrals over $[0, \Theta]$ with $\Theta \to \frac{\pi}{2}$.

Checks

- Scaling: replacing x, α, β by kx, $k\alpha$, $k\beta$ leaves the first integral unchanged and divides the second by k. Both answers, π and $\pi/\sqrt{\alpha\beta}$, behave this way.
- Let $\alpha \to \beta$. Then $1/x \approx 1/\beta$ across the whole (tiny) interval, so the second integral should be about π/β, and indeed $\pi/\sqrt{\alpha\beta} \to \pi/\beta$.
- Numerically, with $\alpha = 2$, $\beta = 7$, quadrature gives 0.83963 for the second integral, matching $\pi/\sqrt{14} = 0.83963$. Random choices of $\alpha < \beta$ all gave π for the first integral, and swapping the endpoints gave $-\pi$.

Common slips

- Answering π again when $\alpha > \beta$. The integrand is positive, but the limits run backwards, so the answer is $-\pi$.
- Writing $\sqrt{(\beta-\alpha)^2\sin^2\theta\cos^2\theta} = (\beta-\alpha)\sin\theta\cos\theta$ without checking the signs of the three factors.

Beyond STEP

The substitution has a neat geometric meaning. Since $x = \frac{\alpha+\beta}{2} - \frac{\beta-\alpha}{2}\cos 2\theta$, it parametrises the semicircle on diameter $[\alpha, \beta]$, and $\sqrt{(x-\alpha)(\beta-x)}$ is the height of that semicircle above x. The second integral is a first taste of evaluating integrals by contour integration: in complex analysis, both results drop out of the residue theorem in a couple of lines. Needham's *Visual Complex Analysis* explains this beautifully, and the careful treatment of improper integrals is in Abbott's *Understanding Analysis*.

WORKED SOLUTION

STEP I 1987, Question 6

Pure • Areas and Inverse Functions

STEP I 1987, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q6

The key idea

$\mathrm{F}(s)$ is the area *under* the curve, between it and the x-axis; $\mathrm{G}(t)$ is the area to the *left* of the curve, between it and the y-axis. Drawn on the same picture, these two regions together cover the rectangle with corners $(0,0)$ and (s,t), with something to spare unless the corner (s,t) sits exactly on the curve. For the second part we choose s to make that happen, and then the hard integral $\int \sin^{-1} y \,\mathrm{d}y$ comes for free from the easy one, $\int \sin x \,\mathrm{d}x$.

Solution

The inequality $(*)$

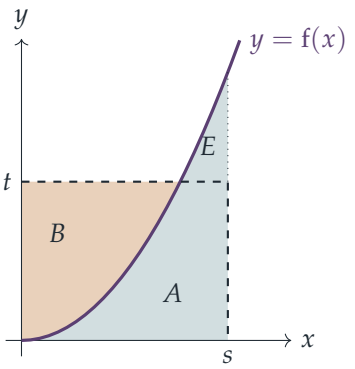
First, a fact about f. It is continuous and takes each value at most once, so it is strictly monotonic (a continuous function that went up and then down would take some value twice, by the intermediate value theorem). Since $\mathrm{f}(0) = 0$ and $\mathrm{f}(x) \geq 0$, it must be strictly *increasing*. So g is increasing too, and for $0 \leq x \leq a$, $0 \leq y \leq \mathrm{f}(a)$,

$$y \leq \mathrm{f}(x) \iff \mathrm{g}(y) \leq x.$$

Define two regions:

$$A = \{(x,y) : 0 \leq x \leq s,\ 0 \leq y \leq \mathrm{f}(x)\}, \qquad \text{area } \mathrm{F}(s),$$
$$B = \{(x,y) : 0 \leq y \leq t,\ 0 \leq x \leq \mathrm{g}(y)\}, \qquad \text{area } \mathrm{G}(t),$$

and let R be the rectangle $0 \leq x \leq s$, $0 \leq y \leq t$, of area st. The area of B is $\mathrm{G}(t)$ because we are integrating horizontal strips of length $\mathrm{g}(y)$ ("area with respect to y").



Every point of R lies in A or in B: if $(x, y) \in R$ then either $y \le \mathrm{f}(x)$, so the point is in A, or $y > \mathrm{f}(x)$, which means $x < \mathrm{g}(y)$, so the point is in B. The overlap $A \cap B$ is just the arc of the curve itself, which has zero area. Hence

$$\mathrm{F}(s) + \mathrm{G}(t) = \text{area}(A) + \text{area}(B) \ge \text{area}(A \cup B) \ge \text{area}(R) = st,$$

which is $(*)$.

Ask yourself Exactly what has been counted that isn't in the rectangle?

The excess is the part E of $A \cup B$ sticking out of R. If $t < \mathrm{f}(s)$ (as in the figure), B stays inside R, but A pokes out above $y = t$ in the region $\mathrm{g}(t) < x \le s$, $t < y \le \mathrm{f}(x)$. Since $\mathrm{f}(x) > t$ for every $x > \mathrm{g}(t)$ and f is continuous, this region has positive area, so the inequality is strict. If $t > \mathrm{f}(s)$, the same happens with the roles swapped: B pokes out to the right of $x = s$. If $t = \mathrm{f}(s)$, the corner (s, t) lies on the curve, $A \cup B$ is exactly R, and there is no excess.

So **equality holds in** $(*)$ **if and only if** $t = \mathrm{f}(s)$, that is, when (s, t) lies on the curve.

The integral of $\sin^{-1} y$

Take $\mathrm{f}(x) = \sin x$ on $[0, \frac{1}{2}\pi]$. It is continuous, increasing, passes through the origin and stays in the first quadrant, so everything above applies with $a = \frac{1}{2}\pi$, $\mathrm{f}(a) = 1$ and $\mathrm{g}(y) = \sin^{-1} y$.

Given $t \in [0, 1]$, choose $s = \sin^{-1} t$, which lies in $[0, \frac{1}{2}\pi]$. Then $t = \sin s = \mathrm{f}(s)$, so equality holds in $(*)$:

$$\int_0^t \sin^{-1} y \, \mathrm{d}y = \mathrm{G}(t) = st - \mathrm{F}(s).$$

Now the easy integral:

$$\mathrm{F}(s) = \int_0^s \sin x \, \mathrm{d}x = 1 - \cos s.$$

Substituting $s = \sin^{-1} t$,

$$\int_0^t \sin^{-1} y \, \mathrm{d}y = t \sin^{-1} t - \big(1 - \cos(\sin^{-1} t)\big),$$

as required.

Checking by differentiation

Since $\sin^{-1} t \in [0, \frac{1}{2}\pi]$, its cosine is non-negative, so $\cos(\sin^{-1} t) = \sqrt{1-t^2}$. The right-hand side is therefore $t\sin^{-1} t - 1 + \sqrt{1-t^2}$. For $0 \le t < 1$, using $\dfrac{\mathrm{d}}{\mathrm{d}t}\sin^{-1} t = \dfrac{1}{\sqrt{1-t^2}}$ and the product rule,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(t\sin^{-1} t - 1 + \sqrt{1-t^2}\right) &= \sin^{-1} t + \frac{t}{\sqrt{1-t^2}} - \frac{t}{\sqrt{1-t^2}} \\ &= \sin^{-1} t,\end{aligned}$$

which, by the fundamental theorem of calculus, is the derivative of the left-hand side. Both sides are 0 at $t = 0$, so they agree on $[0, 1)$, and by continuity at $t = 1$ as well.

Checks

- At $t = 1$: the formula gives $\frac{1}{2}\pi - 1 \approx 0.5708$. Geometrically, the area left of $y = \sin x$ up to height 1 is the rectangle $\frac{1}{2}\pi \times 1$ minus the area under the arch, 1. A numerical integration of $\sin^{-1} y$ over $[0, 1]$ gives 0.57080.
- At $t = \frac{1}{2}$: the formula gives $\frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2} \approx 0.12782$, matching Simpson's rule.
- For $(*)$, try $\mathrm{f}(x) = x^2$, $s = 1$, $t = 4$: $\mathrm{F} + \mathrm{G} = \frac{1}{3} + \frac{16}{3} = \frac{17}{3} > 4$, while with $t = 1 = \mathrm{f}(1)$ we get $\frac{1}{3} + \frac{2}{3} = 1 = st$.

Common slips

- Writing $\cos(\sin^{-1} t) = \pm\sqrt{1-t^2}$ without choosing the sign: the range of $\sin^{-1}$ makes it positive.
- Saying only "the regions cover the rectangle" without explaining why the inequality is strict when $t \ne \mathrm{f}(s)$; the equality question needs that.
- Forgetting that f must be increasing: the picture relies on it.

Beyond STEP

The inequality $(*)$ is *Young's inequality*. With $\mathrm{f}(x) = x^{p-1}$ it becomes $st \le \frac{s^p}{p} + \frac{t^q}{q}$ where $\frac{1}{p} + \frac{1}{q} = 1$, the key step in proving Hölder's inequality, one of the workhorses of analysis. The pairing of F and G is an early glimpse of the Legendre transform, which links Lagrangian and Hamiltonian mechanics. For the analysis behind all this, try Bartle and Sherbert, *Introduction to Real Analysis*.

WORKED SOLUTION

STEP I 1987, Question 7

Pure • Trigonometric Series

STEP I 1987, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q7

The key idea

Both series are sums of sines whose angles go up in equal steps. Multiplying a term like $\sin\big((2k+1)\theta\big)$ by $2\sin\theta$ or $2\cos\theta$ turns it, by a product-to-sum formula, into a *difference* or *sum* of two neighbouring terms, so the whole series telescopes. The only choice is which factor to use: $2\sin\theta$ suits the plain sum, and $2\cos\theta$ suits the alternating one.

Solution

Setting up

Write $\theta = \dfrac{2\pi}{23}$. The angles are $2\pi/23$, $6\pi/23$, $10\pi/23, \ldots, 42\pi/23$, increasing by $4\pi/23 = 2\theta$ each time, so they are

$$(2k+1)\theta \qquad \text{for } k = 0, 1, \ldots, 10,$$

since $21\theta = 42\pi/23$. That makes 11 terms, and in the alternating series term k has sign $(-1)^k$ (the last term, $k = 10$, is $+$, as it should be). So we want

$$S_1 = \sum_{k=0}^{10} \sin\big((2k+1)\theta\big), \qquad S_2 = \sum_{k=0}^{10} (-1)^k \sin\big((2k+1)\theta\big).$$

We shall use $22\theta = \dfrac{44\pi}{23} = 2\pi - \dfrac{2\pi}{23}$, so that

$$\cos 22\theta = \cos\theta, \qquad \sin 22\theta = -\sin\theta.$$

The first series

By the product-to-sum formula $2\sin A \sin B = \cos(A-B) - \cos(A+B)$,

$$2\sin\theta\,\sin\big((2k+1)\theta\big) = \cos(2k\theta) - \cos\big((2k+2)\theta\big).$$

Summing over $k = 0, \ldots, 10$, each $\cos(2k\theta)$ with $1 \le k \le 10$ appears once with each sign and cancels, leaving

$$2\sin\theta\, S_1 = \cos 0 - \cos 22\theta = 1 - \cos\theta.$$

Since $0 < \theta < \pi$, $\sin\theta \neq 0$ and we may divide:

$$S_1 = \frac{1-\cos\theta}{2\sin\theta} = \frac{2\sin^2(\theta/2)}{4\sin(\theta/2)\cos(\theta/2)} = \tfrac{1}{2}\tan\frac{\theta}{2},$$

using $1 - \cos\theta = 2\sin^2(\theta/2)$ and $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$, and cancelling $\sin(\theta/2) \neq 0$. With $\theta/2 = \pi/23$,

$$\boxed{\sin\tfrac{2\pi}{23} + \sin\tfrac{6\pi}{23} + \cdots + \sin\tfrac{42\pi}{23} = \tfrac{1}{2}\tan\tfrac{\pi}{23}.}$$

Ask yourself Why multiply by $2\cos\theta$ for the alternating series, rather than $2\sin\theta$ again?

With $2\sin\theta$ we get differences $\cos(2k\theta) - \cos((2k+2)\theta)$, and the alternating signs would make neighbouring terms *add* rather than cancel. With $2\cos\theta$ we get *sums* $\sin((2k+2)\theta) + \sin(2k\theta)$, and now it is the alternating sign that produces the cancellation.

The second series

By $2\cos A\sin B = \sin(B+A) + \sin(B-A)$,

$$2\cos\theta\,\sin\big((2k+1)\theta\big) = \sin\big((2k+2)\theta\big) + \sin(2k\theta).$$

Multiply by $(-1)^k$ and sum. For $1 \le k \le 10$, $\sin(2k\theta)$ appears from term $k-1$ with sign $(-1)^{k-1}$ and from term k with sign $(-1)^k$, so it cancels. What survives is $\sin 0$ from $k = 0$ and $(-1)^{10}\sin 22\theta$ from $k = 10$:

$$2\cos\theta\, S_2 = \sin 22\theta = -\sin\theta.$$

Since $0 < \theta < \pi/2$, $\cos\theta \neq 0$, so

$$S_2 = -\frac{\sin\theta}{2\cos\theta} = -\tfrac{1}{2}\tan\theta,$$

that is,

$$\boxed{\sin\tfrac{2\pi}{23} - \sin\tfrac{6\pi}{23} + \cdots - \sin\tfrac{38\pi}{23} + \sin\tfrac{42\pi}{23} = -\tfrac{1}{2}\tan\tfrac{2\pi}{23}.}$$

If you prefer a positive-looking form, this is also $\frac{1}{2}\tan\frac{21\pi}{23}$, since $\tan(\pi - x) = -\tan x$.

Checks

- Numerically, $\frac{1}{2}\tan(\pi/23) \approx 0.068723$ and $-\frac{1}{2}\tan(2\pi/23) \approx -0.140093$; adding up the eleven terms directly gives the same values to 15 significant figures.

- The negative sign of the second answer is believable: the partial sums swing between about ± 0.5, and the last term added, $\sin(42\pi/23) \approx -0.27$, is itself negative.
- The same method gives the general formulas $\sum_{k=0}^{n-1} \sin((2k+1)\theta) = \dfrac{\sin^2 n\theta}{\sin\theta}$ and $\sum_{k=0}^{n-1} (-1)^k \sin((2k+1)\theta) = (-1)^{n-1} \dfrac{\sin 2n\theta}{2\cos\theta}$; try $n = 1$, where both reduce to $\sin\theta$.

Common slips

- Miscounting the terms: the angles step by $4\pi/23$, not $2\pi/23$, so there are 11 terms, not 21.
- Expecting both answers to be positive. Angles beyond π have negative sines, and $\sin(44\pi/23) = -\sin(2\pi/23)$ is where the minus sign comes from.

Beyond STEP

These sums are the imaginary parts of geometric series: $S_1 = \operatorname{Im} \sum \mathrm{e}^{\mathrm{i}(2k+1)\theta}$, and S_2 is the same with ratio $-\mathrm{e}^{2\mathrm{i}\theta}$, which gives a second route to both answers. Sums of this kind, such as the Dirichlet kernel, are the starting point of Fourier analysis, where a function is built from sines and cosines. Needham's *Visual Complex Analysis* shows why such sums behave as they do, using pictures of rotating vectors.

WORKED SOLUTION

STEP I 1987, Question 8

Pure • Integration by Substitution

STEP I 1987, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q8

The key idea

A substitution is only a licence to change variables if the new variable moves smoothly and in one direction through an interval as the old one does. Here $t = 1/x$ blows up at $x = 0$, which sits right in the middle of $[-1, 1]$, so the "new limits" -1 and 1 describe a completely different set of t-values. A quick look at the signs of the two integrands confirms something has gone wrong.

Solution

Why the substitution fails

Formally, with $x = 1/t$ we have $\mathrm{d}x = -\frac{1}{t^2}\,\mathrm{d}t$ and

$$\frac{1}{(1+x^2)^2} = \frac{1}{\left(1+\frac{1}{t^2}\right)^2} = \frac{t^4}{(1+t^2)^2}, \qquad \text{so} \qquad \frac{\mathrm{d}x}{(1+x^2)^2} = \frac{-t^2}{(1+t^2)^2}\,\mathrm{d}t,$$

and $x = \pm 1$ corresponds to $t = \pm 1$. This is how one would "show" that the integrals are equal. But it can't be right: the first integrand is positive everywhere, so the first integral is positive, while the second integrand is ≤ 0 everywhere, so the second integral is negative.

Ask yourself As x runs from -1 to 1, which values does $t = 1/x$ actually take?

As x goes from -1 up to 0^-, t goes from -1 down to $-\infty$; at $x = 0$ there is no t at all; and as x goes from 0^+ to 1, t comes down from $+\infty$ to 1. So the x-interval $[-1, 1]$ corresponds to $t \in (-\infty, -1] \cup [1, \infty)$, the *outside* of $[-1, 1]$, not $[-1, 1]$ itself. The substitution rule needs $x = \phi(t)$ with ϕ differentiable on the whole t-interval between the new limits, and $\phi(t) = 1/t$ is not even defined at $t = 0 \in [-1, 1]$. The values of t in $(-1, 1)$ correspond to $|x| > 1$, which lie outside the original range.

Used correctly, on each half separately, the substitution gives for example

$$\int_0^1 \frac{\mathrm{d}x}{(1+x^2)^2} = \int_\infty^1 \frac{-t^2}{(1+t^2)^2}\,\mathrm{d}t = \int_1^\infty \frac{t^2}{(1+t^2)^2}\,\mathrm{d}t,$$

which is a true statement (we check it below), but it says nothing about the integral over $[-1,1]$ in t.

The first integral

Put $x = \tan\theta$, so $\mathrm{d}x = \sec^2\theta\,\mathrm{d}\theta$ and $1+x^2 = \sec^2\theta$; this substitution is fine, since tan is smooth and increasing on $[-\frac{\pi}{4}, \frac{\pi}{4}]$. Then

$$\begin{aligned}\int \frac{\mathrm{d}x}{(1+x^2)^2} &= \int \frac{\sec^2\theta}{\sec^4\theta}\,\mathrm{d}\theta = \int \cos^2\theta\,\mathrm{d}\theta = \int \tfrac{1}{2}(1+\cos 2\theta)\,\mathrm{d}\theta \\ &= \tfrac{1}{2}\theta + \tfrac{1}{4}\sin 2\theta = \tfrac{1}{2}\theta + \tfrac{1}{2}\cdot\frac{\tan\theta}{1+\tan^2\theta} = \tfrac{1}{2}\arctan x + \frac{x}{2(1+x^2)},\end{aligned}$$

using $\sin 2\theta = 2\sin\theta\cos\theta = 2\tan\theta\cos^2\theta$. Hence

$$\int_{-1}^1 \frac{\mathrm{d}x}{(1+x^2)^2} = \left[\tfrac{1}{2}\arctan x + \frac{x}{2(1+x^2)}\right]_{-1}^1 = 2\left(\frac{\pi}{8}+\frac{1}{4}\right) = \boxed{\frac{\pi}{4}+\frac{1}{2}}.$$

The second integral

Write the numerator as $-t^2 = 1-(1+t^2)$:

$$\frac{-t^2}{(1+t^2)^2} = \frac{1}{(1+t^2)^2} - \frac{1}{1+t^2}.$$

The first piece integrates to $\frac{\pi}{4}+\frac{1}{2}$ over $[-1,1]$, by the previous part, and $\int_{-1}^1 \frac{\mathrm{d}t}{1+t^2} = \left[\arctan t\right]_{-1}^1 = \frac{\pi}{2}$. So

$$\int_{-1}^1 \frac{-t^2}{(1+t^2)^2}\,\mathrm{d}t = \frac{\pi}{4}+\frac{1}{2}-\frac{\pi}{2} = \boxed{\frac{1}{2}-\frac{\pi}{4}},$$

which is negative (≈ -0.285), as the sign of the integrand demands, while the first integral is about 1.285.

Checks

- Simpson's rule gives 1.2853981… and −0.2853981… for the two integrals, agreeing with $\frac{\pi}{4}+\frac{1}{2}$ and $\frac{1}{2}-\frac{\pi}{4}$.
- The two answers add up to 1. There's a reason: $\frac{1}{(1+x^2)^2} - \frac{x^2}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$, which is the derivative of $\frac{x}{1+x^2}$, and $\left[\frac{x}{1+x^2}\right]_{-1}^1 = 1$.
- The correct half-range statement checks out: $\int_1^\infty \frac{t^2}{(1+t^2)^2}\,\mathrm{d}t = \frac{\pi}{4} - \left(\frac{\pi}{4}-\frac{\pi}{8}-\frac{1}{4}\right) = \frac{\pi}{8}+\frac{1}{4} = \int_0^1 \frac{\mathrm{d}x}{(1+x^2)^2}$.

Common slips

- Saying only "$1/t$ is undefined at $t = 0$" without explaining that the x-range $[-1, 1]$ corresponds to $|t| \geq 1$: the real point is that the two integrals cover different sets of values.
- Getting the sign of $\mathrm{d}x = -\frac{1}{t^2}\,\mathrm{d}t$ wrong, which hides the tell-tale sign contradiction.
- Forgetting the $\frac{1}{4}\sin 2\theta$ term when integrating $\cos^2\theta$.

Beyond STEP

The substitution rule $\int_a^b f(\phi(t))\phi'(t)\,\mathrm{d}t = \int_{\phi(a)}^{\phi(b)} f(x)\,\mathrm{d}x$ is proved in a first analysis course from the chain rule and the Fundamental Theorem of Calculus, and the proof shows exactly what is needed: ϕ must be continuously differentiable on all of $[a, b]$. Here it fails at one point, and that one point is enough to wreck the result. Abbott's *Understanding Analysis* develops the Riemann integral and these theorems with care.

WORKED SOLUTION

STEP I 1987, Question 9

Pure • Barycentric Coordinates

STEP I 1987, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q9

The key idea

Moving from A towards B is a matter of adding a fraction of the vector $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$. Doing this once gives points on a segment. Doing it twice, first along AB and then from that point towards C, sweeps out the whole triangle. For the sketches, the trick is to rewrite each case so it looks like something we already understand: either a flip of the triangle case, or a point A plus multiples of two edge vectors.

Solution

Points on AB

A point P on the segment AB is reached from A by moving a fraction t of the way along $\overrightarrow{AB}$, with $0 \le t \le 1$ ($t = 0$ gives A and $t = 1$ gives B). So

$$\mathbf{p} = \mathbf{a} + t(\mathbf{b} - \mathbf{a}) = (1 - t)\mathbf{a} + t\,\mathbf{b}.$$

Put $\rho = 1 - t$ and $\sigma = t$. Then $\rho \ge 0$ and $\sigma \ge 0$ because $0 \le t \le 1$, and $\rho + \sigma = 1$. This is the required form. Conversely, every such pair ρ, σ gives a point of the segment, by taking $t = \sigma$.

Points on PC, and points inside the triangle

In exactly the same way, a point Q on the segment PC has position vector

$$\mathbf{q} = (1 - s)\mathbf{p} + s\,\mathbf{c}, \qquad 0 \le s \le 1.$$

Substituting $\mathbf{p} = \rho\mathbf{a} + \sigma\mathbf{b}$,

$$\mathbf{q} = (1 - s)\rho\,\mathbf{a} + (1 - s)\sigma\,\mathbf{b} + s\,\mathbf{c}.$$

Let $\lambda = (1 - s)\rho$, $\mu = (1 - s)\sigma$ and $\nu = s$. Each is a product of non-negative numbers, so $\lambda, \mu, \nu \ge 0$. Also

$$\lambda + \mu + \nu = (1 - s)(\rho + \sigma) + s = (1 - s) + s = 1.$$

Ask yourself Why does every point inside the triangle lie on some segment PC?

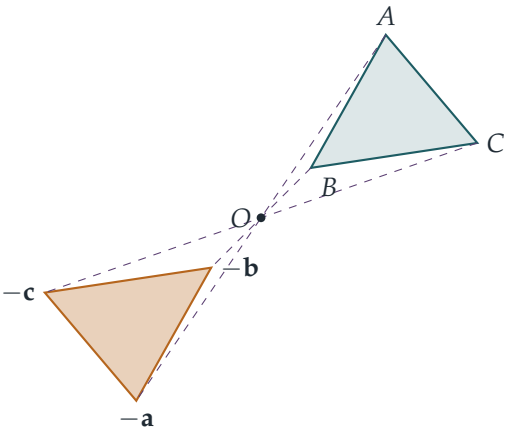
Take a point X inside the triangle (or on its boundary). If $X = C$ there is nothing to prove: take $s = 1$. Otherwise draw the line from C through X and carry it on beyond X. Because X lies inside the angle ACB, this line stays between CA and CB, so it must cross the opposite side AB, at a point P say. Since X is in the triangle, it lies between C and P, so X is on the segment PC. By the working above, $\mathbf{x} = \lambda\mathbf{a} + \mu\mathbf{b} + \nu\mathbf{c}$ with $\lambda, \mu, \nu \geq 0$ and $\lambda + \mu + \nu = 1$, as required.

Part (i)

Write $\lambda = -\lambda'$, $\mu = -\mu'$ and $\nu = -\nu'$. The conditions become $\lambda', \mu', \nu' \geq 0$ and $\lambda' + \mu' + \nu' = 1$, and the point is

$$\lambda\mathbf{a} + \mu\mathbf{b} + \nu\mathbf{c} = -\left(\lambda'\mathbf{a} + \mu'\mathbf{b} + \nu'\mathbf{c}\right).$$

The bracket runs over every point of the triangle ABC, by what we have just shown (with the converse from the first part, every such combination is in the triangle). So the region is the set of points $-\mathbf{x}$ with X in triangle ABC. In other words, it is the triangle with vertices $-\mathbf{a}$, $-\mathbf{b}$, $-\mathbf{c}$, together with its interior: the image of triangle ABC under a half-turn about the origin O.



The shaded amber triangle is the answer to (i). Each vertex is the reflection of a vertex of ABC through O, shown by the dashed lines.

Part (ii)

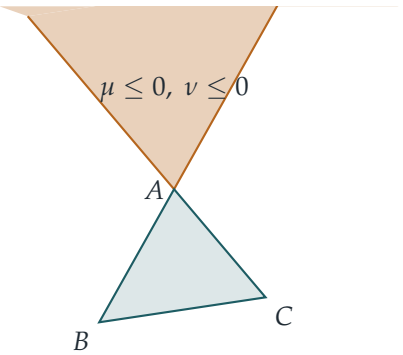
Now $\lambda = 1 - \mu - \nu$, so

$$\lambda\mathbf{a} + \mu\mathbf{b} + \nu\mathbf{c} = \mathbf{a} + \mu(\mathbf{b} - \mathbf{a}) + \nu(\mathbf{c} - \mathbf{a}).$$

We start at A and add multiples of $\overrightarrow{AB}$ and $\overrightarrow{AC}$. Writing $\mu = -m$ and $\nu = -n$ with $m, n \geq 0$, the point is

$$\mathbf{a} + m(\mathbf{a} - \mathbf{b}) + n(\mathbf{a} - \mathbf{c}), \qquad m \geq 0,\ n \geq 0.$$

The vector $\mathbf{a} - \mathbf{b}$ points along BA continued beyond A, and $\mathbf{a} - \mathbf{c}$ points along CA continued beyond A. Non-negative combinations of two non-parallel vectors fill exactly the angle between them (since $\mathbf{b} - \mathbf{a}$ and $\mathbf{c} - \mathbf{a}$ are not parallel, every point of the plane is $\mathbf{a} + m(\mathbf{a} - \mathbf{b}) + n(\mathbf{a} - \mathbf{c})$ for exactly one pair m, n). So the region is the infinite wedge with vertex A bounded by the extensions of BA and CA beyond A, boundary rays included. It is the angle vertically opposite to angle BAC. Note that λ isn't restricted in the question, but here it is automatically at least 1.



The amber wedge (continuing without end) is the answer to (ii). Its edges are the sides BA and CA extended through A. The teal triangle ABC is shown for reference.

Checks

- Vertices: in (ii), $(\lambda, \mu, \nu) = (1, 0, 0)$ gives A itself, the tip of the wedge; $(2, -1, 0)$ gives $2\mathbf{a} - \mathbf{b}$, the reflection of B in A, which lies on one edge. In (i), $(-1, 0, 0)$ gives $-\mathbf{a}$, a vertex of the reflected triangle.
- Centroids: in (i), $\lambda = \mu = \nu = -\frac{1}{3}$ gives minus the centroid of ABC, which is the centroid of the reflected triangle, as it should be.
- Numerically, a computer took 20,000 random triangles and random coefficients satisfying each set of conditions. It confirmed that every point from (i) lies in the triangle $(-\mathbf{a}, -\mathbf{b}, -\mathbf{c})$, that every point from (ii) has non-negative coordinates along $\mathbf{a} - \mathbf{b}$ and $\mathbf{a} - \mathbf{c}$ from A, and that every point of ABC splits as a point of some segment PC.

Common slips

- In (i), drawing a region near ABC. The coefficients sum to -1, not 1, so the region is the half-turn of ABC about the origin and depends on where O is.

- In (ii), drawing only a triangle. There is no lower bound on μ and ν, so the region is unbounded.
- Shading the angle next to BAC (for example between AB and CA extended). That would need one of μ, ν positive.

Beyond STEP

The numbers (λ, μ, ν) with $\lambda + \mu + \nu = 1$ are the *barycentric coordinates* of a point with respect to the triangle. When the sum is 1 the point doesn't depend on where the origin is: shifting every position vector by $\mathbf{d}$ shifts $\lambda\mathbf{a} + \mu\mathbf{b} + \nu\mathbf{c}$ by $(\lambda + \mu + \nu)\mathbf{d} = \mathbf{d}$ too. That is why the answer to (ii) is a genuine geometric region, while (i) depends on O. The triangle itself is the *convex hull* of A, B, C, an idea at the heart of linear programming and convex optimisation. The signs of the barycentric coordinates split the plane into seven regions, of which (ii) is one. For a first-year view of these affine ideas, see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP I 1987, Question 10

Mechanics • Elastic Strings

STEP I 1987, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q10

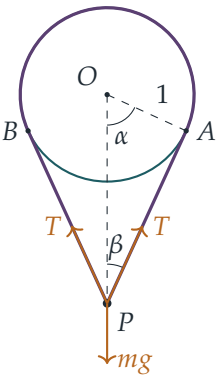
The key idea

The particle drags the bottom of the band down off the cylinder, so the band has two parts: an arc that still hugs the cylinder over the top, and two straight pieces that leave the cylinder *along tangents* and meet at the particle. The stretched length is then pure geometry: right-angled triangles give the straight pieces and the angle subtended at the axis. Because the cylinder is smooth, the tension is the same all the way round, and resolving at the particle links that tension to mg; Hooke's law then gives λ.

Solution

The model

We model the band as a light elastic string of natural length 2π obeying Hooke's law, the cylinder as smooth, and the particle as a point mass. We work in a vertical cross-section: the cylinder is a unit circle with centre O on the axis, and the particle is at P, a distance x vertically below O. For the band to leave the cylinder at all we need $x > 1$.



By symmetry about the vertical through O, the band leaves the cylinder at two points

A and B, one on each side. A taut string that comes off a smooth curve leaves it along the tangent, so PA is tangent to the circle at A, which means $OA \perp AP$. Let $\alpha = \angle AOP$ and $\beta = \angle APO$.

The stretched length

Triangle OAP has a right angle at A, with $OA = 1$ and hypotenuse $OP = x$. Hence

$$AP = \sqrt{x^2 - 1}, \qquad \cos\alpha = \frac{1}{x}, \qquad \sin\beta = \frac{1}{x}.$$

The band is in contact with the cylinder everywhere except between A and B through the bottom, an angle of 2α at O. So the arc in contact subtends $2\pi - 2\alpha$, and on a unit circle its length is $2\pi - 2\alpha$. Adding the two straight pieces, the stretched length is

$$\boxed{L = 2\sqrt{x^2 - 1} + 2\pi - 2\arccos\frac{1}{x}}$$

where $\arccos(1/x)$ lies between 0 and $\frac{\pi}{2}$, since $x > 1$. The extension is therefore

$$L - 2\pi = 2\sqrt{x^2 - 1} - 2\arccos\frac{1}{x}.$$

Ask yourself The band is curved round the cylinder. Why is the tension the same in the straight pieces as in the arc?

On a small piece of band in contact with a smooth surface, the only force from the cylinder is normal to the surface, so there is nothing to balance a change in tension along the band. Hence the tension T is constant along the whole band, and Hooke's law applies to the band as a whole:

$$T = \frac{\lambda\,(L - 2\pi)}{2\pi}.$$

(If the cylinder were rough, the tension could vary round the arc, and this step would fail.)

Equilibrium of the particle

The particle is acted on by its weight mg and by tensions T along PA and PB, each at angle β to the upward vertical. The horizontal components cancel by symmetry. Resolving vertically,

$$2T\cos\beta = mg, \qquad \text{with} \qquad \cos\beta = \frac{\sqrt{x^2 - 1}}{x},$$

so

$$T = \frac{mgx}{2\sqrt{x^2 - 1}}.$$

The case $x = 2$

Now $\cos\alpha = \frac{1}{2}$, so $\alpha = \frac{\pi}{3}$, and $\sqrt{x^2-1} = \sqrt{3}$. The stretched length is

$$L = 2\sqrt{3} + 2\pi - \frac{2\pi}{3} = 2\sqrt{3} + \frac{4\pi}{3}, \qquad \text{so} \qquad L - 2\pi = 2\sqrt{3} - \frac{2\pi}{3}.$$

The tension is $T = \dfrac{2mg}{2\sqrt{3}} = \dfrac{mg}{\sqrt{3}}$. Hooke's law gives

$$\lambda = \frac{2\pi T}{L - 2\pi} = \frac{2\pi mg/\sqrt{3}}{2\sqrt{3} - 2\pi/3} = \frac{2\pi mg}{6 - 2\pi/\sqrt{3}} = \frac{\pi mg}{3 - \pi/\sqrt{3}},$$

and multiplying top and bottom by $\sqrt{3}$,

$$\boxed{\lambda = \frac{\sqrt{3}\,\pi\, mg}{3\sqrt{3} - \pi} \approx 2.65\, mg.}$$

Checks

- As $x \to 1^+$, $\arccos(1/x) \to 0$ and $\sqrt{x^2-1} \to 0$, so $L \to 2\pi$: the band just touches the bottom of the cylinder and is unstretched, as it should be. Meanwhile $T \to \infty$, which matches the fact that two nearly horizontal strings can only hold up a weight with a huge tension.
- At $x = 2$, $L = 2\sqrt{3} + 4\pi/3 \approx 7.653$, a little more than $2\pi \approx 6.283$, and λ has the units of a force, like mg.
- Independently, we minimised the total potential energy $-mgx + \lambda(L - 2\pi)^2/(4\pi)$ numerically with $\lambda \approx 2.6484\, mg$; the minimum is at $x = 2$, which confirms both the geometry and the tension.

Common slips

- Taking the straight pieces from P to the ends of the horizontal diameter. The band leaves along tangents, so it comes off *below* the level of the axis, at angle $\alpha = \arccos(1/x)$ from the downward vertical.
- Using the extension of one straight piece in Hooke's law. The tension is uniform, so it's the extension of the whole band, $L - 2\pi$, divided by the whole natural length 2π.

Beyond STEP

The claim that tension is constant round a smooth surface is the frictionless case of the capstan equation: with coefficient of friction μ, the tension can change by a factor of up to $e^{\mu\theta}$ around an angle θ. That is why a rope wrapped a few times round a

post can hold a ship. The energy check above is also a first taste of the variational view of mechanics, in which equilibria are stationary points of potential energy. For a first-year treatment of forces, friction and potential energy in this spirit, read Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 1987, Question 11

Mechanics • Circular Motion and Energy

STEP I 1987, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q11

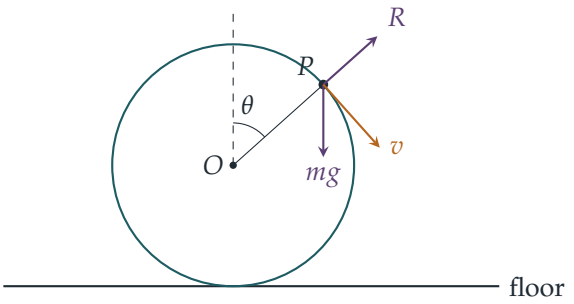
The key idea

While the particle is on the sphere it moves in a circle, so two equations govern it: energy conservation gives its speed, and Newton's second law along the radius gives the normal reaction. Contact is lost exactly when the reaction falls to zero. After that the particle is a projectile, so its horizontal velocity never changes; combining that with energy conservation all the way down to the floor gives the vertical velocity without solving for the trajectory.

Solution

Model

Treat the particle as a point mass on a smooth, fixed sphere with centre O, a height r above the floor (the sphere rests on the floor). "Displaced gently" means it starts at the top with negligible speed. Let θ be the angle between OP and the upward vertical, where P is the particle, and let v be its speed. The only forces are the weight mg and the normal reaction R along OP, outwards.



Where contact is lost

The particle has fallen a height $r - r\cos\theta$, so by conservation of energy (the reaction does no work, being perpendicular to the motion)

$$\tfrac{1}{2}mv^2 = mgr(1-\cos\theta), \qquad \text{so} \qquad v^2 = 2gr(1-\cos\theta).$$

While in contact the particle moves on a circle of radius r, so its acceleration towards O is v^2/r. Resolving towards O:

$$mg\cos\theta - R = \frac{mv^2}{r} = 2mg(1-\cos\theta), \qquad \text{so} \qquad R = mg(3\cos\theta - 2).$$

At the top $R = mg > 0$, and R decreases as θ increases. Contact is lost when $R = 0$, that is when

$$\cos\theta = \tfrac{2}{3}, \qquad v^2 = \tfrac{2}{3}gr.$$

Ask yourself Which angle is the angle the velocity makes with the horizontal?

The velocity is along the tangent to the circle, which is perpendicular to the radius OP. The radius makes angle θ with the vertical, so the tangent makes the same angle θ with the horizontal (turn both lines through a right angle). So at the moment it leaves the sphere the particle is moving downwards at an angle

$$\cos^{-1}\tfrac{2}{3} \approx 48.2^\circ$$

below the horizontal.

Striking the floor

At the moment of leaving, the horizontal component of velocity is

$$v\cos\theta = \sqrt{\tfrac{2}{3}gr}\cdot\tfrac{2}{3}, \qquad \text{so} \qquad (v\cos\theta)^2 = \tfrac{8}{27}gr.$$

From then on the only force is gravity, so this horizontal component is unchanged until the particle reaches the floor. (The particle does not touch the sphere again: once R would be negative the sphere is curving away faster than the particle falls, and the particle lands at a horizontal distance of about $1.46r$ from the line through O, well clear of the sphere.)

Now apply energy conservation to the whole motion, from rest at the top (height $2r$) down to the floor. The particle hits the floor with speed V where

$$\tfrac{1}{2}mV^2 = mg(2r), \qquad \text{so} \qquad V^2 = 4gr.$$

If V_y is the vertical component of velocity on impact, then $V^2 = (v\cos\theta)^2 + V_y^2$, so

$$V_y^2 = 4gr - \tfrac{8}{27}gr = \tfrac{100}{27}gr, \qquad V_y = \frac{10}{3}\sqrt{\frac{gr}{3}}.$$

The vertical component of momentum as it strikes the floor therefore has magnitude

$$mV_y = \frac{10m}{3}\sqrt{\frac{gr}{3}} = \frac{10m\sqrt{3gr}}{9},$$

directed downwards.

Checks

- Using projectile motion directly: the particle leaves at height $r + r\cos\theta = \frac{5}{3}r$ with downward vertical speed $v\sin\theta$, where $\sin\theta = \frac{\sqrt{5}}{3}$, so $(v\sin\theta)^2 = \frac{10}{27}gr$. Falling a further $\frac{5}{3}r$ adds $2g \cdot \frac{5}{3}r = \frac{90}{27}gr$, giving $V_y^2 = \frac{100}{27}gr$ again.
- $V_y^2 = \frac{100}{27}gr \approx 3.70gr$ is a bit less than $V^2 = 4gr$, as it must be, since a little of the energy stays in horizontal motion.
- Stepping the motion numerically (with $g = 9.81$, $r = 1.3$, $m = 2$) gives contact lost at 48.19° and vertical momentum 13.745, matching $\frac{10m\sqrt{3gr}}{9}$; the trajectory stays outside the sphere.

Common slips

- Measuring the fall from the floor rather than from the top of the sphere: on the sphere the drop is $r(1 - \cos\theta)$.
- Forgetting that the horizontal velocity is fixed after leaving, and writing $V_y^2 = 4gr$ as if all the kinetic energy were vertical.
- Taking the velocity to make angle $90^\circ - \theta$ with the horizontal; that is the angle of the radius, not the tangent.

Beyond STEP

The condition "contact lost when the reaction vanishes" is a first example of a *constraint force*: a force that adjusts itself to keep a particle on a surface, and which can only push. In Lagrangian mechanics such constraints are built in from the start, and the reaction appears as a Lagrange multiplier. Kleppner and Kolenkow, *An Introduction to Mechanics*, is an excellent place to practise circular motion and energy methods at the next level.

WORKED SOLUTION

STEP I 1987, Question 12

Mechanics • Projectiles with Restitution

STEP I 1987, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q12

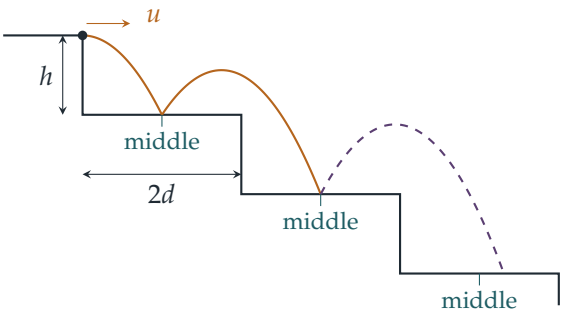
The key idea

Between bounces the particle is a projectile, and (with smooth steps) a bounce reverses and scales only the vertical component of velocity, so the horizontal speed never changes. That means the horizontal distances fix the flight times, and the flight times together with the drop of one step fix the vertical speeds. Everything can be measured in terms of the time of the first fall, which keeps the algebra tiny.

Solution

Model

Treat the particle as a point mass, ignore air resistance, and take the steps to be smooth and horizontal, so an impact leaves the horizontal velocity unchanged and multiplies the vertical velocity by $-e$. Let the particle be kicked with horizontal speed u, and let t_1 be the time of the first flight.



The solid arcs are the first two flights; the dashed arc is what actually happens next (see the final part).

The coefficient of restitution

First flight. The particle starts with no vertical velocity, falls h and travels d horizontally (half a step width, to the middle of the first step). So

$$h = \tfrac{1}{2}gt_1^2, \qquad d = ut_1.$$

It strikes the first step with downward vertical speed gt_1, and leaves it moving upwards with vertical speed egt_1, still with horizontal speed u.

Second flight. From the middle of the first step to the middle of the second is a horizontal distance of $2d$. Since the horizontal speed is still $u = d/t_1$, this flight lasts

$$T = \frac{2d}{u} = 2t_1.$$

In that time the particle must end up a height h lower. Taking upwards as positive,

$$\begin{aligned} -h &= (egt_1)(2t_1) - \tfrac{1}{2}g(2t_1)^2 \\ &= (2e-2)\,gt_1^2 \\ &= (2e-2)(2h), \end{aligned}$$

using $gt_1^2 = 2h$. Dividing by h gives $-1 = 4e - 4$, so

$$e = \tfrac{3}{4}.$$

(On this flight the particle is a height $egt_1 \cdot t_1 - \frac{1}{2}gt_1^2 = \frac{1}{2}h$ above the first step when it passes over that step's edge, so it does clear the edge.)

Can it keep going?

> **Ask yourself** What would every later flight have to look like if the pattern continued?

Every flight from the middle of one step to the middle of the next covers $2d$ horizontally at speed u, so each lasts $2t_1$, and each must drop exactly h. The calculation above then shows that the upward vertical speed w just after each bounce must satisfy

$$-h = w(2t_1) - 2gt_1^2 = 2wt_1 - 4h, \qquad \text{so} \qquad w = \frac{3h}{2t_1} = \tfrac{3}{4}gt_1.$$

So after every bounce the particle must leave with vertical speed $\frac{3}{4}gt_1$; since $e = \frac{3}{4}$, it must arrive at every step with vertical speed exactly gt_1.

That holds at the first step, but not at the second. Arriving there after a flight of time $2t_1$, its vertical velocity is

$$\tfrac{3}{4}gt_1 - g(2t_1) = -\tfrac{5}{4}gt_1,$$

so it hits the second step with downward speed $\frac{5}{4}gt_1$, not gt_1, and rebounds with upward speed $\frac{3}{4} \cdot \frac{5}{4}gt_1 = \frac{15}{16}gt_1 > \frac{3}{4}gt_1$. So it is **not possible**: the particle cannot hit the

middle of every step. In fact it overshoots the middle of the third step. If the third flight lasts st_1, then

$$-h = \tfrac{15}{16}gt_1(st_1) - \tfrac{1}{2}gs^2t_1^2 \quad\Longrightarrow\quad s^2 - \tfrac{15}{8}s - 1 = 0 \quad\Longrightarrow\quad s = \frac{15+\sqrt{481}}{16} \approx 2.31,$$

so it travels about $2.31d$ horizontally and lands on the third step about $0.31d$ beyond its middle (still on that step, since the far edge is $3d$ away).

Checks

- $e = \frac{3}{4}$ does not depend on g, h or d, as it shouldn't: scaling the staircase changes nothing about the shape of the path.
- A numerical simulation with $g = 9.8$, $h = d = 1$ and $e = 0.75$ lands at $x = 1.000$ and $x = 3.000$ (the middles of the first two steps) and then at $x \approx 5.308$, against $3 + 2.308$ predicted above.
- Without a bounce ($e = 0$) the second flight would fall $4h$ in time $2t_1$; the rebound must make up $3h$ of that, giving $2t_1 \cdot w = 3h$, as found.

Common slips

- Taking the first horizontal distance as $2d$ rather than d: the particle starts at the edge and lands in the middle of the first step.
- Changing the horizontal speed at the bounce. On a smooth surface the impulse is vertical, so only the vertical component changes.
- Concluding "yes" because each flight looks identical: the arrival speeds grow, so the rebounds grow too.

Beyond STEP

A bouncing particle is a simple example of a *discrete dynamical system*: the state just after one bounce determines the state after the next through a fixed map, and asking whether the motion can repeat is asking for a fixed point of that map. Here the fixed point (rebound $\frac{6}{7}gt_1$, which would land only $\frac{4}{7}h$ lower) is incompatible with the staircase, so no periodic motion fits. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and impulse carefully.

WORKED SOLUTION

STEP I 1987, Question 13

Mechanics • Variable Forces

STEP I 1987, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q13

The key idea

The force law changes at $x = 0$, so the motion splits into stages, each governed by a simple linear differential equation. On the positive side the force is a restoring force proportional to displacement, which is simple harmonic motion; on the negative side it is a resistance proportional to velocity, which gives exponential decay of the speed. The position and velocity at the end of one stage are the initial conditions for the next. The real question is how many stages there are: does the particle ever come back?

Solution

Model

Treat the body as a particle on the x-axis acted on only by the given force, and write $\dot{x} = \mathrm{d}x/\mathrm{d}t$. Newton's second law, $m\ddot{x} = \mathrm{F}$, gives

$$\ddot{x} = -\mu^2 x \quad (x \geq 0), \qquad \ddot{x} = -\kappa\dot{x} \quad (x < 0).$$

The mass cancels, so it plays no part in the answer.

Stage 1: simple harmonic motion while $x \geq 0$

Since $v_0 > 0$, the particle immediately enters $x > 0$, where $\ddot{x} = -\mu^2 x$. This is simple harmonic motion with angular frequency μ, so its general solution is

$$x = A\cos\mu t + B\sin\mu t.$$

The condition $x = 0$ at $t = 0$ gives $A = 0$, and $\dot{x} = B\mu = v_0$ at $t = 0$ gives $B = v_0/\mu$. So

$$x = \frac{v_0}{\mu}\sin\mu t, \qquad \dot{x} = v_0\cos\mu t.$$

This stays valid while $x \geq 0$, that is while $\sin \mu t \geq 0$, so for $0 \leq t \leq \pi/\mu$. The particle reaches a greatest displacement v_0/μ at $t = \pi/(2\mu)$ and returns to the origin at

$$t = T = \frac{\pi}{\mu}, \qquad \text{with velocity} \qquad \dot{x} = v_0 \cos \pi = -v_0.$$

It comes back with the same speed it started with, as energy is conserved on this side (the force $-m\mu^2 x$ comes from the potential energy $\frac{1}{2}m\mu^2 x^2$, which is zero at the origin).

Stage 2: resisted motion while $x < 0$

Moving with velocity $-v_0$, the particle now enters $x < 0$, where $\ddot{x} = -\kappa \dot{x}$. Writing $u = \dot{x}$, this is the first-order equation $\mathrm{d}u/\mathrm{d}t = -\kappa u$, whose solution with $u = -v_0$ at $t = T$ is

$$\dot{x} = -v_0 \, \mathrm{e}^{-\kappa(t-T)}.$$

Integrating, with $x = 0$ at $t = T$:

$$x = \int_T^t -v_0 \, \mathrm{e}^{-\kappa(s-T)} \, \mathrm{d}s = -\frac{v_0}{\kappa} \left(1 - \mathrm{e}^{-\kappa(t-T)}\right).$$

Ask yourself Does the particle ever return to $x \geq 0$, starting a third stage?

No. The velocity $-v_0 \mathrm{e}^{-\kappa(t-T)}$ is negative for every $t > T$, so x decreases throughout this stage; it is strictly negative for all $t > T$ and the formula above holds for ever. The resistance slows the particle but never reverses it. As $t \to \infty$, $\mathrm{e}^{-\kappa(t-T)} \to 0$, so the particle approaches the point $x = -v_0/\kappa$ without ever reaching it.

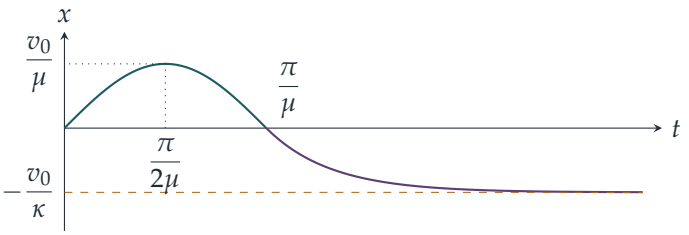
Answer

$$x = \begin{cases} \dfrac{v_0}{\mu} \sin \mu t, & 0 \leq t \leq \dfrac{\pi}{\mu}, \\ -\dfrac{v_0}{\kappa} \left(1 - \mathrm{e}^{-\kappa(t-\pi/\mu)}\right), & t > \dfrac{\pi}{\mu}. \end{cases}$$

Both pieces have $x = 0$ and $\dot{x} = -v_0$ at $t = \pi/\mu$, so the graph is smooth there (with a continuous gradient), although $\ddot{x}$ jumps from 0 to κv_0.

Sketch

The graph is half an arch of a sine curve above the axis, followed by a curve that leaves the origin with the same slope $-v_0$ and levels off towards the asymptote $x = -v_0/\kappa$. (The sketch uses $\mu = 1$, $\kappa = 1$.)



In general the peak height v_0/μ and the depth of the asymptote v_0/κ are independent: a weak resistance (small κ) lets the particle drift a long way to the left.

Checks

- Units: μ and κ both have units of s^{-1}, so v_0/μ and v_0/κ are lengths, as they must be.
- Limiting case: as $\kappa \to 0$ there is no resistance, and indeed $-\frac{v_0}{\kappa}(1 - \mathrm{e}^{-\kappa s}) \to -v_0 s$, uniform motion at speed v_0.
- Numerically: stepping the equation of motion with a Runge–Kutta method for $\mu = 1.7$, $\kappa = 0.9$, $v_0 = 2.2$ matched the formula to within 3×10^{-5} for $0 \le t \le 15$; the particle peaked at $x = 1.2941 = v_0/\mu$ and settled at $x = -2.4444 = -v_0/\kappa$.

Common slips

- Measuring time in the second stage from $t = 0$ rather than from $t = \pi/\mu$: the exponent must be $-\kappa(t - \pi/\mu)$.
- Forgetting to check whether the particle re-enters $x \ge 0$; the question asks for the *whole* subsequent motion, so the claim that there is no third stage needs its one-line justification.

Beyond STEP

Here the force is a different linear expression on each side of $x = 0$, a piecewise-linear system; the trick of gluing solutions together, matching position and velocity at the switching point, is the standard way such systems (and impact and friction problems) are handled in a first course in differential equations. Adding both terms at once, $\ddot{x} + \kappa\dot{x} + \mu^2 x = 0$, gives the damped harmonic oscillator, whose underdamped, critically damped and overdamped regimes are a central example in any first mechanics course. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats the damped oscillator carefully.

WORKED SOLUTION

STEP I 1987, Question 14

Probability and Statistics • Geometric Series of Probabilities

STEP I 1987, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q14

The key idea

Once the first game has been played, an undecided tournament has no freedom left at all. The winner of each game faces the referee, and if the tournament is to stay undecided the referee must win, so the next winner is forced. So everything is decided by who wins game 1 and by the single moment when a player first wins twice in a row. The probabilities then reduce to geometric series that cycle with period 3.

Solution

Throughout, in any game the winner of the previous game plays against its referee. The previous loser sits out as referee and *cannot win* the current game.

Part (i)

Game 1: A beats B, and B referees game 2. Game 2 is A against C, and C wins, which is possible. Game 3 is C against B, and B wins, which is also possible. But no one has won two games in a row, so after ACB the tournament is still undecided and must continue. Hence ACB is **not** the record of a complete tournament (it is only the start of one).

Part (ii)

After A wins game 1, game 2 is A against C with B refereeing. So B cannot win game 2, and ABB is **impossible**.

Part (iii)

As in (i), the first three games ACB are possible, and game 3 was C against B with A refereeing. So game 4 is B against A, and B can win it. Now B has won games 3 and 4 in a row, so the tournament ends. Hence $ACBB$ **is** a complete tournament, won by B in four games.

Undecided after 5 games

Suppose X wins game 1, where X is A or B, and let Y be the other one. If the tournament is to remain undecided, each game must be won by the player who was *not* the previous winner, and that can only be the previous referee. This forces the record

$$X\,C\,Y\,X\,C\,Y\,X\,\ldots$$

because each new winner is the previous game's referee. Concretely, game 2 is X v C (Y refereeing), so C must win. Game 3 is C v Y (X refereeing), so Y must win. Game 4 is Y v X, so X must win, and so on with period 3.

Game 1 can go either way. After that, each further game that keeps things undecided happens with probability $\frac{1}{2}$, since the games are independent and each player has an even chance. So

$$P(\text{undecided after } n \text{ games}) = \left(\tfrac{1}{2}\right)^{n-1}, \qquad P(\text{undecided after 5 games}) = \tfrac{1}{16}.$$

Who wins, and when

The tournament ends at game n (for $n \geq 2$) when it is undecided after $n-1$ games and the winner of game $n-1$ wins again. This has probability $\left(\frac{1}{2}\right)^{n-2} \cdot \frac{1}{2} = \left(\frac{1}{2}\right)^{n-1}$. The champion is the winner of game $n-1$, that is, the $(n-1)$th letter of the forced sequence. Write $k = n-1$. Then:

k mod 3	1	2	0
champion if A won game 1	A	C	B
champion if B won game 1	B	C	A

So the tournament ends at game $k+1$ with probability $\left(\frac{1}{2}\right)^{k}$. If $k \equiv 2 \pmod 3$ this whole probability belongs to C. Otherwise it is split equally between A and B, giving $\left(\frac{1}{2}\right)^{k+1}$ each.

> **Ask yourself** Why does C get the whole probability at $k \equiv 2$, but only half at other times?

C is always the second letter of the forced sequence, whoever won game 1. So both branches give the win to C at the same moments. A and B swap roles between the two branches, so each of them collects only the branch in which they occupy that position.

Winning in 5 or fewer games

Here $n = 2, 3, 4, 5$, so $k = 1, 2, 3, 4$:

$$P(A \text{ wins in} \le 5) = \underbrace{\tfrac{1}{4}}_{k=1} + \underbrace{\tfrac{1}{16}}_{k=3} + \underbrace{\tfrac{1}{32}}_{k=4} = \tfrac{11}{32},$$
$$P(B \text{ wins in} \le 5) = \tfrac{11}{32} \quad \text{(by the symmetry between } A \text{ and } B\text{)},$$
$$P(C \text{ wins in} \le 5) = \underbrace{\tfrac{1}{4}}_{k=2}.$$

As a check, $\frac{11}{32} + \frac{11}{32} + \frac{8}{32} = \frac{15}{16} = 1 - \frac{1}{16}$, which agrees with the probability of being undecided after 5 games.

Eventual winner

Because $P(\text{undecided after } n \text{ games}) = \left(\frac{1}{2}\right)^{n-1} \to 0$, the tournament ends with probability 1. So we just add over all $k \ge 1$. Each residue class gives a geometric series with ratio $\frac{1}{8}$:

$$P(C \text{ wins}) = \tfrac{1}{4} + \tfrac{1}{32} + \tfrac{1}{256} + \cdots = \frac{1/4}{1 - 1/8} = \frac{2}{7},$$
$$P(A \text{ wins}) = \underbrace{\left(\tfrac{1}{4} + \tfrac{1}{32} + \cdots\right)}_{k \equiv 1} + \underbrace{\left(\tfrac{1}{16} + \tfrac{1}{128} + \cdots\right)}_{k \equiv 0} = \frac{1/4}{7/8} + \frac{1/16}{7/8} = \frac{2}{7} + \frac{1}{14} = \frac{5}{14},$$

as required. By symmetry $P(B \text{ wins}) = \frac{5}{14}$ as well, and indeed $\frac{5}{14} + \frac{5}{14} + \frac{2}{7} = 1$.

So C, who sits out the first game, is at a disadvantage: $\frac{2}{7} \approx 0.286$ against $\frac{5}{14} \approx 0.357$ for each of the others.

Checks

- The three probabilities of winning within 5 games add to $\frac{15}{16}$, and the eventual ones add to 1.
- An exact enumeration over all possible game sequences, using fractions, confirms $\frac{1}{16}$, $\frac{11}{32}$, $\frac{11}{32}$, $\frac{1}{4}$. Run to 60 games, it gives $0.357142\ldots$ for A and B and $0.285714\ldots$ for C.
- A quick alternative for C: let c be the probability that C wins. After game 1, C wins at once with probability $\frac{1}{4}$. If the play gets back to the same situation three games later (probability $\frac{1}{8}$), we start again, so $c = \frac{1}{4} + \frac{1}{8}c$, giving $c = \frac{2}{7}$.

Common slips

- Saying ACB is a valid record: every game in it is possible, but nobody has won twice running, so the tournament isn't over.

- Forgetting that the previous loser referees, and so can't win the next game (this rules out ABB).
- Giving C only half the probability at $k \equiv 2$: C wins at those moments in *both* branches.

Beyond STEP

The tournament is a Markov chain: the state after each game is just "who won last, and who is refereeing", and the champion is decided when the chain is absorbed. The first-step argument in the Checks section, writing an equation for a probability by conditioning on what happens next, is the standard way to find absorption probabilities, and it avoids summing series. Grimmett and Welsh, *Probability: An Introduction*, introduces Markov chains and absorption at first-year level.

WORKED SOLUTION

STEP I 1987, Question 15

Probability and Statistics • Continuous Random Variables

STEP I 1987, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q15

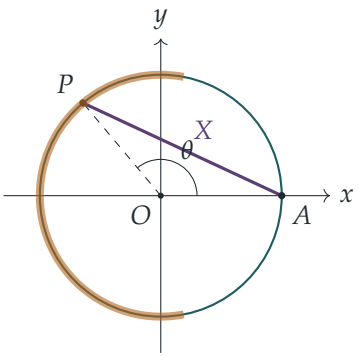
The key idea

"Uniform on the circle" means the *angle* that P makes at the centre is uniformly distributed, so the natural move is to write $P = (\cos\theta, \sin\theta)$ with θ uniform on $[0, 2\pi)$. The distance from $(1, 0)$ is then a chord of the unit circle, and the half-angle formula turns it into a single sine. After that, the mean is one integral, the variance needs almost no integration at all, and the probability is a question about an interval of angles.

Solution

Setting up

Let θ be the angle from the positive x-axis to OP, where O is the origin. Choosing P uniformly on the circle means choosing arc length uniformly, so θ is uniform on $[0, 2\pi)$ with density $\frac{1}{2\pi}$. Write $A = (1, 0)$ and $P = (\cos\theta, \sin\theta)$.



Then

$$X^2 = (\cos\theta - 1)^2 + \sin^2\theta = 2 - 2\cos\theta = 4\sin^2\tfrac{\theta}{2},$$

using $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$. Since $0 \le \frac{\theta}{2} < \pi$, we have $\sin\frac{\theta}{2} \ge 0$, so taking the positive

square root is safe:

$$X = 2\sin\tfrac{\theta}{2}.$$

The mean

$$\mathrm{E}(X) = \frac{1}{2\pi}\int_0^{2\pi} 2\sin\tfrac{\theta}{2}\,\mathrm{d}\theta = \frac{1}{2\pi}\Big[-4\cos\tfrac{\theta}{2}\Big]_0^{2\pi} = \frac{1}{2\pi}(4+4) = \frac{4}{\pi}.$$

So $\mathrm{E}(X) = \dfrac{4}{\pi} \approx 1.273$.

The variance

Ask yourself Do we need to integrate $4\sin^2\frac{\theta}{2}$ to find $\mathrm{E}(X^2)$?

No: the unsquared form is easier. $X^2 = 2 - 2\cos\theta$, and $\cos\theta$ averages to zero over a full turn, since $\int_0^{2\pi}\cos\theta\,\mathrm{d}\theta = 0$. Hence $\mathrm{E}(X^2) = 2$, and

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \big(\mathrm{E}(X)\big)^2 = 2 - \frac{16}{\pi^2} \approx 0.379.$$

The probability that X exceeds its mean

We need $2\sin\frac{\theta}{2} > \frac{4}{\pi}$, that is $\sin\frac{\theta}{2} > \frac{2}{\pi}$. Note $0 < \frac{2}{\pi} < 1$, so this has solutions. Let $\alpha = \arcsin\frac{2}{\pi}$, which lies in $(0, \frac{\pi}{2})$. As $\frac{\theta}{2}$ runs over $[0, \pi)$, the sine rises from 0 to 1 and falls back, and it exceeds $\sin\alpha$ exactly when

$$\alpha < \tfrac{\theta}{2} < \pi - \alpha, \qquad \text{that is} \qquad 2\alpha < \theta < 2\pi - 2\alpha.$$

This is the thickened arc in the diagram: the points on the far side of the circle from A. Since θ is uniform, the probability is the length of this interval divided by 2π:

$$\mathrm{P}\big(X > \mathrm{E}(X)\big) = \frac{2\pi - 4\alpha}{2\pi} = \boxed{1 - \frac{2}{\pi}\arcsin\frac{2}{\pi}} \approx 0.561.$$

Checks

- X ranges over $[0, 2]$, and the mean $4/\pi \approx 1.27$ sits above the midpoint 1. That fits the picture: the chord length is near its maximum for a large share of the circle (as P passes the far side, X changes slowly), so the distribution is weighted towards large values.
- For the same reason the probability of exceeding the mean is a little more than $\frac{1}{2}$, and 0.561 is.
- Numerically, a midpoint-rule average over 2×10^5 equally spaced angles gives mean 1.273240, variance 0.378861 and $\mathrm{P} = 0.56066$, and a 10^6-point simulation agrees to within sampling error.

Common slips

- Taking $X = 2\sin\frac{\theta}{2}$ with $\theta \in (-\pi, \pi]$. Then $\sin\frac{\theta}{2}$ can be negative, and you need $X = 2\left|\sin\frac{\theta}{2}\right|$; using $[0, 2\pi)$ avoids the modulus.
- Treating the x-coordinate of P as uniform on $[-1, 1]$. Uniform on the circle means uniform in angle (arc length), not in x.

Beyond STEP

This is a first example of transforming a random variable: the density of X is $f(x) = \dfrac{2}{\pi\sqrt{4 - x^2}}$ on $[0, 2]$, an arcsine-type law, and you can recover every answer above from it. The question also hints at how slippery "at random" can be: picking a random chord of a circle in different natural ways gives different answers (Bertrand's paradox). Both ideas are developed in Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP I 1987, Question 16

Probability and Statistics • Bayes and Normal Approximation

STEP I 1987, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q16

The key idea

Part (i) is Bayes' theorem in disguise. Voting the same way twice is *evidence* about which party the member belongs to, because a Preservative always does it and a Progressive only sometimes does. So we first update the chance that the member is a Preservative, and then predict the third vote. In part (ii) the Preservatives' votes are fixed numbers, so all the randomness comes from the 40 Progressives. Their number of votes in favour is binomial, and that is what we approximate by a normal distribution.

Solution

Part (i)

Let R be the event that the chosen member is a Preservative, so $P(R) = \frac{60}{100} = \frac{3}{5}$ and $P(R') = \frac{2}{5}$. Let S_2 be the event that the member voted the same way on the first two occasions, and S_3 the event that the third vote also matches.

A Preservative never changes their mind, so $P(S_2 \mid R) = 1$. A Progressive votes at random, each vote independently for or against with probability $\frac{1}{2}$. Whatever the first vote was, the second matches it with probability $\frac{1}{2}$, so $P(S_2 \mid R') = \frac{1}{2}$. Hence

$$P(S_2) = \tfrac{3}{5} \cdot 1 + \tfrac{2}{5} \cdot \tfrac{1}{2} = \tfrac{4}{5}, \qquad P(R \mid S_2) = \frac{P(R)\,P(S_2 \mid R)}{P(S_2)} = \frac{3/5}{4/5} = \frac{3}{4}.$$

So, given what we have seen, the member is a Preservative with probability $\frac{3}{4}$ and a Progressive with probability $\frac{1}{4}$.

Ask yourself Why can't we just say $P(R \mid S_2) = \frac{3}{5}$, since the member was chosen at random?

Because the information S_2 is more likely for a Preservative than for a Progressive, it shifts the odds. Before, the odds were $3 : 2$ in favour of a Preservative. After seeing S_2 they are $3 \cdot 1 : 2 \cdot \frac{1}{2} = 3 : 1$.

Now, given the member's party, the third vote does not depend on the first two votes beyond what they tell us about the party. A Preservative repeats the vote for certain, and a Progressive matches it with probability $\frac{1}{2}$. So

$$P(S_3 \mid S_2) = \tfrac{3}{4} \cdot 1 + \tfrac{1}{4} \cdot \tfrac{1}{2} = \frac{7}{8}.$$

(Equivalently, $P(S_3 \mid S_2) = \dfrac{P(S_2 \cap S_3)}{P(S_2)} = \dfrac{\frac{3}{5} + \frac{2}{5} \cdot \frac{1}{4}}{\frac{4}{5}} = \dfrac{7/10}{4/5} = \dfrac{7}{8}$.)

Part (ii)

Consider a vote on an item of Preservative policy, and assume all 100 members vote. A proportion α of the 60 Preservatives, that is 60α members, always vote against, and the remaining $60(1-\alpha)$ always vote for. Let X be the number of Progressives voting for. Each Progressive votes for with probability $\frac{1}{2}$, independently, so

$$X \sim \mathrm{B}\left(40, \tfrac{1}{2}\right), \qquad E(X) = 20, \qquad \mathrm{Var}(X) = 40 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = 10.$$

A simple majority of 100 votes means at least 51 in favour:

$$60(1-\alpha) + X \geq 51 \iff X \geq 60\alpha - 9.$$

Since $n = 40$ is fairly large and $p = \frac{1}{2}$ makes the binomial symmetric, we approximate X by $Y \sim \mathrm{N}(20, 10)$. Here $60\alpha - 9$ is a whole number (a number of members), so with a continuity correction

$$P(\text{majority}) = P(X \geq 60\alpha - 9) \approx P(Y > 60\alpha - 9.5) = P\left(Z > \frac{60\alpha - 29.5}{\sqrt{10}}\right),$$

where $Z \sim \mathrm{N}(0,1)$.

This probability decreases as α increases, so an election is called exactly when α is large enough to push it below 0.8. From tables, $P(Z > -0.8416) = 0.8$, so we need

$$\begin{aligned} \frac{60\alpha - 29.5}{\sqrt{10}} > -0.8416 &\iff 60\alpha > 29.5 - 0.8416\sqrt{10} = 26.84 \\ &\iff \alpha > 0.447. \end{aligned}$$

Since 60α must be a whole number of members, the smallest possible value is $60\alpha = 27$. So the least value of α that results in an election is

$$\alpha = \frac{27}{60} = 0.45.$$

To see that this is right: with 26 rebels the approximate chance of a majority is $P(Z > -1.107) \approx 0.866$, and with 27 rebels it is $P(Z > -0.791) \approx 0.785 < 0.8$.

Checks

- In (i), $\frac{7}{8}$ lies between $\frac{1}{2}$ (a known Progressive) and 1 (a known Preservative), as it must. A simulation of 400,000 random members gave 0.875.
- In (ii), the exact binomial probabilities are $P(X \geq 17) = 0.8659$ (26 rebels) and $P(X \geq 18) = 0.7852$ (27 rebels). These agree with the normal values to three decimal places, and confirm that the crossover is between 26 and 27 rebels, i.e. at $\alpha = 0.45$.

Common slips

- Answering $\frac{3}{5} \cdot 1 + \frac{2}{5} \cdot \frac{1}{2} = \frac{4}{5}$ in (i). This ignores the information that the member has already voted the same way twice.
- Treating a simple majority as "at least 50". A tie of 50–50 is not a majority, so 51 votes are needed.
- Leaving out the continuity correction. Without it the threshold becomes $\alpha > 0.456$, which leads to 28 rebels instead of 27, and that is wrong according to the exact binomial calculation.

Beyond STEP

Part (i) is a small example of *Bayesian updating*: data change our belief about a hidden "type", and predictions are then averages over the updated beliefs (the posterior predictive distribution). The normal approximation in part (ii) is a case of the central limit theorem, and the near-perfect agreement here is no accident, since for $p = \frac{1}{2}$ the binomial is symmetric. Both ideas are developed carefully in Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP II 1987, Question 1

Pure • Algebraic Identities

STEP II 1987, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q1

The key idea

Don't try to substitute for one of the letters. Expand the right-hand side instead. The squares a^2, b^2, c^2 appear on both sides and cancel, so what is left is x times a linear expression, and that expression is exactly a multiple of the given condition. Once you see where the numbers 7 and 11 come from, the general result follows.

Solution

It helps to do the expansion once, in general. For any numbers p, q, r,

$$(px-a)^2+(qx-b)^2+(rx-c)^2=(p^2+q^2+r^2)x^2-2x(pa+qb+rc)+a^2+b^2+c^2,$$

so

$$(px-a)^2+(qx-b)^2+(rx-c)^2-(a^2+b^2+c^2)=x\big[(p^2+q^2+r^2)x-2(pa+qb+rc)\big]. \qquad (*)$$

Part (i)

Here $(p,q,r)=(1,2,3)$, so $p^2+q^2+r^2=14$ and $(*)$ becomes

$$(x-a)^2+(2x-b)^2+(3x-c)^2-(a^2+b^2+c^2)=x\big[14x-2(a+2b+3c)\big].$$

If $a+2b+3c=7x$, the bracket is $14x-14x=0$. So the two sides are equal, as required.

Part (ii)

Now $(p,q,r)=(2,3,3)$, so $p^2+q^2+r^2=4+9+9=22$ and

$$(2x-a)^2+(3x-b)^2+(3x-c)^2-(a^2+b^2+c^2)=x\big[22x-2(2a+3b+3c)\big].$$

If $2a+3b+3c=11x$, the bracket is $22x-22x=0$, which proves (ii).

Ask yourself Why 7 in (i) and 11 in (ii)?

They are half of $p^2+q^2+r^2$: $7=\frac{1}{2}(1+4+9)$ and $11=\frac{1}{2}(4+9+9)$. That tells us what the general statement must be.

The general result

Identity $(*)$ proves the following directly.

For any real numbers p,q,r,a,b,c,x, if

$$2(pa+qb+rc)=(p^2+q^2+r^2)\,x,$$

then

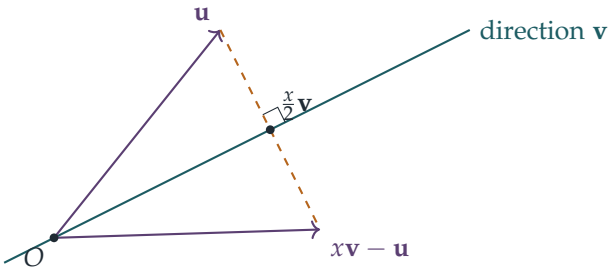
$$a^2+b^2+c^2=(px-a)^2+(qx-b)^2+(rx-c)^2.$$

Part (i) is the case $(p,q,r)=(1,2,3)$ and part (ii) is the case $(p,q,r)=(2,3,3)$.

The same expansion works with any number of variables. If $2\sum_{i=1}^{n}p_ia_i=\left(\sum_{i=1}^{n}p_i^2\right)x$, then $\sum a_i^2=\sum(p_ix-a_i)^2$. From $(*)$ we also get a converse: if $x\neq 0$ and the identity holds, then the condition must hold.

What is going on geometrically

Write $\mathbf{u}=(a,b,c)$ and $\mathbf{v}=(p,q,r)$. The condition reads $\mathbf{u}\cdot\mathbf{v}=\frac{x}{2}|\mathbf{v}|^2$, which says that $\frac{x}{2}\mathbf{v}$ is the foot of the perpendicular from $\mathbf{u}$ to the line through O in direction $\mathbf{v}$. The vector $x\mathbf{v}-\mathbf{u}=2\cdot\frac{x}{2}\mathbf{v}-\mathbf{u}$ is then the reflection of $\mathbf{u}$ in that line. A reflection preserves length, so $|x\mathbf{v}-\mathbf{u}|=|\mathbf{u}|$, and that is the identity.



Checks

Try (i) with $x=1$, $a=b=0$, $c=\frac{7}{3}$. The left side is $\frac{49}{9}$ and the right side is $1+4+\left(\frac{2}{3}\right)^2=5+\frac{4}{9}=\frac{49}{9}$. We also tested both parts and the general result in exact rational arithmetic, on 2000 random cases each (up to six variables in the general case), and confirmed that the identity fails whenever $x\neq 0$ and the condition fails.

Common slips

- Substituting $c = \frac{1}{3}(7x - a - 2b)$ straight away. It works, but the algebra is messy and hides the structure.
- Giving the general result as "$pa + qb + rc = kx$" with no link between k and p, q, r. The identity needs $k = \frac{1}{2}(p^2 + q^2 + r^2)$.

Beyond STEP

The geometric picture is the start of linear algebra. The map $\mathbf{u} \mapsto 2\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} - \mathbf{u}$ is a reflection, an orthogonal linear map, and orthogonal maps are exactly the linear maps that preserve lengths. Projections and reflections like this one underlie least squares and the Householder transformations used in numerical computation. Strang's *Introduction to Linear Algebra* develops these ideas with the same geometric flavour.

WORKED SOLUTION

STEP II 1987, Question 2

Pure • Trigonometric Identities

STEP II 1987, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q2

The key idea

The expression only involves $\sin^2 A$, $\sin^2 B$ and $\sin^2 C$, and it is a *quadratic* in $\sin^2 C$. The condition on $A \pm B \pm C$ says that C is tied to $A + B$ or $A - B$, so it is natural to guess that the two roots of that quadratic are $\sin^2(A+B)$ and $\sin^2(A-B)$. Checking the sum and product of the roots confirms the guess, and the result drops out.

Solution

Write $x = \sin^2 A$, $y = \sin^2 B$, $z = \sin^2 C$, and call the left-hand side E. Then

$$E = x^2 + y^2 + z^2 - 2yz - 2zx - 2xy + 4xyz.$$

Collecting powers of z,

$$\begin{aligned} E &= z^2 - 2z(x + y - 2xy) + (x^2 - 2xy + y^2) \\ &= z^2 - 2(x + y - 2xy)\,z + (x - y)^2. \end{aligned}$$

The roots of the quadratic

We claim that

$$E = \big(z - \sin^2(A+B)\big)\big(z - \sin^2(A-B)\big).$$

Since both sides are monic quadratics in z, it is enough to check that $\sin^2(A+B)$ and $\sin^2(A-B)$ have the right sum and product.

Sum. Expanding $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and squaring, the cross terms cancel when we add:

$$\begin{aligned} \sin^2(A+B) + \sin^2(A-B) &= 2\sin^2 A \cos^2 B + 2\cos^2 A \sin^2 B \\ &= 2x(1-y) + 2(1-x)y \\ &= 2(x + y - 2xy). \end{aligned}$$

Product. Using the difference of two squares,

$$\sin(A+B)\sin(A-B) = \sin^2 A\cos^2 B - \cos^2 A\sin^2 B = x(1-y) - (1-x)y = x - y,$$

so $\sin^2(A+B)\sin^2(A-B) = (x-y)^2$.

Both match the coefficients of E, which proves the claim.

> **Ask yourself** Why is it enough to know that $\sin^2 C$ equals $\sin^2(A+B)$ or $\sin^2(A-B)$?

Because then z is one of the two roots, so one factor of E vanishes. It remains to show that the condition on the angles gives exactly this.

Using the condition

Suppose $A + \varepsilon_1 B + \varepsilon_2 C = k\pi$ for some integer k and some signs $\varepsilon_1, \varepsilon_2 \in \{1, -1\}$. Then

$$C = \varepsilon_2\big(k\pi - (A + \varepsilon_1 B)\big).$$

The function $\sin^2$ is even and has period π, so neither the sign ε_2 nor the $k\pi$ changes its value:

$$\sin^2 C = \sin^2(A + \varepsilon_1 B).$$

So $z = \sin^2(A+B)$ (when $\varepsilon_1 = 1$) or $z = \sin^2(A-B)$ (when $\varepsilon_1 = -1$). In either case one factor of E is zero, and therefore

$$E = 0,$$

as required. ■

A neater form (and the converse)

Each factor is a difference of two squares of sines, and $\sin^2 P - \sin^2 Q = \sin(P+Q)\sin(P-Q)$ (the product formula above with $A = P$, $B = Q$). So

$$E = \sin(A+B+C)\,\sin(C-A-B)\,\sin(C+A-B)\,\sin(C-A+B).$$

The four angles here are, up to sign, exactly $A+B+C$, $A+B-C$, $A-B+C$ and $A-B-C$. This makes the result transparent: $E = 0$ *if and only if* one of the four angles $A \pm B \pm C$ is a multiple of π.

Checks

- Take $A = \frac{\pi}{6}$, $B = \frac{\pi}{3}$, $C = \frac{\pi}{2}$, so $A + B - C = 0$. Then $x = \frac{1}{4}$, $y = \frac{3}{4}$, $z = 1$ and $E = \frac{1}{16} + \frac{9}{16} + 1 - \frac{3}{2} - \frac{1}{2} - \frac{3}{8} + \frac{3}{4} = 0$.
- With $A = 0.3$, $B = 0.7$, $C = 1.1$ (no special relation), both E and the four-sine product evaluate to 0.05538, so the expression is genuinely non-zero away from the condition.

- A random test over 10^5 triples confirmed the factorisation, and that $E = 0$ to within 10^{-14} whenever one of $A \pm B \pm C$ is forced to be a multiple of π.

Common slips

- Treating only the case $A + B + C = \pi$ (a triangle). The question allows any of the four sign patterns and any multiple of π; the evenness and π-periodicity of $\sin^2$ handle them all at once.
- Solving the quadratic with the formula and getting tangled in $\sqrt{xy(1-x)(1-y)}$. Checking the sum and product of the guessed roots avoids square roots and sign worries entirely.

Beyond STEP

The expression $x^2 + y^2 + z^2 - 2yz - 2zx - 2xy$ appears all over mathematics: with x, y, z replaced by a^2, b^2, c^2 it is -16 times the square of the area of a triangle with sides a, b, c (Heron's formula), and it is also the discriminant-like quantity that decides whether three lengths can form a triangle. Seeing a symmetric polynomial as a quadratic in one variable, and guessing its roots, is a technique that recurs throughout algebra and invariant theory. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 1987, Question 3

Pure • Conjugate Surds

STEP II 1987, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q3

The key idea

Write $\alpha = a + \sqrt{a^2 - b}$ and $\beta = a - \sqrt{a^2 - b}$. These are conjugates: their sum $2a$ and product b are integers, so $\alpha^n + \beta^n$ is always an integer. If the condition on b forces the small conjugate β to lie strictly between 0 and 1, then β^n is the fractional part we need, and everything follows.

Solution

Throughout, let

$$\alpha = a + \sqrt{a^2 - b}, \qquad \beta = a - \sqrt{a^2 - b}, \qquad \text{so} \qquad \alpha + \beta = 2a, \quad \alpha\beta = a^2 - (a^2 - b) = b.$$

Where β lies. First, the square root is real. Since $b < 2a - 1$,

$$a^2 - b > a^2 - 2a + 1 = (a-1)^2 \geq 0.$$

Taking positive square roots of this strict inequality gives $\sqrt{a^2 - b} > a - 1$. Also $b > 0$, so $a^2 - b < a^2$ and $\sqrt{a^2 - b} < a$. Combining these,

$$a - 1 < \sqrt{a^2 - b} < a \quad \Longrightarrow \quad 0 < \beta < 1.$$

(The condition is sharp: if $b = 2a - 1$ then $\sqrt{a^2 - b} = a - 1$ and $\beta = 1$ exactly.)

Part (i)

Since $0 < \beta < 1$, we have $0 < \beta^n < 1$ for every positive integer n. Now M is an integer and $\beta^n = M + s$ with $0 \leq s < 1$, so M is the integer part of β^n, which is 0. Hence

$$M = 0 \qquad \text{and} \qquad s = \beta^n.$$

Part (ii)

Let $S_n = \alpha^n + \beta^n$. We claim S_n is an integer. Expanding both powers by the binomial theorem, with $d = \sqrt{a^2 - b}$,

$$S_n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}\left(d^k + (-d)^k\right) = 2 \sum_{k \text{ even}} \binom{n}{k} a^{n-k}(a^2 - b)^{k/2},$$

because the terms with odd k cancel. Every surviving term is an integer, so $S_n \in \mathbb{Z}$. (Equivalently, α, β are roots of $t^2 - 2at + b = 0$, so $S_n = 2aS_{n-1} - bS_{n-2}$ with $S_0 = 2$ and $S_1 = 2a$.)

Therefore

$$\alpha^n = S_n - \beta^n, \qquad S_n \in \mathbb{Z}, \quad 0 < \beta^n < 1.$$

? **Ask yourself** Why does this force $N = S_n$ and $r = \beta^n$?

The representation $\alpha^n = N - r$ with N an integer and $0 \le r < 1$ is unique: N must be the least integer that is at least α^n. If $N' - r' = N - r$ with both $r, r' \in [0, 1)$, then $N - N' = r - r'$ is an integer strictly between -1 and 1, so it is 0. Our expression $S_n - \beta^n$ is of that form, so

$$N = S_n = \alpha^n + \beta^n, \qquad r = \beta^n = s.$$

Part (iii)

Using $r = \beta^n$, $N = \alpha^n + \beta^n$ and $(\alpha\beta)^n = b^n$,

$$\begin{aligned} r^2 - Nr + b^n &= \beta^{2n} - (\alpha^n + \beta^n)\beta^n + \alpha^n\beta^n \\ &= \beta^{2n} - \alpha^n\beta^n - \beta^{2n} + \alpha^n\beta^n \\ &= 0, \end{aligned}$$

as required. In other words, r is the smaller root of $t^2 - Nt + b^n = 0$, whose roots are α^n and β^n.

The final part

Take $a = 8$ and $a^2 - b = 63$, so $b = 1$ and $\sqrt{63} = 3\sqrt{7}$. The condition $b < 2a - 1 = 15$ holds, so all of the above applies with $\alpha = 8 + 3\sqrt{7}$:

$$\left(8 + 3\sqrt{7}\right)^n = N - r, \qquad N \in \mathbb{Z},$$

and α^n falls short of the integer N by r. Part (iii) with $b = 1$ gives $r(N - r) = 1$, that is,

$$r = \frac{1}{N - r} = \frac{1}{\left(8 + 3\sqrt{7}\right)^n} = \left(8 - 3\sqrt{7}\right)^n.$$

In particular $r \neq 0$, so the power is never itself an integer, and $r \to 0$. Now $3\sqrt{7} = \sqrt{63}$ is just below $\sqrt{64} = 8$, so

$$8 + 3\sqrt{7} \approx 16 = 2^4, \qquad \text{hence} \qquad r = \left(8 + 3\sqrt{7}\right)^{-n} \approx 16^{-n} = 2^{-4n}.$$

So for large n, $\left(8+3\sqrt{7}\right)^n$ is within about 2^{-4n} of an integer, lying just below it.

How good is "about"? Since $8+3\sqrt{7} \approx 15.937$, we have $r = 2^{-cn}$ with $c = \log_2(8+3\sqrt{7}) \approx 3.994$. The exponent agrees with $4n$ to within 0.2%, which is the sense in which the estimate holds. The ratio $r/2^{-4n} = \left(16/(8+3\sqrt{7})\right)^n$ creeps up slowly, from about 1.004 at $n=1$ to 1.04 at $n=10$.

Checks

- $n=1$: $8+3\sqrt{7} \approx 15.93725$, which is $16-0.06275$, and $8-3\sqrt{7} \approx 0.06275$, as predicted. Also $r^2-16r+1=0$ has smaller root $8-\sqrt{63}$.
- $n=2$: $(8+3\sqrt{7})^2 = 127+48\sqrt{7} \approx 253.99606$, so $N = 254 = S_2 = 2\cdot 8\cdot 16 - 2$ and $r \approx 0.003937$, against $2^{-8} \approx 0.003906$.
- A computer check over thousands of random pairs (a,b) with $b < 2a-1$ and $n \le 7$ confirmed $0 < \beta^n < 1$, $N-\alpha^n = \beta^n$ and $r^2 - Nr + b^n = 0$. A high-precision check for $a=8$, $b=1$, $n \le 10$ confirmed $N-\alpha^n = (8-3\sqrt{7})^n$; at $n=10$ this is 9.46×10^{-13}, against $2^{-40} \approx 9.09\times 10^{-13}$.

Common slips

- Showing only $\beta < 1$ and forgetting $\beta > 0$ (which needs $b > 0$). Both are needed for $M = 0$.
- Asserting $N = \alpha^n + \beta^n$ without saying why the integer and fractional parts in $N - r$ are unique.
- Expecting $r/2^{-4n} \to 1$. It doesn't: the approximation is in the exponent, since $8+3\sqrt{7}$ is close to 16 but not equal to it.

Beyond STEP

The numbers $\alpha^n + \beta^n$ are an example of a trace: α is an algebraic integer whose conjugate lies inside the unit interval, so its powers get exponentially close to integers. Such numbers are called Pisot numbers; the golden ratio is another, which is why φ^n is close to the Lucas number L_n. The case here is also tied to Pell's equation: $8^2 - 7\cdot 3^2 = 1$, so $(8,3)$ is the smallest positive solution of $x^2 - 7y^2 = 1$. Davenport's *The Higher Arithmetic* treats Pell's equation and continued fractions beautifully.

WORKED SOLUTION

STEP II 1987, Question 4

Pure • Complex Numbers and Geometry

STEP II 1987, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q4

The key idea

Multiplying by $e^{i\theta}$ is a rotation, and an equilateral triangle is exactly what you get when one side is rotated through 60° about a vertex. That turns the geometry into a linear equation in α, β, γ. Allowing either orientation means *one of two* linear expressions vanishes, and their product is the symmetric quadratic in the question. Since it is symmetric, it can be written in terms of the coefficients of the cubic.

Solution

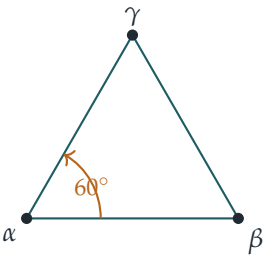
Throughout, $\omega = e^{2\pi i/3}$, so $\omega^3 = 1$, $\bar{\omega} = \omega^2$ and $1 + \omega + \omega^2 = 0$.

Multiplication by $e^{i\theta}$

If $z = re^{i\phi}$ then $ze^{i\theta} = re^{i(\phi+\theta)}$: the modulus is unchanged and the argument goes up by θ. So the point $ze^{i\theta}$ is the image of the point z under an anticlockwise rotation through angle θ about the origin.

More generally, $\gamma - \alpha = e^{i\theta}(\beta - \alpha)$ says that the vector from α to γ is the vector from α to β rotated anticlockwise through θ; that is, γ is the image of β under rotation through θ about α.

Anticlockwise equilateral triangles



Distinct points α, β, γ form an equilateral triangle in anticlockwise order exactly when γ is obtained from β by rotating through $\frac{\pi}{3}$ anticlockwise about α, i.e. when

$$\gamma - \alpha = e^{i\pi/3}(\beta - \alpha).$$

(Necessity: in such a triangle $|\alpha\gamma| = |\alpha\beta|$ and the angle at α is $\frac{\pi}{3}$, measured anticlockwise from $\alpha\beta$ to $\alpha\gamma$. Sufficiency: the equation gives $|\alpha\gamma| = |\alpha\beta|$ with angle $\frac{\pi}{3}$ between them, so the triangle is isosceles with apex angle $60°$, hence equilateral, and the orientation is anticlockwise.)

It is neater to write this symmetrically. Since $e^{i\pi/3} = -\omega^2$ (because $-\omega^2 = e^{i\pi}e^{4\pi i/3} = e^{i\pi/3}$, using $e^{7\pi i/3} = e^{i\pi/3}$), the condition becomes

$$\begin{aligned} & & \gamma - \alpha &= -\omega^2\beta + \omega^2\alpha \\ \iff & & \gamma + \omega^2\beta - (1+\omega^2)\alpha &= 0 \\ \iff & & \gamma + \omega^2\beta + \omega\alpha &= 0, \end{aligned}$$

using $1 + \omega^2 = -\omega$. Multiplying by ω^2 (which is non-zero, so this is reversible) gives the equivalent condition

$$\alpha + \omega\beta + \omega^2\gamma = 0.$$

A quick sanity check: $1, \omega, \omega^2$ are anticlockwise vertices of an equilateral triangle, and $1 + \omega \cdot \omega + \omega^2 \cdot \omega^2 = 1 + \omega^2 + \omega = 0$.

Either orientation

Any three distinct points forming an equilateral triangle are anticlockwise either in the order α, β, γ or in the order α, γ, β (cyclic re-orderings do not change the orientation). By the previous part, for distinct α, β, γ,

$$\begin{aligned} \text{equilateral} &\iff \alpha + \omega\beta + \omega^2\gamma = 0 \ \text{ or } \ \alpha + \omega\gamma + \omega^2\beta = 0 \\ &\iff (\alpha + \omega\beta + \omega^2\gamma)(\alpha + \omega^2\beta + \omega\gamma) = 0. \end{aligned}$$

? **Ask yourself** How do we see that this product is the expression we want?

Just expand it. The squared terms give $\alpha^2+\omega^3\beta^2+\omega^3\gamma^2=\alpha^2+\beta^2+\gamma^2$. The cross terms are

$$\begin{aligned}\alpha\beta(\omega^2+\omega)+\alpha\gamma(\omega+\omega^2)+\beta\gamma(\omega^2+\omega^4) &= \alpha\beta(\omega+\omega^2)+\gamma\alpha(\omega+\omega^2)+\beta\gamma(\omega^2+\omega)\\ &= -\alpha\beta-\gamma\alpha-\beta\gamma,\end{aligned}$$

using $\omega^4=\omega$ and $\omega+\omega^2=-1$. Hence

$$(\alpha+\omega\beta+\omega^2\gamma)(\alpha+\omega^2\beta+\omega\gamma)=\alpha^2+\beta^2+\gamma^2-\beta\gamma-\gamma\alpha-\alpha\beta,$$

and so the distinct points α,β,γ form an equilateral triangle (in some order) if and only if

$$\alpha^2+\beta^2+\gamma^2-\beta\gamma-\gamma\alpha-\alpha\beta=0,$$

as required.

It is worth noting what happens without distinctness: if, say, $\alpha=\beta$, the expression is $(\alpha-\gamma)^2$, which vanishes only when $\gamma=\alpha$ too. So the expression is zero precisely when the points form an equilateral triangle *or all three coincide*. We need this below.

The cubic

Let the roots of $z^3+az^2+bz+c=0$ be α,β,γ. Then

$$\alpha+\beta+\gamma=-a,\qquad \beta\gamma+\gamma\alpha+\alpha\beta=b,\qquad \alpha\beta\gamma=-c.$$

The expression is symmetric, so we write it in terms of these:

$$\begin{aligned}\alpha^2+\beta^2+\gamma^2-(\beta\gamma+\gamma\alpha+\alpha\beta) &= (\alpha+\beta+\gamma)^2-3(\beta\gamma+\gamma\alpha+\alpha\beta)\\ &= a^2-3b.\end{aligned}$$

By the remark above, $a^2=3b$ exactly when the roots form an equilateral triangle or are all equal. All three roots equal t means $z^3+az^2+bz+c=(z-t)^3$, so $a=-3t$, $b=3t^2$, $c=-t^3$; that is, $b=\frac{a^2}{3}$ and $c=\frac{a^3}{27}$. So the roots lie at the vertices of a (genuine) equilateral triangle if and only if

$$\boxed{a^2=3b \quad\text{and}\quad 27c\neq a^3}$$

If one is willing to count three coincident points as a degenerate equilateral triangle, the condition is simply $a^2=3b$.

There is a nice way to see this directly. Substituting $z=w-\frac{a}{3}$ removes the z^2 term; when $a^2=3b$ the linear term vanishes too and the equation becomes $w^3=\frac{a^3}{27}-c$. Its roots are the three cube roots of a non-zero number, which form an equilateral triangle centred at $w=0$, i.e. at $z=-\frac{a}{3}$, the centroid of the roots.

Checks

- The roots of $z^3-1=0$ ($a=b=0$, $c=-1$) are $1,\omega,\omega^2$: equilateral, and indeed $a^2=0=3b$ and $27c=-27\neq 0$.

- The roots of $z^3 - z = 0$ are $0, \pm 1$, collinear; here $a^2 - 3b = 3 \neq 0$.
- Numerically, for 1000 random triangles α, β, $\gamma = \alpha + e^{i\pi/3}(\beta - \alpha)$, we checked that the triangle is anticlockwise and equilateral, that $\alpha + \omega\beta + \omega^2\gamma$ and the quadratic expression vanish (in every ordering), and that the product identity and $a^2 - 3b$ formula hold for random complex numbers to within 10^{-8}.

Common slips

- Getting the sign or orientation wrong: $e^{i\pi/3}$ is $-\omega^2$, not ω; rotating by ω (that is, $120°$) about a vertex does not give an equilateral triangle.
- Forgetting the triple-root case: $a^2 = 3b$ alone also allows $z^3 + az^2 + bz + c = (z + \frac{a}{3})^3$, where the three "vertices" coincide.

Beyond STEP

The condition $\alpha + \omega\beta + \omega^2\gamma = 0$ is one component of a discrete Fourier transform of the triple (α, β, γ), and the trick of removing the z^2 term is the first step of solving the cubic by radicals, which leads into Galois theory. Geometrically, the result says the roots are equilateral exactly when the derivative $3z^2 + 2az + b$ has a double root, at the centroid; this is a special case of Marden's theorem about the roots of the derivative. Needham's *Visual Complex Analysis* is a beautiful place to explore the geometry of complex multiplication further.

WORKED SOLUTION

STEP II 1987, Question 5

Pure • Lagrange Inversion Series

STEP II 1987, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q5

The key idea

Lagrange's identity is a machine. You feed it the function $\phi(y) = y - \mathrm{f}(y)$, raise it to the nth power, differentiate $n-1$ times, and add up. All the work is in setting the problem up so that ϕ is simple, ideally a single power of y or a single exponential, because then the $(n-1)$th derivative of ϕ^n can be written down at once. In each part we choose f so that the equation to be solved reads $\mathrm{f}(x) = y$ for a convenient value of y.

Solution

Throughout, write $\phi(y) = y - \mathrm{f}(y)$, so the identity says

$$\mathrm{f}^{-1}(y) = y + \sum_{n=1}^{\infty} \frac{1}{n!} \frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}} \left[\phi(y)^n\right].$$

Note that the $n = 1$ term involves the zeroth derivative, which is just $\phi(y)$ itself.

The one fact we use repeatedly is that, for a whole number $m \geq n - 1$,

$$\frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}} y^m = \frac{m!}{(m-n+1)!} y^{m-n+1},$$

since each differentiation brings down the current power and lowers it by one.

Part (i)

With $\mathrm{f}(x) = \alpha x$ we have $\phi(y) = (1-\alpha)y$, so $\phi(y)^n = (1-\alpha)^n y^n$. Differentiating $n-1$ times, with $m = n$ above,

$$\frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}} \left[(1-\alpha)^n y^n\right] = (1-\alpha)^n \, n! \, y.$$

The $n!$ cancels the $\frac{1}{n!}$, so the right-hand side of the identity is

$$y + \sum_{n=1}^{\infty} (1-\alpha)^n y = y \sum_{n=0}^{\infty} (1-\alpha)^n.$$

This is a geometric series with common ratio $1-\alpha$. Since $0<\alpha<2$ we have $-1<1-\alpha<1$, so it converges, and its sum is

$$y\cdot\frac{1}{1-(1-\alpha)}=\frac{y}{\alpha}.$$

On the other hand, $y=\alpha x$ gives $x=y/\alpha$ directly, so $\mathrm{f}^{-1}(y)=y/\alpha$. The two agree, which verifies the identity in this case. The condition $0<\alpha<2$ is exactly the condition for the series to converge.

Part (ii)

The equation is $\mathrm{f}(x)=\frac{1}{2}$ with

$$\mathrm{f}(x)=x-\tfrac{1}{4}x^3,\qquad\text{so}\qquad\phi(y)=y-\mathrm{f}(y)=\tfrac{1}{4}y^3.$$

Then $\phi(y)^n=4^{-n}y^{3n}$, and using the derivative formula with $m=3n$ (which is at least $n-1$),

$$\frac{1}{n!}\frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}}\left[4^{-n}y^{3n}\right]=\frac{(3n)!}{n!\,(2n+1)!\,4^n}\,y^{2n+1}.$$

So Lagrange's identity gives

$$\mathrm{f}^{-1}(y)=y+\sum_{n=1}^{\infty}\frac{(3n)!}{n!\,(2n+1)!\,4^n}\,y^{2n+1}.$$

The leading y is exactly what this general term gives when $n=0$, since $\frac{0!}{0!\,1!\,4^0}\,y^1=y$. So we can absorb it into the sum and start from $n=0$:

$$\mathrm{f}^{-1}(y)=\sum_{n=0}^{\infty}\frac{(3n)!}{n!\,(2n+1)!\,4^n}\,y^{2n+1}.$$

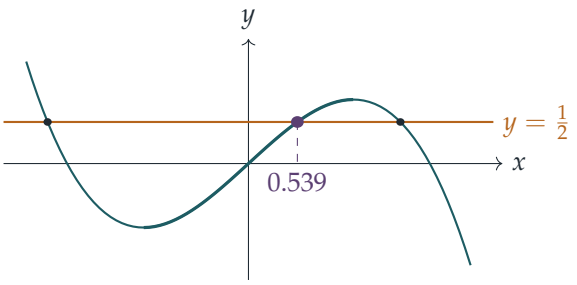
Now put $y=\frac{1}{2}$. Then $4^n\cdot 2^{2n+1}=2^{2n}\cdot 2^{2n+1}=2^{4n+1}$, so

$$x=\mathrm{f}^{-1}\left(\tfrac{1}{2}\right)=\sum_{n=0}^{\infty}\frac{(3n)!}{n!\,(2n+1)!\,2^{4n+1}},$$

and since $\mathrm{f}(x)=\frac{1}{2}$ this x is a root of $\frac{1}{2}=x-\frac{1}{4}x^3$, as required.

Ask yourself The cubic has three real roots. Which one is the series?

The function f isn't one-to-one on the whole real line, so "f^{-1}" means the branch the series describes. The series is a power series in y that starts $y+\cdots$, so it gives $\mathrm{f}^{-1}(0)=0$ and $\mathrm{f}^{-1}(y)\approx y$ for small y. It is the inverse of f restricted to the increasing piece through the origin, where $\mathrm{f}'(x)=1-\frac{3}{4}x^2>0$, that is, $|x|<2/\sqrt{3}$. The first few terms are $\frac{1}{2}+\frac{1}{32}+\frac{3}{512}+\cdots\approx 0.537$, and the sum is 0.5392 to four decimal places: the smallest positive root, marked in the sketch below. The other roots, about 1.675 and -2.214, lie on the decreasing parts of the curve.



The thicker stretch of the curve is the increasing branch through the origin; the series gives the root on it, shown in plum.

Part (iii)

Here the trick is to put the whole equation into the form $f(x) = y$ with something simple for ϕ. Rearranging $x = e^{\lambda x}$ as

$$x - e^{\lambda x} = 0,$$

we take

$$f(x) = x - e^{\lambda x}, \qquad \phi(y) = y - f(y) = e^{\lambda y},$$

and we want $x = f^{-1}(0)$. Now $\phi(y)^n = e^{n\lambda y}$, and each differentiation multiplies this by $n\lambda$, so

$$\frac{d^{n-1}}{dy^{n-1}} e^{n\lambda y} = (n\lambda)^{n-1} e^{n\lambda y}.$$

(For $n = 1$ this reads $e^{\lambda y}$, correctly.) Lagrange's identity gives

$$f^{-1}(y) = y + \sum_{n=1}^{\infty} \frac{(n\lambda)^{n-1}}{n!} e^{n\lambda y},$$

and putting $y = 0$,

$$x = \sum_{n=1}^{\infty} \frac{n^{n-1}}{n!} \lambda^{n-1} = 1 + \lambda + \frac{3}{2}\lambda^2 + \frac{8}{3}\lambda^3 + \frac{125}{24}\lambda^4 + \cdots.$$

Equivalently, shifting the index, $x = \sum_{n=0}^{\infty} \frac{(n+1)^{n-1}}{n!} \lambda^n$.

For which λ does this converge? The ratio of consecutive terms is

$$\frac{(n+1)^n \lambda^n / (n+1)!}{n^{n-1}\lambda^{n-1}/n!} = \lambda \left(1 + \frac{1}{n}\right)^{n-1} \to e\lambda,$$

so by the ratio test the series converges for $|\lambda| < 1/e$. This is a natural boundary: $y = x$ is tangent to $y = e^{\lambda x}$ exactly when $\lambda = 1/e$ (at $x = e$), and for $\lambda > 1/e$ there is no real root at all. For $0 < \lambda < 1/e$ there are two positive roots, and the series gives the smaller one, the one that tends to 1 as $\lambda \to 0$.

Checks

- In (ii), the first four terms are $\frac{1}{2}$, $\frac{1}{32}$, $\frac{3}{512}$ and $\frac{3}{2048}$. Summing 200 terms numerically gives 0.539189, and Newton's method on $x^3 - 4x + 2 = 0$ gives the root $0.5391888\ldots$, in agreement to machine precision.
- In (iii), put $\lambda = 0$: the series gives $x = 1$, and indeed $x = \mathrm{e}^0 = 1$. Summing the series for $\lambda = 0.2$ gives $x \approx 1.29586$, and $\mathrm{e}^{0.2 \times 1.29586} = 1.29586$. For $\lambda = -0.3$ it gives 0.78918, which also satisfies the equation. For $\lambda = 0.2$ the other root is about 12.71, far from the series value.
- In (i), try $\alpha = 1$: then $\phi = 0$, every term of the sum vanishes, and $\mathrm{f}^{-1}(y) = y$, as it should.

Common slips

- Forgetting that the $n = 1$ term has no differentiation (the zeroth derivative), and so dropping or mangling it.
- In (ii), not noticing that the standalone y is the $n = 0$ term of the general formula, and so failing to match the sum, which starts from $n = 0$.
- In (iii), trying $\mathrm{f}(x) = x\mathrm{e}^{-\lambda x}$ or similar. That is legitimate, but then ϕ is a difference of two terms and the derivatives are a mess. Choosing $\phi = \mathrm{e}^{\lambda y}$ makes them trivial.

Beyond STEP

This is the *Lagrange inversion theorem*, which is usually proved with complex analysis (Cauchy's residue theorem) or with formal power series. The coefficients $(n+1)^{n-1}/n!$ in (iii) belong to the Lambert W function, and n^{n-1} counts rooted labelled trees on n vertices. Similarly, the numbers $\frac{(3n)!}{n!\,(2n+1)!}$ in (ii) count ternary trees, a cousin of the Catalan numbers. So Lagrange inversion is a central tool of enumerative combinatorics. Needham's *Visual Complex Analysis* gives the complex-analytic background, and Siklos's *Advanced Problems in Mathematics* has more STEP practice with series.

WORKED SOLUTION

STEP II 1987, Question 6

Pure • Definite Integrals by Symmetry

STEP II 1987, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q6

The key idea

The interval $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$ is symmetric, so the reflection $\theta \mapsto -\theta$ is the natural first move: it turns $\sin\theta$ into $-\sin\theta$ and gives the second form for free. Averaging the two forms removes the odd part and leaves a denominator in $\sin^2\theta$ only, which a little algebra splits into a constant plus a standard $\tan\theta$ integral. The final part is about one innocent-looking step: $\sqrt{\cos^2 2\alpha}$ is $|\cos 2\alpha|$, not $\cos 2\alpha$.

Solution

Throughout, write $s = \sin 2\alpha$ and $c = \cos 2\alpha$, so $s^2 + c^2 = 1$. For $0 < \alpha < \frac{1}{4}\pi$ we have $0 < s < 1$ and $0 < c < 1$. Since $s < 1$, the denominator $1 \mp s\sin\theta$ is positive on the whole interval, so all the integrals are proper.

The second form of I

Substitute $\theta = -\phi$, so $\mathrm{d}\theta = -\mathrm{d}\phi$ and the limits swap:

$$I = \int_{\frac{1}{2}\pi}^{-\frac{1}{2}\pi} \frac{\cos^2(-\phi)}{1 - \sin(-\phi)\, s}\,(-\mathrm{d}\phi) = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\cos^2\phi}{1 + s\sin\phi}\,\mathrm{d}\phi,$$

using $\cos(-\phi) = \cos\phi$ and $\sin(-\phi) = -\sin\phi$. Renaming ϕ as θ gives the required form.

The expression involving $\cot^2 2\alpha$

Adding the two expressions for I,

$$2I = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \cos^2\theta \left(\frac{1}{1 - s\sin\theta} + \frac{1}{1 + s\sin\theta}\right)\mathrm{d}\theta = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{2\cos^2\theta}{1 - s^2\sin^2\theta}\,\mathrm{d}\theta,$$

so

$$I = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\cos^2\theta}{1 - s^2\sin^2\theta}\,\mathrm{d}\theta.$$

Now rewrite the denominator using $1 = \cos^2\theta + \sin^2\theta$ and $1 - s^2 = c^2$:

$$1 - s^2\sin^2\theta = \cos^2\theta + c^2\sin^2\theta.$$

Ask yourself How do we split $\dfrac{\cos^2\theta}{\cos^2\theta + c^2\sin^2\theta}$ into a constant plus something we can integrate?

Aim to write the numerator as a multiple of the denominator plus a remainder with no θ-dependence after dividing by $\cos^2\theta$. Since

$$(\cos^2\theta + c^2\sin^2\theta) - c^2(\cos^2\theta + \sin^2\theta) = (1 - c^2)\cos^2\theta = s^2\cos^2\theta,$$

we have, dividing by $s^2(\cos^2\theta + c^2\sin^2\theta)$ (allowed, as $s \neq 0$),

$$\frac{\cos^2\theta}{\cos^2\theta + c^2\sin^2\theta} = \frac{1}{s^2} - \frac{c^2}{s^2}\cdot\frac{1}{\cos^2\theta + c^2\sin^2\theta} = \frac{1}{s^2} - \frac{c^2}{s^2}\cdot\frac{\sec^2\theta}{1 + c^2\tan^2\theta},$$

where in the last step we divided top and bottom by $\cos^2\theta$. Integrating over an interval of length π, and using $c^2/s^2 = \cot^2 2\alpha$,

$$I = \frac{\pi}{\sin^2 2\alpha} - \cot^2 2\alpha \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\sec^2\theta}{1 + \cos^2 2\alpha\tan^2\theta}\,\mathrm{d}\theta,$$

as required.

Evaluating I

Call the remaining integral J. Substitute $t = \tan\theta$, so $\mathrm{d}t = \sec^2\theta\,\mathrm{d}\theta$; as θ runs from $-\frac{1}{2}\pi$ to $\frac{1}{2}\pi$, t runs over the whole real line. Then, for $c \neq 0$,

$$J = \int_{-\infty}^{\infty} \frac{\mathrm{d}t}{1 + c^2t^2} = \left[\frac{1}{|c|}\arctan\big(|c|\,t\big)\right]_{-\infty}^{\infty} = \frac{\pi}{|c|}.$$

(The substitution $u = |c|t$ is what produces $|c|$: it keeps the limits running from $-\infty$ to ∞ in the right order.)

For $0 < \alpha < \frac{1}{4}\pi$ we have $c > 0$, so $J = \pi/c$ and

$$I = \frac{\pi}{s^2} - \frac{c^2}{s^2}\cdot\frac{\pi}{c} = \frac{\pi(1-c)}{1-c^2} = \frac{\pi}{1+c} = \frac{\pi}{1+\cos 2\alpha} = \frac{\pi}{2\cos^2\alpha} = \tfrac{1}{2}\pi\sec^2\alpha,$$

cancelling $1 - c \neq 0$ and using $1 + \cos 2\alpha = 2\cos^2\alpha$. This is the required result.

The case $\frac{1}{4}\pi < \alpha < \frac{1}{2}\pi$

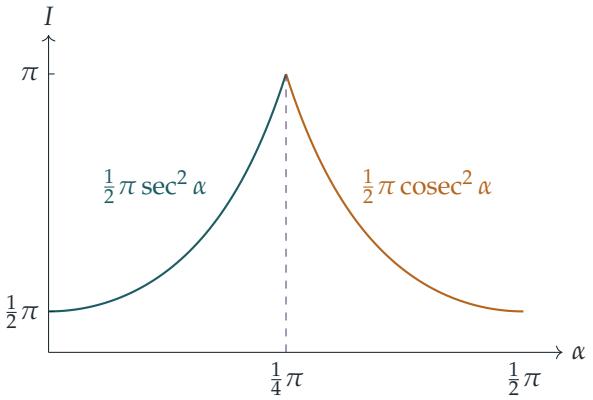
Ask yourself Does anything in the argument change when α passes $\frac{1}{4}\pi$?

Almost nothing: we still have $0 < s < 1$, so every step up to the formula for I in terms of J holds unchanged. The one difference is that now $c = \cos 2\alpha < 0$, so $|c| = -c$ and $J = -\pi/c$. Hence

$$I = \frac{\pi}{s^2} + \frac{c^2}{s^2}\cdot\frac{\pi}{c} = \frac{\pi(1+c)}{1-c^2} = \frac{\pi}{1-c} = \frac{\pi}{2\sin^2\alpha} = \tfrac{1}{2}\pi\,\mathrm{cosec}^2\,\alpha,$$

using $1 - \cos 2\alpha = 2\sin^2\alpha$. Blindly copying the first answer would give the wrong value.

The graph of I against α shows the two formulae meeting at $\alpha = \frac{1}{4}\pi$, where both equal π:



Checks

- At $\alpha = \frac{1}{4}\pi$ we have $s = 1$ and the integrand is $\cos^2\theta/(1-\sin\theta) = 1+\sin\theta$, which integrates to π. Both formulae give π there, as they should.
- At $\alpha = \frac{1}{6}\pi$: $s = \frac{\sqrt{3}}{2}$, $c = \frac{1}{2}$, so $I = \pi/(1+c) = \frac{2}{3}\pi$, and $\frac{1}{2}\pi\sec^2\frac{1}{6}\pi = \frac{1}{2}\pi\cdot\frac{4}{3} = \frac{2}{3}\pi$.
- Simpson's rule on the original integral at ten random values of α on both sides of $\frac{1}{4}\pi$ agrees with the formulae to better than 10^{-6}; for example $\alpha \approx 0.6308$ gives $I \approx 2.408759$ both ways.
- As $\alpha \to 0$ the integrand tends to $\cos^2\theta$, whose integral is $\frac{1}{2}\pi$, matching $\frac{1}{2}\pi\sec^2 0$.

Common slips

- Writing $\int_{-\infty}^{\infty}\frac{\mathrm{d}t}{1+c^2t^2} = \frac{\pi}{c}$ without the modulus. For $c < 0$ this is negative, which is impossible for a positive integrand.
- Losing the minus sign from $\mathrm{d}\theta = -\mathrm{d}\phi$, or forgetting that it is cancelled by swapping the limits.

- Worrying that $t = \tan\theta$ is undefined at $\theta = \pm\frac{1}{2}\pi$: the t-integral is improper but converges, which is all we need.

Beyond STEP

The tidy symmetry $\alpha \leftrightarrow \frac{1}{2}\pi - \alpha$ in the answer is no accident: I depends on α only through $\sin 2\alpha$, which is unchanged by that swap, while a formula like $\frac{1}{2}\pi\sec^2\alpha$ is not. Algebra that silently squares and square-roots can hide exactly this kind of issue. Integrals of rational functions of $\sin\theta$ and $\cos\theta$ over a full period (such as our averaged integrand, which has period π) are routinely done at university by contour integration around the unit circle, where the residue theorem makes them almost mechanical. For a beautifully geometric introduction, see Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP II 1987, Question 7

Pure • Trapezium Rule and Factorials

STEP II 1987, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q7

The key idea

The trapezium sum for $\ln x$ with unit steps is almost exactly $\ln n!$, because the heights it adds are $\ln 1, \ln 2, \ldots, \ln n$. Since $\ln x$ is concave, every chord lies below the curve, so the trapezium rule always *underestimates* the integral; that one fact gives the direction of every inequality. For the last part, the trick is to rescale the fine-grid error so that it becomes a unit-step error on a different interval, which can then be compared directly with the error behind $\mathrm{g}(kn)$.

Solution

Justifying the rule

Split $[x_0, x_N]$ into the N strips $[x_r, x_{r+1}]$, each of width h. On each strip, replace the curve by the straight line (chord) joining $(x_r, \mathrm{f}(x_r))$ to $(x_{r+1}, \mathrm{f}(x_{r+1}))$. The region under the chord is a trapezium of area $\frac{1}{2}h\{\mathrm{f}(x_r) + \mathrm{f}(x_{r+1})\}$. Adding over $r = 0, 1, \ldots, N-1$, every interior ordinate $\mathrm{f}(x_1), \ldots, \mathrm{f}(x_{N-1})$ appears in two neighbouring trapezia and the end ordinates appear once each, which gives the stated formula. The approximation is good when h is small, as the chord then stays close to the curve.

Unit steps and $n! \approx \mathrm{g}(n)$

Integrating by parts with $u = \ln x$, $\mathrm{d}v = \mathrm{d}x$,

$$I_n = \int_1^n \ln x \,\mathrm{d}x = \Big[x \ln x\Big]_1^n - \int_1^n 1 \,\mathrm{d}x = n \ln n - n + 1.$$

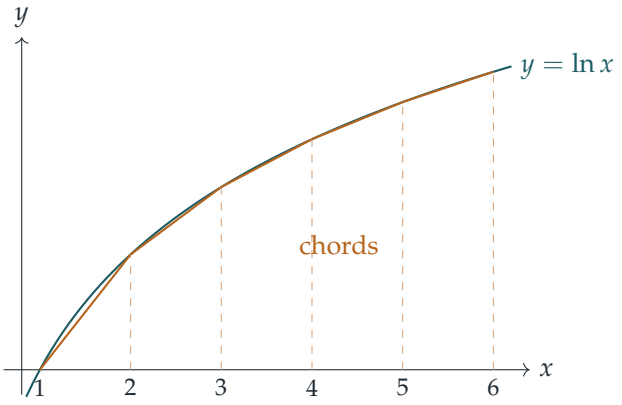
With $h = 1$ the ordinates are at $x = 1, 2, \ldots, n$, so the trapezium sum is

$$\begin{aligned} T_n &= \tfrac{1}{2}\{\ln 1 + 2\ln 2 + \cdots + 2\ln(n-1) + \ln n\} \\ &= \ln 2 + \ln 3 + \cdots + \ln n - \tfrac{1}{2}\ln n = \ln n! - \tfrac{1}{2}\ln n, \end{aligned}$$

using $\ln 1 = 0$. Setting $T_n \approx I_n$,

$$\ln n! \approx \left(n + \tfrac{1}{2}\right)\ln n + 1 - n, \qquad \text{so} \qquad n! \approx n^{n+\frac{1}{2}}\mathrm{e}^{1-n} = \mathrm{g}(n).$$

Why $n! < \mathrm{g}(n)$



Since $\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}\ln x = -\dfrac{1}{x^2} < 0$, the curve $y = \ln x$ is strictly concave: on each strip $[m, m+1]$ the chord lies strictly below the curve except at its endpoints (see the sketch). So each trapezium has strictly less area than the region under the curve on that strip, and adding the $n - 1 \geq 1$ strips,

$$T_n < I_n \quad \Longrightarrow \quad \ln n! - \tfrac{1}{2}\ln n < n\ln n - n + 1 \quad \Longrightarrow \quad \ln n! < \ln \mathrm{g}(n).$$

As exp is increasing, $n! < \mathrm{g}(n)$.

Strips of width $1/k$

Now $h = 1/k$, $N = k(n-1)$ and $x_r = 1 + \frac{r}{k} = \frac{k+r}{k}$ for $r = 0, 1, \ldots, k(n-1)$. The last ordinate is $x_N = n$. Writing the trapezium sum as "h times the sum of all ordinates, minus half of each end ordinate", and using $\ln x_0 = 0$,

$$T_n^{(k)} = \frac{1}{k}\left\{\sum_{r=0}^{k(n-1)} \ln\frac{k+r}{k} - \tfrac{1}{2}\ln n\right\}.$$

The numerators $k + r$ run through $k, k+1, \ldots, kn$, and there are $k(n-1)+1$ factors of k in the denominators, so

$$\sum_{r=0}^{k(n-1)} \ln\frac{k+r}{k} = \ln\frac{(kn)!}{(k-1)!} - \big(k(n-1)+1\big)\ln k = \ln\frac{(kn)!}{k!} - k(n-1)\ln k,$$

since $(k-1)!\,k = k!$. Setting $kT_n^{(k)} \approx kI_n = kn\ln n - kn + k$ and rearranging,

$$\begin{aligned}\ln(kn)! &\approx \ln k! + k(n-1)\ln k + \tfrac{1}{2}\ln n + kn\ln n + k(1-n)\\ &= \ln k! + \left(kn + \tfrac{1}{2}\right)\ln n + k(1-n)(1-\ln k),\end{aligned}$$

that is,

$$(kn)! \approx k!\, n^{kn+\frac{1}{2}} \left(\frac{\mathrm{e}}{k}\right)^{k(1-n)} =: A,$$

as required.

Which is closer?

Write $E(a,b)$ for the error $\displaystyle\int_a^b \ln t \,\mathrm{d}t$ − (unit-step trapezium sum on $[a,b]$), for integers $a < b$. By the concavity argument above, $E(a,b) > 0$, and errors add over adjacent intervals: $E(1,b) = E(1,a) + E(a,b)$.

The derivation of A was exact apart from replacing $T_n^{(k)}$ by I_n, so

$$\ln A - \ln(kn)! = k\big(I_n - T_n^{(k)}\big).$$

Similarly, from the unit-step work with kn in place of n, $\ln \mathrm{g}(kn) - \ln(kn)! = E(1,kn)$.

> **Ask yourself** How can an error on a grid of width $1/k$ be compared with an error on a grid of width 1?

Substitute $x = t/k$. The ordinates $x_r = (k+r)/k$ become $t = k, k+1, \ldots, kn$, and $\mathrm{d}x = \mathrm{d}t/k$, so

$$k\big(I_n - T_n^{(k)}\big) = \int_k^{kn} \ln\frac{t}{k}\,\mathrm{d}t \;-\; \big(\text{unit-step trapezium sum of } \ln\tfrac{t}{k} \text{ on } [k,kn]\big).$$

Since $\ln\frac{t}{k} = \ln t - \ln k$ and the trapezium rule is exact for the constant $\ln k$, the constant cancels and this is just $E(k,kn)$. Therefore

$$\ln \mathrm{g}(kn) - \ln(kn)! = E(1,kn) = E(1,k) + E(k,kn) = E(1,k) + \big(\ln A - \ln(kn)!\big).$$

Both A and $\mathrm{g}(kn)$ exceed $(kn)!$, and for $k \geq 2$ we have $E(1,k) > 0$, so

$$(kn)! < A < \mathrm{g}(kn) \qquad (k \geq 2).$$

So the new approximation A is closer to $(kn)!$ than $\mathrm{g}(kn)$ is, whether you measure by the difference or the ratio. When $k = 1$ the two are identical, as $A = 1!\, n^{n+\frac{1}{2}}\mathrm{e}^{1-n} = \mathrm{g}(n)$. In words, A uses the *exact* value $k!$ for the first k factors, and only approximates the rest, so it avoids the error $E(1,k)$ from the start of the curve, where ln bends most.

Checks

- $k = 1$ in the final formula gives back $\mathrm{g}(n)$, as it must.
- Take $n = 5$: $5! = 120$ and $\mathrm{g}(5) = 5^{5.5}\mathrm{e}^{-4} \approx 128.0$, so indeed $n! < \mathrm{g}(n)$.
- Take $k = 3$, $n = 5$: $15! \approx 1.3077 \times 10^{12}$, $A \approx 1.3369 \times 10^{12}$ and $\mathrm{g}(15) \approx 1.4102 \times 10^{12}$, confirming $(kn)! < A < \mathrm{g}(kn)$.
- The ratio $\mathrm{g}(kn)/A = \exp E(1,k)$ depends only on k, not n; for $k = 3$ it is $\mathrm{g}(3)/3! \approx 1.0548$, matching the numbers above.

Common slips

- Miscounting the factors of k in the width-$1/k$ sum: there are $k(n-1)+1$ ordinates, and the extra $\ln k$ is what turns $(k-1)!$ into $k!$.
- Arguing "$n! < \mathrm{g}(n)$" from a sketch with the chords above the curve. For a concave function the chords lie below, so the trapezium rule underestimates.
- Forgetting that the error in $\ln A$ is k times the fine-grid error, not the fine-grid error itself.

Beyond STEP

Here $\mathrm{g}(n)$ is a crude form of Stirling's formula, $n! \sim \sqrt{2\pi n}\, n^n \mathrm{e}^{-n}$. Since the errors $E(1,n)$ increase with n and are bounded (each strip's error is at most a constant times $1/m^2$), they converge, which explains why $n!/\mathrm{g}(n)$ tends to the constant $\sqrt{2\pi}/\mathrm{e} \approx 0.922$. The Euler–Maclaurin formula makes this precise and gives the full asymptotic series. Bartle and Sherbert, *Introduction to Real Analysis*, covers the Riemann integral and the error estimates for numerical rules that underpin this.

WORKED SOLUTION

STEP II 1987, Question 8

Pure • Vector Loci in 3D

STEP II 1987, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q8

The key idea

Each equation becomes recognisable once it is written in terms of distances from the points A and B with position vectors $\mathbf{a}$ and $\mathbf{b}$. For the equivalence, everything can be expressed using just two numbers, $P = |\mathbf{r}-\mathbf{a}|^2$ and $Q = |\mathbf{r}-\mathbf{b}|^2$. Then (i) and (ii) become two simple linear equations in P and Q, and so do (iii) and (iv).

Solution

Throughout, A, B and R are the points with position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{r}$, and M is the midpoint of AB, with position vector $\mathbf{m} = \frac{1}{2}(\mathbf{a}+\mathbf{b})$. We assume $\mathbf{a} \neq \mathbf{b}$ (the case $\mathbf{a} = \mathbf{b}$ is dealt with at the end), and write $d = |\mathbf{a}-\mathbf{b}| = AB$.

Part (i)

Expanding,

$$\begin{aligned} |\mathbf{r}-\mathbf{a}|^2 - |\mathbf{r}-\mathbf{b}|^2 &= \left(|\mathbf{r}|^2 - 2\mathbf{a}\cdot\mathbf{r} + |\mathbf{a}|^2\right) - \left(|\mathbf{r}|^2 - 2\mathbf{b}\cdot\mathbf{r} + |\mathbf{b}|^2\right) \\ &= -2\Big[(\mathbf{a}-\mathbf{b})\cdot\mathbf{r} - \tfrac{1}{2}\left(|\mathbf{a}|^2 - |\mathbf{b}|^2\right)\Big]. \end{aligned}$$

So (i) holds exactly when $|\mathbf{r}-\mathbf{a}| = |\mathbf{r}-\mathbf{b}|$, that is, when $RA = RB$. The locus is the **plane that perpendicularly bisects** AB: the plane through M with normal $\mathbf{a}-\mathbf{b}$. (We can see it is a plane directly, since (i) has the form $\mathbf{n}\cdot\mathbf{r} = \text{constant}$ with $\mathbf{n} = \mathbf{a}-\mathbf{b} \neq \mathbf{0}$, and $\mathbf{r} = \mathbf{m}$ satisfies it because $(\mathbf{a}-\mathbf{b})\cdot(\mathbf{a}+\mathbf{b}) = |\mathbf{a}|^2 - |\mathbf{b}|^2$.)

Part (ii)

Equation (ii) says that $\overrightarrow{RA}$ and $\overrightarrow{RB}$ are perpendicular, or that R is A or B. To identify the locus, complete the square:

$$\begin{aligned}|\mathbf{r}-\mathbf{m}|^2 &= |\mathbf{r}|^2 - (\mathbf{a}+\mathbf{b})\cdot\mathbf{r} + \tfrac{1}{4}|\mathbf{a}+\mathbf{b}|^2 \\ &= (\mathbf{a}-\mathbf{r})\cdot(\mathbf{b}-\mathbf{r}) - \mathbf{a}\cdot\mathbf{b} + \tfrac{1}{4}|\mathbf{a}+\mathbf{b}|^2 \\ &= (\mathbf{a}-\mathbf{r})\cdot(\mathbf{b}-\mathbf{r}) + \tfrac{1}{4}|\mathbf{a}-\mathbf{b}|^2.\end{aligned}$$

So (ii) is equivalent to $|\mathbf{r}-\mathbf{m}| = \frac{1}{2}d$. The locus is the **sphere with AB as a diameter**: centre M, radius $\frac{1}{2}AB$. This is the three-dimensional version of "the angle in a semicircle is a right angle".

Parts (iii) and (iv)

Equation (iii) says $RA = d/\sqrt{2}$. Its locus is the **sphere with centre A and radius $AB/\sqrt{2}$**. In the same way, the locus of (iv) is the **sphere with centre B and radius $AB/\sqrt{2}$**.

The equivalence

Let $P = |\mathbf{r}-\mathbf{a}|^2$ and $Q = |\mathbf{r}-\mathbf{b}|^2$. Part (i) showed that

$$\text{(i)} \iff P = Q.$$

For (ii), write $\mathbf{a}-\mathbf{b} = (\mathbf{a}-\mathbf{r}) - (\mathbf{b}-\mathbf{r})$ and square:

$$d^2 = |\mathbf{a}-\mathbf{r}|^2 + |\mathbf{b}-\mathbf{r}|^2 - 2(\mathbf{a}-\mathbf{r})\cdot(\mathbf{b}-\mathbf{r}) = P + Q - 2(\mathbf{a}-\mathbf{r})\cdot(\mathbf{b}-\mathbf{r}).$$

Hence

$$\text{(ii)} \iff P + Q = d^2.$$

Equations (iii) and (iv) say $P = \frac{1}{2}d^2$ and $Q = \frac{1}{2}d^2$. Now

$$\left(P = Q \text{ and } P + Q = d^2\right) \iff P = Q = \tfrac{1}{2}d^2.$$

The forward direction holds because substituting $Q = P$ gives $2P = d^2$. For the reverse, if $P = Q = \frac{1}{2}d^2$ then certainly $P = Q$ and $P + Q = d^2$. Every step above is an "if and only if", so (i) and (ii) together are equivalent to (iii) and (iv) together, as required.

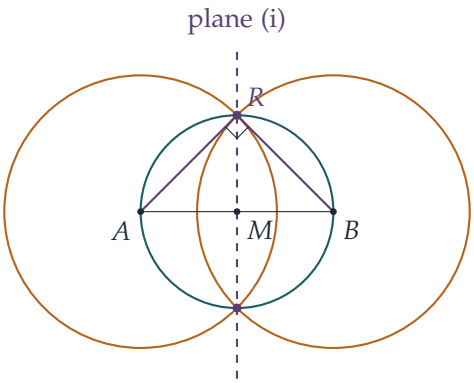
> **Ask yourself** How can a plane meeting a sphere give the same set as two spheres meeting?

Both pairs describe the same circle. The plane (i) passes through M, the centre of the sphere (ii), so it cuts that sphere in a great circle: centre M, radius $\frac{1}{2}AB$, lying in the perpendicular bisector plane of AB. The spheres (iii) and (iv) have equal radii $AB/\sqrt{2}$, and these sum to more than AB, so they do meet. By symmetry they meet in a circle in the perpendicular bisector plane, centred at M. Its radius is $\sqrt{\frac{1}{2}AB^2 - \frac{1}{4}AB^2} = \frac{1}{2}AB$, by Pythagoras in triangle AMR, so it is the same circle.

Geometrical meaning

A point R lies on this circle exactly when triangle ARB is *right-angled and isosceles*, with the right angle at R. The pair (i) and (ii) describes this through its properties: $RA = RB$ (isosceles) and $\angle ARB = 90^\circ$ (right angle). The pair (iii) and (iv) describes it through its side lengths: in a right isosceles triangle with hypotenuse AB, both legs have length $AB/\sqrt{2}$. The algebra above is Pythagoras' theorem and its converse in vector form. The identity $d^2 = P + Q - 2(\mathbf{a} - \mathbf{r}) \cdot (\mathbf{b} - \mathbf{r})$ is the cosine rule in triangle ARB.

The diagram shows a cross-section by any plane containing A and B. The bisecting plane appears as the dashed line, the sphere (ii) as the teal circle, and the spheres (iii) and (iv) as the amber circles. All four meet at the two points where the circle of solutions crosses the page.



The degenerate case

If $\mathbf{a} = \mathbf{b}$, then (i) reads $0 = 0$ and holds for all $\mathbf{r}$, while (ii), (iii) and (iv) each say $\mathbf{r} = \mathbf{a}$. The equivalence still holds, but every locus collapses to the single point A (or to all of space for (i)).

Checks

Take $\mathbf{a} = (1, 0, 0)$ and $\mathbf{b} = (-1, 0, 0)$. Then (i) is the plane $x = 0$, and (ii) is $x^2 + y^2 + z^2 = 1$. Equations (iii) and (iv) are $(x \mp 1)^2 + y^2 + z^2 = 2$, and subtracting them gives $x = 0$ again. All four meet in the unit circle $y^2 + z^2 = 1$, $x = 0$. For example, $R = (0, 1, 0)$ gives $RA = RB = \sqrt{2}$ and $\overrightarrow{RA} \cdot \overrightarrow{RB} = -1 + 1 = 0$. We also checked all the identities above exactly, in rational arithmetic, for 5000 random choices of $\mathbf{a}$, $\mathbf{b}$, $\mathbf{r}$. We confirmed that random points on the predicted circle satisfy all four equations.

Common slips

- Describing (ii) as "a circle". In three dimensions it is a sphere, and the locus of (i) is a plane, not a line.

- Proving only one direction. The question asks for an equivalence, so make sure each step reverses, or prove both directions.
- Saying the two pairs "both give a circle" without checking it is the *same* circle: same plane, same centre, same radius.

Beyond STEP

Each of these loci is a level set of a quadratic function of $\mathbf{r}$, and the trick of subtracting two sphere equations to get a plane is the idea of the *radical plane* of two spheres. More generally, describing the same geometric object by different systems of equations is the starting point of algebraic geometry. Here, a circle in space is cut out by two equations, and many different pairs of equations do the job. Strang's *Introduction to Linear Algebra* is a good place to strengthen the vector algebra, in particular the link between dot products, lengths and angles.

WORKED SOLUTION

STEP II 1987, Question 9

Pure • Orthogonal Matrices

STEP II 1987, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q9

The key idea

The condition $\mathbf{B}^{\mathrm{T}}\mathbf{B} = \mathbf{I}$ has inverses on both outside ends once we transpose $\mathbf{B}$. Because those inverse matrices are invertible, we can multiply through by $\mathbf{I} - \mathbf{A}^{\mathrm{T}}$ on the left and $\mathbf{I} - \mathbf{A}$ on the right without losing information. What's left is a polynomial identity in $\mathbf{A}$ and $\mathbf{A}^{\mathrm{T}}$ that collapses almost at once. The thing to be careful about is that every step must be reversible, since we need "if and only if".

Solution

The standard results we use are:

- $(\mathbf{XY})^{\mathrm{T}} = \mathbf{Y}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}$, and transposition is linear, so $(\mathbf{I} \pm \mathbf{A})^{\mathrm{T}} = \mathbf{I} \pm \mathbf{A}^{\mathrm{T}}$ (as $\mathbf{I}^{\mathrm{T}} = \mathbf{I}$);
- if $\mathbf{X}$ is non-singular then so is $\mathbf{X}^{\mathrm{T}}$, and $(\mathbf{X}^{-1})^{\mathrm{T}} = (\mathbf{X}^{\mathrm{T}})^{-1}$.

Step 1: the transpose of B. Since $\mathbf{I} - \mathbf{A}$ is non-singular, so is its transpose $\mathbf{I} - \mathbf{A}^{\mathrm{T}}$. Reversing the order of the product,

$$\mathbf{B}^{\mathrm{T}} = \left((\mathbf{I} - \mathbf{A})^{-1}\right)^{\mathrm{T}}(\mathbf{I} + \mathbf{A})^{\mathrm{T}} = (\mathbf{I} - \mathbf{A}^{\mathrm{T}})^{-1}(\mathbf{I} + \mathbf{A}^{\mathrm{T}}).$$

Hence

$$\mathbf{B}^{\mathrm{T}}\mathbf{B} = (\mathbf{I} - \mathbf{A}^{\mathrm{T}})^{-1}\,(\mathbf{I} + \mathbf{A}^{\mathrm{T}})(\mathbf{I} + \mathbf{A})\,(\mathbf{I} - \mathbf{A})^{-1}.$$

Step 2: clear the inverses, reversibly. Write $\mathbf{P} = \mathbf{I} - \mathbf{A}^{\mathrm{T}}$, $\mathbf{Q} = \mathbf{I} - \mathbf{A}$ and $\mathbf{M} = (\mathbf{I} + \mathbf{A}^{\mathrm{T}})(\mathbf{I} + \mathbf{A})$, so that $\mathbf{B}^{\mathrm{T}}\mathbf{B} = \mathbf{P}^{-1}\mathbf{M}\mathbf{Q}^{-1}$. We claim

$$\mathbf{P}^{-1}\mathbf{M}\mathbf{Q}^{-1} = \mathbf{I} \quad \Longleftrightarrow \quad \mathbf{M} = \mathbf{PQ}.$$

($\Rightarrow$) Multiply on the left by $\mathbf{P}$ and on the right by $\mathbf{Q}$: $\mathbf{M} = \mathbf{PIQ} = \mathbf{PQ}$.
($\Leftarrow$) Multiply on the left by $\mathbf{P}^{-1}$ and on the right by $\mathbf{Q}^{-1}$: $\mathbf{P}^{-1}\mathbf{M}\mathbf{Q}^{-1} = \mathbf{P}^{-1}\mathbf{PQQ}^{-1} = \mathbf{I}$.

Ask yourself Why can't we just "multiply both sides" and stop there?

Multiplying an equation by a matrix always gives a true equation, but in general it can't be undone: from $\mathbf{XY} = \mathbf{XZ}$ we can't conclude $\mathbf{Y} = \mathbf{Z}$ unless $\mathbf{X}$ is invertible. Here the reverse direction works only because $\mathbf{P}$ and $\mathbf{Q}$ have inverses, which is exactly where the hypothesis that $\mathbf{I} - \mathbf{A}$ is non-singular (and hence so is $\mathbf{I} - \mathbf{A}^{\mathrm{T}}$) is used. Note too that we must keep $\mathbf{P}$ on the left and $\mathbf{Q}$ on the right, as matrix multiplication isn't commutative.

Step 3: expand. Expanding both sides of $\mathbf{M} = \mathbf{PQ}$, carefully keeping the order of factors,

$$(\mathbf{I} + \mathbf{A}^{\mathrm{T}})(\mathbf{I} + \mathbf{A}) = \mathbf{I} + \mathbf{A} + \mathbf{A}^{\mathrm{T}} + \mathbf{A}^{\mathrm{T}}\mathbf{A},$$
$$(\mathbf{I} - \mathbf{A}^{\mathrm{T}})(\mathbf{I} - \mathbf{A}) = \mathbf{I} - \mathbf{A} - \mathbf{A}^{\mathrm{T}} + \mathbf{A}^{\mathrm{T}}\mathbf{A}.$$

The $\mathbf{I}$ and $\mathbf{A}^{\mathrm{T}}\mathbf{A}$ terms appear on both sides, so subtracting,

$$\mathbf{M} = \mathbf{PQ} \iff 2(\mathbf{A} + \mathbf{A}^{\mathrm{T}}) = \mathbf{O} \iff \mathbf{A} + \mathbf{A}^{\mathrm{T}} = \mathbf{O}.$$

Each of these is reversible: adding or subtracting the same matrix on both sides, and multiplying by the non-zero scalar 2 or $\frac{1}{2}$.

Conclusion. Chaining Steps 1–3, every link being an equivalence,

$$\mathbf{B}^{\mathrm{T}}\mathbf{B} = \mathbf{I} \iff (\mathbf{I} + \mathbf{A}^{\mathrm{T}})(\mathbf{I} + \mathbf{A}) = (\mathbf{I} - \mathbf{A}^{\mathrm{T}})(\mathbf{I} - \mathbf{A}) \iff \mathbf{A} + \mathbf{A}^{\mathrm{T}} = \mathbf{O},$$

which is what we had to prove. In words: $\mathbf{B}$ is orthogonal exactly when $\mathbf{A}$ is skew-symmetric.

Checks

- Take $\mathbf{A} = \begin{pmatrix} 0 & t \\ -t & 0 \end{pmatrix}$, which is skew-symmetric. Then $\det(\mathbf{I} - \mathbf{A}) = 1 + t^2 \neq 0$ and

$$\mathbf{B} = \frac{1}{1+t^2}\begin{pmatrix} 1 - t^2 & 2t \\ -2t & 1 - t^2 \end{pmatrix},$$

  a rotation matrix (put $t = \tan\frac{\theta}{2}$), so $\mathbf{B}^{\mathrm{T}}\mathbf{B} = \mathbf{I}$ as expected.
- The 1×1 case: $\mathbf{A} = (a)$ with $a \neq 1$ gives $\mathbf{B} = \left(\frac{1+a}{1-a}\right)$, and $\mathbf{B}^2 = 1$ forces $1 + a = \pm(1 - a)$, i.e. $a = 0$ (the other sign is impossible). This matches $2a = 0$.
- We tested about 2000 random matrices of sizes 1 to 5 with exact rational arithmetic: every skew-symmetric $\mathbf{A}$ gave $\mathbf{B}^{\mathrm{T}}\mathbf{B} = \mathbf{I}$ exactly, and none of the others did.

Common slips

- Writing $\mathbf{B}^{\mathrm{T}} = (\mathbf{I} + \mathbf{A}^{\mathrm{T}})(\mathbf{I} - \mathbf{A}^{\mathrm{T}})^{-1}$ without reversing the order. (It happens to be true, because $\mathbf{I} + \mathbf{A}^{\mathrm{T}}$ and $(\mathbf{I} - \mathbf{A}^{\mathrm{T}})^{-1}$ commute, but that needs its own justification.)
- Expanding $(\mathbf{I} + \mathbf{A}^{\mathrm{T}})(\mathbf{I} + \mathbf{A})$ with $\mathbf{A}\mathbf{A}^{\mathrm{T}}$ instead of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$. Here it cancels either way, but only if both sides are expanded consistently.
- Proving only one direction. The question asks for "if and only if", so say why each step can be reversed.

Beyond STEP

The map $\mathbf{A} \mapsto (\mathbf{I} + \mathbf{A})(\mathbf{I} - \mathbf{A})^{-1}$ is the *Cayley transform*. It turns skew-symmetric matrices, which form a vector space, into orthogonal matrices, which form a group, giving a rational parametrisation of rotations. (One can show that $\mathbf{I} - \mathbf{A}$ is automatically invertible when $\mathbf{A}$ is real and skew-symmetric, and that the orthogonal matrices reached are exactly those without eigenvalue -1.) This is a first glimpse of how a Lie group is related to its Lie algebra. Strang's *Introduction to Linear Algebra* is a good place to read more about orthogonal and skew-symmetric matrices.

WORKED SOLUTION

STEP II 1987, Question 10

Pure • Group Axioms

STEP II 1987, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q10

The key idea

Everything happens in the subscripts, so translate each operation into ordinary arithmetic on the indices $1, 2, \ldots, N$: the first operation is "take the maximum" and the second is "take the distance apart, plus one". Then test the four axioms (closure, associativity, identity, inverses) one at a time. A failing axiom needs only one explicit counterexample, and a holding axiom needs a general argument.

Solution

Throughout, write $j \circ k$ for the index of $a_j \circ a_k$, and similarly for $*$. Two elements are equal exactly when their indices are equal.

The operation $\circ$

Here $a_j \circ a_k = a_{\max(j,k)}$.

Closure holds. If $1 \leq j, k \leq N$ then $\max(j,k)$ is one of j and k, so it lies in $\{1, \ldots, N\}$ and $a_j \circ a_k \in S$.

Associativity holds. For any j, k, l,

$$(a_j \circ a_k) \circ a_l = a_{\max(\max(j,k),l)} = a_{\max(j,k,l)} = a_{\max(j,\max(k,l))} = a_j \circ (a_k \circ a_l),$$

since both bracketings give the largest of the three indices.

An identity exists: it is a_1. Every index is at least 1, so $\max(1,k) = \max(k,1) = k$, giving $a_1 \circ a_k = a_k \circ a_1 = a_k$ for every k.

Inverses fail. An inverse of a_j would be some a_k with $a_j \circ a_k = a_1$, that is $\max(j,k) = 1$. But $\max(j,k) \geq j$, so for $j \geq 2$ this is impossible. For example $a_2 \circ a_k$ is a_2 or something later for every k, never a_1. Only a_1 has an inverse (itself).

So S under $\circ$ satisfies closure, associativity and identity, but not inverses, and it is not a group.

Ask yourself Could some other element be the identity, so that the inverse argument has to be redone?

No: an identity is unique. If e and e' were both identities then $e = e \circ e' = e'$. So the only candidate for "$a_j \circ a_k$ equals the identity" is a_1, and the argument above is complete.

The operation $*$

Here $a_j * a_k = a_{|j-k|+1}$. For $N = 4$ the table of indices is

$*$	1	2	3	4
1	1	2	3	4
2	2	1	2	3
3	3	2	1	2
4	4	3	2	1

and the patterns visible in it hold for every N.

Closure holds. If $1 \le j, k \le N$ then $0 \le |j-k| \le N-1$, so $1 \le |j-k|+1 \le N$ and $a_j * a_k \in S$.

An identity exists: it is a_1. Since $k \ge 1$, $|1-k| = k-1$, so $a_1 * a_k = a_k$, and $a_k * a_1 = a_k$ likewise (the operation is commutative because $|j-k| = |k-j|$).

Inverses hold. $a_j * a_j = a_{0+1} = a_1$, the identity, so every element is its own inverse. (This is the diagonal of 1s in the table.)

Associativity fails. Take a_2, a_2, a_3, which needs $N \ge 3$:

$$(a_2 * a_2) * a_3 = a_1 * a_3 = a_3,$$
$$a_2 * (a_2 * a_3) = a_2 * a_{|2-3|+1} = a_2 * a_2 = a_1.$$

These differ, so $*$ is not associative.

So S under $*$ satisfies closure, identity and inverses, but not associativity, and again it is not a group.

Ask yourself Where was $N > 2$ used?

Only in the counterexample to associativity, which uses a_3. When $N = 2$ the table is that of the group of order 2 (a_1 the identity, $a_2 * a_2 = a_1$), and $*$ is associative; for $\circ$ the inverse argument needs only a_2, so $\circ$ fails for $N = 2$ as well.

Checks

Small cases by brute force: for every N from 2 to 9 we listed all triples to test associativity and searched for identities and inverses directly. For $\circ$: closed, associative, identity a_1, and only a_1 invertible. For $*$: closed, identity a_1, every element self-inverse, and

associativity failing (including at (a_2, a_2, a_3)) for every $N \geq 3$ but holding for $N = 2$. You can do the $N = 3$ case by hand from a 3×3 table in a minute.

Common slips

- Claiming $\circ$ has no identity because "max" is not like multiplication. Check the smallest element: a_1 works.
- Trying to prove associativity of $*$ in general, or testing only triples involving a_1 (which always associate, since a_1 is the identity). A single explicit counterexample is what's needed.
- Forgetting closure. It is easy to assume, but here it needs the bound $|j - k| \leq N - 1$.

Beyond STEP

A set with an associative operation and an identity, like S under $\circ$, is called a *monoid*; "max" on a totally ordered set is a standard example, and it is also a *semilattice*. A set with closure, identity and inverses but no associativity, like S under $*$, shows that associativity is a genuinely independent axiom: it cannot be deduced from the other three. For how the group axioms work together, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 1987, Question 11

Mechanics • Friction and Moments

STEP II 1987, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q11

The key idea

The pivot force is unknown and we do not want it, so we take moments about the pivot O, where it has no moment. The rod is three-dimensional, but the plane of the ring gives two natural directions: perpendicular to the plane (where the normal reaction acts) and within it (where friction acts). Splitting the weight the same way, one moment equation gives the normal reaction and another gives the friction. Their ratio is μ.

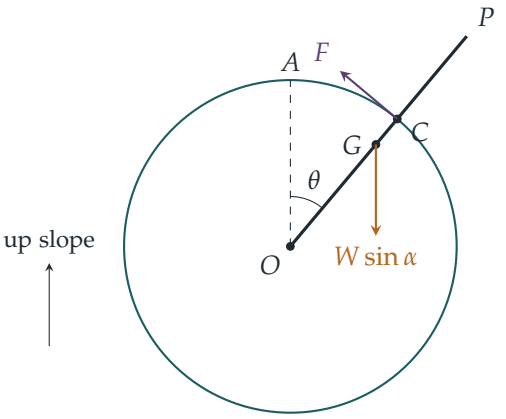
Solution

The model. The rod is a uniform rod OP of weight W, with its centre of mass G at its midpoint, so $OG = \frac{1}{2}b$. It is smoothly pivoted at O, the centre of the ring. It lies across the ring at a point C of the circumference, so $OC = a$. Since O and C both lie in the plane of the ring, the rod lies in that plane. Let A be the highest point of the ring, so OA points straight up the line of greatest slope, and $\angle AOC = \theta$.

The forces. The forces on the rod are:

- its weight W at G;
- the force at the smooth pivot O, which is unknown but has no moment about O;
- the contact force at C. The rod and the ring's wire cross at C. The rod runs along the radius and the wire runs along the tangent, and both lie in the plane, so their common normal is perpendicular to the plane. The normal reaction R is therefore perpendicular to the plane of the ring. Friction F lies in the plane. The rod can only move by turning about O, which sends C along the tangent to the ring. So friction acts along that tangent, perpendicular to the rod.

Resolve the weight into $W\cos\alpha$ perpendicular to the plane (into it) and $W\sin\alpha$ down the line of greatest slope, within the plane. The diagram is drawn looking straight down onto the plane, with A at the top. R points out of the page at C.



Moments about O. Because the pivot is smooth, the total moment about O of the remaining forces is zero about *every* axis through O. Two axes give useful equations.

Axis in the plane, through O, perpendicular to the rod. Only the forces perpendicular to the plane have a moment about this axis. These are $W\cos\alpha$ at distance $\frac{1}{2}b$ and R at distance a. The in-plane forces F and $W\sin\alpha$ meet this axis or are parallel to it, so they have no moment about it. Hence

$$R\,a = W\cos\alpha\cdot\tfrac{1}{2}b, \qquad \text{so} \qquad R = \frac{Wb\cos\alpha}{2a}.$$

Axis perpendicular to the plane, through O. Only in-plane forces matter now. The rod makes angle θ with the up-slope direction, so the component of $W\sin\alpha$ perpendicular to the rod is $W\sin\alpha\sin\theta$, acting at distance $\frac{1}{2}b$. Friction is already perpendicular to the rod, at distance a. Hence

$$F\,a = W\sin\alpha\sin\theta\cdot\tfrac{1}{2}b, \qquad \text{so} \qquad F = \frac{Wb\sin\alpha\sin\theta}{2a}.$$

The in-plane part of the weight tends to swing the rod away from A, down towards the bottom of the ring, and friction opposes this.

? **Ask yourself** Couldn't friction also have a component along the rod?

A force along the rod passes through O, so it has no moment about O. It cannot help or hinder the rotation, and the pivot can supply any force along the rod. The only possible motion is turning about O, so the friction that matters for limiting equilibrium is the component along the tangent, which is F. We take friction to act wholly along the tangent.

Limiting equilibrium. In limiting equilibrium $F = \mu R$:

$$\frac{Wb\sin\alpha\sin\theta}{2a} = \mu\,\frac{Wb\cos\alpha}{2a} \qquad \Longrightarrow \qquad \boxed{\mu = \tan\alpha\,\sin\theta}$$

(we take $0 \le \theta \le \pi$). The rod rests in equilibrium at angle θ whenever $\tan\alpha\sin\theta \le \mu$. Notice that a, b and W have all cancelled.

Checks

- If $\theta = 0$ or $\theta = \pi$, the rod lies along the line of greatest slope. Then the in-plane weight pulls straight along the rod, and no friction is needed: $\mu = 0$. The same holds when $\alpha = 0$, since a horizontal ring gives no in-plane weight.
- Friction is hardest to supply when $\theta = \frac{1}{2}\pi$, with the rod horizontal. Then $\mu = \tan\alpha$, which matches a block on a slope of angle α.
- A numerical check built the forces as 3D vectors for 2000 random choices of α, θ, a, b, W. It solved the vector moment equation about O for R and F, and found F/R agreeing with $\tan\alpha \sin\theta$ to within 4×10^{-15}. For example, $\alpha = 30^\circ$ and $\theta = 60^\circ$ give $\mu = \frac{1}{2}$.

Common slips

- Taking the normal reaction at C to be along the radius, as it would be for a rod resting *inside* a ring in a vertical plane. Here the rod lies across the wire, so the normal is perpendicular to the plane of the ring.
- Resolving forces instead of taking moments. That brings in the three unknown components of the pivot force, which the question doesn't want.

Beyond STEP

Here we took moments about two perpendicular axes through O. This is a shortcut for the single vector equation $\sum \mathbf{r} \times \mathbf{F} = \mathbf{0}$, which says that the total torque about O vanishes. Torque as a vector, and the equilibrium conditions for a rigid body in three dimensions, are standard in first-year mechanics. They lead on to the rotation of rigid bodies about a fixed point. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops torque and angular momentum as vectors in exactly this way.

WORKED SOLUTION

STEP II 1987, Question 12

Mechanics • Equilibrium and Energy

STEP II 1987, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q12

The key idea

"Equilibrium in every position" is a statement about potential energy: if the bead can rest anywhere on the wire, the potential energy of the system can't change as the bead slides along it, so it must be constant on the wire. Writing that energy in terms of the bead's height and its distance from the ring gives the curve at once. The same fact makes the second part neat: with the potential energy constant, conservation of energy says the total kinetic energy of bead and particle is constant, and all that remains is some geometry linking the two speeds.

Solution

We model the bead and the hanging mass as particles, the string as light and inextensible, and the ring and wire as smooth. Put the ring at the origin O, with y measured vertically upwards. Let the bead be at polar coordinates (r, θ), with θ measured anticlockwise from the horizontal, so the bead is at height $y = r\sin\theta$ relative to O. Let L be the length of the string.

The equation of the curve

The string from O to the bead has length r, so the hanging particle is a distance $L - r$ below the ring. Taking O as the zero level, the potential energy of the system is

$$V = mgr\sin\theta - mg(L - r) = mg\,(r + r\sin\theta) - mgL.$$

The wire is smooth, so its reaction on the bead is perpendicular to it and the only forces that could move the system along the wire come from V. The system is in equilibrium at a position exactly when V is stationary there with respect to movement along the wire. Equilibrium at *every* position means $\mathrm{d}V/\mathrm{d}s = 0$ all along the wire (where s is arc length), so V is constant on the wire:

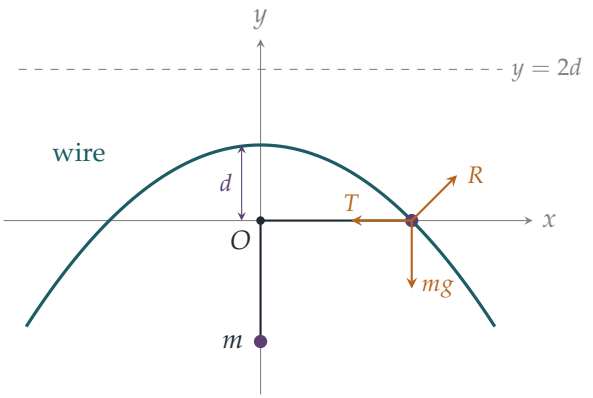
$$r(1 + \sin\theta) = c \quad \text{for some constant } c.$$

You can see this by resolving forces, too. In equilibrium the hanging particle gives tension mg throughout the string (the ring is smooth). On the bead, a tension mg towards O has tangential component $-mg\,\frac{\mathrm{d}r}{\mathrm{d}s}$, and the weight has tangential component $-mg\,\frac{\mathrm{d}y}{\mathrm{d}s}$. These must cancel, so $\frac{\mathrm{d}}{\mathrm{d}s}(r+y)=0$, which is the same condition.

To find c: since $1+\sin\theta \le 2$, we have $r = c/(1+\sin\theta) \ge c/2$, with equality when $\theta = \frac{\pi}{2}$ (the point of the wire directly above the ring). So the shortest distance from the ring to the wire is $c/2$, which gives $c = 2d$, and the curve is

$$\boxed{r(1+\sin\theta) = 2d} \qquad \text{i.e.} \qquad r = \frac{2d}{1+\sin\theta}.$$

(If you measure θ from the downward vertical instead, the same curve reads $r(1-\cos\theta) = 2d$.) Since $r = 2d - y$, every point of the wire is as far from O as it is from the horizontal line $y = 2d$. So the wire is a parabola with its focus at the ring and its directrix a height $2d$ above it; in Cartesian form, $x^2 = 4d(d-y)$, with vertex $(0, d)$.



In the picture the bead is level with the ring; the tension $T = mg$, the weight mg and the normal reaction R balance because the parabola's slope there is exactly -1.

The speed of the bead

Now the bead moves along the wire with speed u. The string stays taut and its length is fixed, so the hanging particle is still a distance $L - r$ below the ring.

> **Ask yourself** How fast is the hanging particle moving?

Its depth below the ring is $L - r$, so its velocity is vertical with speed $|\dot{r}|$. Note that $\dot{r}$ is only the radial part of the bead's velocity, not its whole speed u.

The tension is no longer mg once things move, but it does no net work on the system: the string is inextensible and the ring smooth, so the work it does on the bead is cancelled by the work it does on the hanging particle. The wire's reaction is perpendicular to the bead's motion and does no work. So energy is conserved:

$$\tfrac{1}{2}mu^2 + \tfrac{1}{2}m\dot{r}^2 + V = \text{constant}.$$

But on the wire $V = mg \cdot 2d - mgL$ is constant, by the first part. Hence

$$u^2 + \dot{r}^2 = \text{constant}. \qquad (*)$$

Next we link $\dot{r}$ to u. Differentiating $r = 2d/(1+\sin\theta)$,

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = -\frac{2d\cos\theta}{(1+\sin\theta)^2} = -\frac{r\cos\theta}{1+\sin\theta}, \qquad \text{so} \qquad \dot{r}^2 = r^2\dot{\theta}^2\,\frac{\cos^2\theta}{(1+\sin\theta)^2} = r^2\dot{\theta}^2\,\frac{1-\sin\theta}{1+\sin\theta},$$

using $\cos^2\theta = (1-\sin\theta)(1+\sin\theta)$. From the curve, $1+\sin\theta = 2d/r$ and $1-\sin\theta = 2-2d/r = 2(r-d)/r$, so

$$\frac{1-\sin\theta}{1+\sin\theta} = \frac{r-d}{d}, \qquad \text{giving} \qquad \dot{r}^2 = \frac{r-d}{d}\,r^2\dot{\theta}^2.$$

In polar coordinates the speed satisfies $u^2 = \dot{r}^2 + r^2\dot{\theta}^2$, so

$$u^2 = r^2\dot{\theta}^2\left(\frac{r-d}{d}+1\right) = \frac{r}{d}\,r^2\dot{\theta}^2, \qquad \text{and therefore} \qquad \dot{r}^2 = \frac{r-d}{r}\,u^2.$$

Ask yourself What is $\dot{r}$ when $r = d$?

At $r = d$ the bead is at the vertex, $\theta = \frac{\pi}{2}$, where r takes its minimum value and $\mathrm{d}r/\mathrm{d}\theta = 0$. So $\dot{r} = 0$ there (as the formula $\dot{r}^2 = \frac{r-d}{r}u^2$ also says), and the hanging particle is momentarily at rest. The constant in $(*)$ is therefore v^2:

$$u^2 + \frac{r-d}{r}\,u^2 = v^2 \quad\Longrightarrow\quad u^2\,\frac{2r-d}{r} = v^2 \quad\Longrightarrow\quad u = \left(\frac{r}{2r-d}\right)^{\frac{1}{2}} v,$$

taking the positive root since u and v are speeds. This is the required result.

Checks

- At $r = d$ the formula gives $u = v$, as it must.
- As $r \to \infty$, $u \to v/\sqrt{2}$. Far out, the parabola's tangent points almost straight along the radius, so $\dot{r} \approx u$ and the kinetic energy $\frac{1}{2}mv^2$ is shared equally between bead and particle.
- Equilibrium at the level of the ring: the point $(2d, 0)$ is on the curve, where tension mg (horizontal) and weight mg (vertical) combine to a force at 45°, normal to the parabola's slope of -1 there.
- Numerically, with $d = 1.3$ and $v = 2$, we integrated the equation of motion of the bead-and-particle system along the parabola from the vertex. At $r = 2.8905$ the bead's speed was 1.606309, matching $\sqrt{r/(2r-d)}\,v$ to about 10^{-8}. We also confirmed that the tangential force on a bead at rest vanishes at 1000 random points of the wire.

Common slips

- Forgetting the hanging particle's kinetic energy, which gives $u = v$ for all r. The potential energy is constant, but the kinetic energy is shared.
- Taking the particle's speed to be u instead of $|\dot{r}|$. Only the radial part of the bead's motion pulls string through the ring.
- Measuring θ from a different line and then quoting $r(1 + \sin\theta) = 2d$. Any correct polar form is fine, but say which line θ is measured from.

Beyond STEP

The idea that a system is in equilibrium where its potential energy is stationary is the *principle of virtual work*, and it leads into Lagrangian mechanics. There, kinetic energy is written in terms of one generalised coordinate (here θ), and the equations of motion follow from it without ever finding the tension. Because V is constant on this wire, the bead and particle move as if no forces acted at all, and our relation is just conservation of the kinetic energy $\frac{1}{2}mr^2(2r - d)\dot{\theta}^2/d$. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers energy methods and constrained motion in this spirit.

WORKED SOLUTION

STEP II 1987, Question 13

Mechanics • Oblique Impacts with Walls

STEP II 1987, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q13

The key idea

A smooth cushion can only change the velocity component perpendicular to it. That component is reversed and multiplied by e, and the component along the cushion is unchanged. Two adjacent cushions act on the two different components, so each gets reversed and scaled by e once. For the closed paths, we follow the puck from cushion to cushion with coordinates and choose the one free quantity, the initial slope, so that the puck comes back to where it started.

Solution

Model. The puck is a particle sliding on a smooth horizontal table with smooth cushions, so nothing acts on it between impacts and it moves in straight lines at constant velocity. At an impact the cushion's impulse is perpendicular to the cushion (it is smooth), so the parallel component of velocity is unchanged. By Newton's law of restitution the perpendicular component is reversed and multiplied by e. We take $0 < e \leq 1$.

Put axes along two sides of the table, so that it is $0 \leq x \leq L$, $0 \leq y \leq B$.

Part (i): two adjacent cushions

Let the puck's velocity be (u, v). Adjacent cushions are one cushion of the form $x = \text{const}$ and one of the form $y = \text{const}$, hit in either order. Hitting an x-cushion sends $(u, v) \mapsto (-eu, v)$, and hitting a y-cushion sends $(u, v) \mapsto (u, -ev)$. After one of each, in either order, the velocity is

$$(-eu,\ -ev) = -e\,(u, v).$$

This is a scalar multiple of the original velocity, so the final path is parallel to the original one, as required. The puck is now travelling in the opposite sense, with its speed reduced by a factor e.

Part (ii): a closed path from a cushion

Let the puck rest at $P = (a, 0)$ on the cushion $y = 0$, with $a : (L - a) = z : 1$, so that

$$a = \frac{zL}{1+z}, \qquad L - a = \frac{L}{1+z}, \qquad 0 < a < L.$$

Looking down on the table with x to the right and y upwards, going clockwise from the cushion $y = 0$ we meet $x = 0$, then $y = B$, then $x = L$. If the ratio is measured from the other end, replace z by $1/z$; the argument is unchanged. Hit the puck with velocity $(-u, v)$, where $u, v > 0$, and write $k = v/u$ for the slope. The velocities on the four legs are

$$(-u, v) \xrightarrow{x=0} (eu, v) \xrightarrow{y=B} (eu, -ev) \xrightarrow{x=L} (-e^2u, -ev) \xrightarrow{y=0} \cdots$$

- Leg 1, from $(a, 0)$ to $x = 0$: it takes horizontal distance a and rises ak, so it hits the left cushion at $W_1 = (0,\, ak)$.
- Leg 2, with slope $v/(eu) = k/e$: it rises $B - ak$ and so moves across $e(B - ak)/k$. It meets the top at $W_2 = \big(e(B/k - a),\, B\big)$.
- Leg 3, with slope $-k$: it moves across $L - x_{W_2}$ and falls $k(L - x_{W_2})$, so it meets the right cushion at height $y_{W_3} = B - kL + e(B - ak)$.
- Leg 4, with slope k/e: it falls y_{W_3} and moves back $e\,y_{W_3}/k$, reaching $y = 0$ at

$$x_{\text{end}} = L - \frac{e\,y_{W_3}}{k} = (1+e)L + e^2a - \frac{e(1+e)B}{k}.$$

We want $x_{\text{end}} = a$. That gives $e(1+e)B/k = (1+e)L - (1-e^2)a = (1+e)\big(L - (1-e)a\big)$, so

$$k = \frac{eB}{L - (1-e)a} = \frac{B}{L} \cdot \frac{e(1+z)}{1+ez}.$$

This is a positive number for every $z > 0$, since $L - (1-e)a > L - a > 0$.

> **Ask yourself** Have we finished? We assumed the puck meets the cushions in the order left, top, right, bottom.

No. We still have to check that each impact point really lies on the side we assumed. Write $D = L - (1-e)a = L - a + ea$, so $k = eB/D$. Then

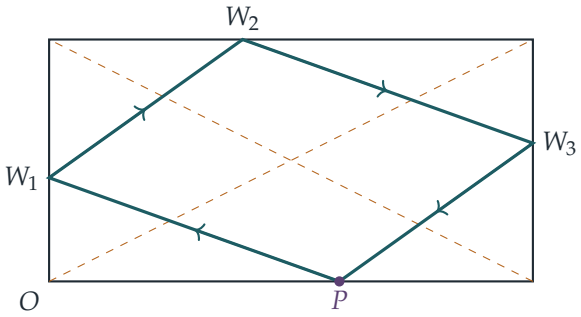
$$y_{W_1} = ak = \frac{eaB}{D},$$

$$x_{W_2} = e\Big(\frac{D}{e} - a\Big) = L - a,$$

$$y_{W_3} = B(1+e) - k(L + ea) = \frac{B\big[(1+e)(L - a + ea) - eL - e^2a\big]}{D} = \frac{B(L-a)}{D}.$$

Since $0 < a < L$ and $e > 0$ we have $0 < ea < D$ and $0 < L - a < D$. So $0 < y_{W_1} < B$, $0 < x_{W_2} < L$ and $0 < y_{W_3} < B$. Each impact is on the open side we assumed, and no corner is hit. So for every z, the shot with slope k above goes left, top, right and comes back to P.

By part (i), legs 1 and 3 are parallel and so are legs 2 and 4, so the closed path is a parallelogram $PW_1W_2W_3$ inscribed in the table.



The dashed lines are the diagonals. The path drawn has $L = 2B$, $e = \frac{1}{2}$ and $z = 1.5$.

Part (iii): two shots from any interior point

Every such shot is one of the part (ii) paths. Suppose the puck starts at an interior point Q with velocity $\mathbf{w}$, bounces at V_1, V_2, V_3, V_4 on the four cushions in clockwise order, and comes back to Q. Apply part (i) twice: after four bounces the velocity is $(-e)^2\mathbf{w} = e^2\mathbf{w}$, so it points the same way as the original. The last leg V_4Q and the first leg QV_1 therefore lie on the same straight line, in the same direction. So $V_4 \to Q \to V_1$ is one straight leg, and the puck's path is a closed path that starts on a cushion at V_4, bounces off the other three clockwise and returns to V_4. The working in part (ii) shows that the slope of such a path is fixed by its starting point. Each such path has one vertex on each side, so all of them are the parallelograms Π_a of part (ii), for $0 < a < L$, and Q lies on a side of Π_a. Conversely, if Q lies on a side of some Π_a, shooting from Q along that side in the direction of travel works. So we must count the pairs (parallelogram, side) through Q.

How the sides sweep the table as z varies. As z runs from 0 to ∞, a runs from 0 to L. From the formulae above, y_{W_1} increases from 0 to B, $x_{W_2} = L - a$ decreases from L to 0, and $y_{W_3} = B/\big(1 + ea/(L-a)\big)$ decreases from B to 0. So Π_a starts as the diagonal through O (at $a \to 0$ the vertices crowd into the corners O and (L, B)) and ends as the other diagonal. Now look at one side, say PW_1. It cuts off the triangle with legs a along the bottom and y_{W_1} up the left side at the corner O. Both legs increase strictly with a, so these triangles are strictly nested and their hypotenuses never cross. They grow continuously from nothing to the half of the table below the diagonal from $(L, 0)$ to $(0, B)$. Precisely, if $Q = (q_1, q_2)$ lies in that half, the function

$$f(a) = \frac{q_1}{a} + \frac{q_2}{y_{W_1}(a)}$$

is continuous and strictly decreasing. It tends to ∞ as $a \to 0$ and to $q_1/L + q_2/B < 1$ as $a \to L$, so $f(a) = 1$ (meaning Q is on PW_1) for exactly one a. If Q is not in that half, $f(a) > 1$ for every a and PW_1 never passes through Q. The same argument applies to each side:

- PW_1 sweeps the half-table below the diagonal from $(L, 0)$ to $(0, B)$ once;
- W_2W_3 sweeps the half-table above that diagonal once;
- W_1W_2 sweeps the half-table above the diagonal from O to (L, B) once;
- W_3P sweeps the half-table below that diagonal once.

A point Q away from the cushions, and not on a diagonal, lies on exactly one side of each diagonal. So exactly two sides, belonging to two different parallelograms, pass through it. Those are the two shots: each one sends the puck along that side, round all four cushions clockwise and back to Q. (If Q is on a diagonal, one of the two shots degenerates into a shot into a corner.)

Checks

- With $e = 1$ the slope is $k = B/L$ for every z, so all the sides are parallel to the diagonals. That is what the usual mirror-image unfolding gives for perfectly elastic cushions.
- Take $L = 2$, $B = 1$, $e = \frac{1}{2}$, $z = 1$: then $a = 1$, $k = \frac{1}{3}$, and the impacts are at $(0, \frac{1}{3})$, $(1, 1)$ and $(2, \frac{2}{3})$. The last leg has $\mathrm{d}x/\mathrm{d}y = e/k = \frac{3}{2}$, so falling $\frac{2}{3}$ moves it back 1, to $(1, 0) = P$.
- In a numerical simulation that steps the puck from wall to wall, all three parts held: random tables, values of e and values of z gave the predicted impact points, and random interior points each lay on exactly two of the closed paths.

Common slips

- Using the mirror-image (unfolding) trick: with $e < 1$ the angle of reflection is not equal to the angle of incidence, so the trick fails.
- Finding k from $x_{\text{end}} = a$ and not checking that each impact lands on the intended side.
- In part (iii), assuming the shot from Q must start along a particular side. The part (i) argument is what joins the first and last legs into one straight leg.

Beyond STEP

Paths like these are *billiards*. With $e = 1$ they are a whole subject of their own, studied with reflections and unfoldings and leading into dynamical systems. Here the cushions lose energy, and the map "velocity before one round $\mapsto$ velocity after" is a pure scaling by e^2. That is why closed orbits exist for every e, even though the puck slows down on each lap. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers impulse and restitution carefully. Siklos, *Advanced Problems in Mathematics*, has more problems in the same spirit.

WORKED SOLUTION

STEP II 1987, Question 14

Mechanics • Elastic Band on Cone

STEP II 1987, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q14

The key idea

Because the band stays a horizontal circle on the cone, its whole state is described by one number: its distance s from the vertex along a generator. So we follow one small element. The inward pull from the tension has a component along the surface of the cone, which drives the element along the generator, and a component into the surface, which the smooth cone cancels. Converting from s to the length x gives simple harmonic motion, and the band goes slack after a quarter of a period.

Solution

The model. We treat the band as a thin elastic string of mass m spread uniformly round a circle. The cone is smooth, so its reaction on each element is normal to the surface, and we ignore gravity. By symmetry every element moves in the same way, so the band stays a circle centred on the axis. Let its distance from the vertex, measured along a generator, be $s(t)$. Its radius is then $s\sin\alpha$, because each generator makes angle α with the axis, and so

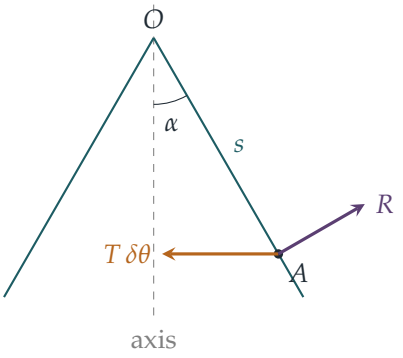
$$x = 2\pi s\sin\alpha.$$

While the band is stretched ($x > l$), Hooke's law gives tension $T = \dfrac{\lambda(x-l)}{l}$.

Forces on one element. Take an element subtending $\delta\theta$ at the axis. It is the fraction $\delta\theta/2\pi$ of the band, so its mass is $\dfrac{m\,\delta\theta}{2\pi}$. Two forces act on it:

- the given force $T\,\delta\theta$ from the tension, perpendicular to the axis and towards it;
- the normal reaction R from the cone, perpendicular to the generator through the element.

The diagram shows the plane containing the axis and the element.



The element is constrained to the surface, so it can move only along the generator OA. We resolve in that direction and perpendicular to it. The force $T\,\delta\theta$ is perpendicular to the axis, and the generator makes angle α with the axis, so the angle between $T\,\delta\theta$ and the direction $\overrightarrow{AO}$ is $90^\circ - \alpha$. Hence:

- along $\overrightarrow{AO}$ (towards the vertex), the component of $T\,\delta\theta$ is $T\,\delta\theta\cos(90^\circ - \alpha) = T\,\delta\theta\sin\alpha$, and R contributes nothing;
- perpendicular to the surface, $R = T\,\delta\theta\cos\alpha$, since there is no motion in that direction.

Since $R \geq 0$, the band really does stay pressed against the cone.

Ask yourself Why can't we just say the inward force $T\,\delta\theta$ makes the radius shrink?

The element can't move straight towards the axis: the cone is in the way. Only the part of $T\,\delta\theta$ along the surface does any accelerating. The rest is cancelled by the reaction. This is where the factor $\sin\alpha$ comes from, and it is the same reason a bead on a smooth slope feels only $mg\sin\theta$ along it.

The equation of motion. Newton's second law for the element along the generator, with s increasing away from the vertex, gives

$$\frac{m\,\delta\theta}{2\pi}\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} = -T\,\delta\theta\sin\alpha \qquad \Longrightarrow \qquad \frac{\mathrm{d}^2 s}{\mathrm{d}t^2} = -\frac{2\pi T\sin\alpha}{m}.$$

The $\delta\theta$ cancels, so every element obeys the same equation, which confirms the band stays circular. Now $x = 2\pi s\sin\alpha$ with α constant, so

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = 2\pi\sin\alpha\,\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} = -\frac{4\pi^2 T\sin^2\alpha}{m} = -\frac{4\pi^2\lambda}{ml}(x-l)\sin^2\alpha,$$

using $T = \lambda(x-l)/l$. This holds while the band is taut, i.e. for $0 < t < t_0$, and rearranges to the required equation. □

Finding t_0. Put $y = x - l$ and $\omega^2 = \dfrac{4\pi^2\lambda\sin^2\alpha}{ml}$, so $\ddot{y} = -\omega^2 y$, which is simple harmonic motion about $y = 0$. Let $x_0 > l$ be the initial length (the band was pushed on,

so it starts stretched). The band is released from rest, so $\dot{y}(0) = 0$, and

$$x - l = (x_0 - l)\cos\omega t.$$

The band becomes slack when x first falls to l, that is, when $\cos\omega t = 0$ for the first time, so $\omega t_0 = \frac{\pi}{2}$. Since $\omega = 2\pi\sin\alpha\sqrt{\lambda/(ml)}$,

$$t_0 = \frac{\pi}{2\omega} = \frac{1}{4\sin\alpha}\sqrt{\frac{ml}{\lambda}}.$$

Notice that t_0 doesn't depend on how far the band was stretched, which is typical of simple harmonic motion: a quarter-period is the same whatever the amplitude.

Checks

- *Units:* λ is a force, so ml/λ has units of $\mathrm{kg\,m/N} = \mathrm{s}^2$, and t_0 is a time.
- *Limiting cases:* as $\alpha \to \frac{\pi}{2}$ the cone flattens into a plane, and the band contracts directly towards the centre with $t_0 = \frac{1}{4}\sqrt{ml/\lambda}$. As $\alpha \to 0$ the cone becomes a cylinder, the tension has no component along the surface, and $t_0 \to \infty$, as it should.
- *Numerical:* with $\alpha = 30°$, $m = 1$, $l = 1$ and $\lambda = 4$ the formula gives $t_0 = \frac{1}{4}$. Stepping the element's equation of motion numerically (and also computing the net tension force on a fine polygonal band directly in three dimensions) reproduces both the factor $\sin\alpha$ and t_0 to many decimal places for random choices of the parameters.

Common slips

- Forgetting the reaction of the cone and writing $\frac{m\,\delta\theta}{2\pi}\ddot{r} = -T\,\delta\theta$ for the radius r. This misses a factor of $\sin^2\alpha$ altogether.
- Using the mass m rather than $\frac{m\,\delta\theta}{2\pi}$ for the element.
- Giving the full period, or half of it, for t_0: the band starts at maximum stretch, so it reaches natural length after a quarter-period.

Beyond STEP

The given force $T\,\delta\theta$ is a small-angle version of a general fact: a string under tension T curving with radius of curvature ρ pushes sideways with force T/ρ per unit length. The same idea, extended to surfaces, explains why the pressure inside a soap bubble exceeds the pressure outside (Laplace's law). Motion constrained to a surface is handled most cleanly with Lagrangian mechanics, where the reaction never needs to appear. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many problems on strings under tension and constrained motion.

WORKED SOLUTION

STEP II 1987, Question 15

Probability and Statistics • Geometric Probability

STEP II 1987, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q15

The key idea

Forget positions and think about *times*. Each vehicle occupies the crossing for an interval of time, and a collision happens exactly when the two intervals overlap. Writing that overlap as two inequalities turns the collision region in the (u_1, u_2)-plane into a wedge between two lines through the origin. With u_1, u_2 uniform on $(0, V)$, the point (u_1, u_2) is uniform on a square, so the probability is just the fraction of the square lying inside the wedge, and the three cases correspond to where the two lines leave the square.

Solution

The collision condition

We model the train and lorry as moving at constant speeds $u_1, u_2 > 0$ along straight roads that cross at a single point (the width of the crossing is neglected, or absorbed into the lengths). A collision occurs if and only if some part of each vehicle is at the crossing at the same instant.

The front of the train reaches the crossing at time d_1/u_1 and its back passes at time $(d_1 + l_1)/u_1$, so the train occupies the crossing for

$$\frac{d_1}{u_1} < t < \frac{d_1 + l_1}{u_1}.$$

Likewise the lorry occupies it for $d_2/u_2 < t < (d_2 + l_2)/u_2$. Two open intervals (a_1, b_1) and (a_2, b_2) overlap exactly when each starts before the other ends: $a_1 < b_2$ and $a_2 < b_1$. So a collision occurs if and only if

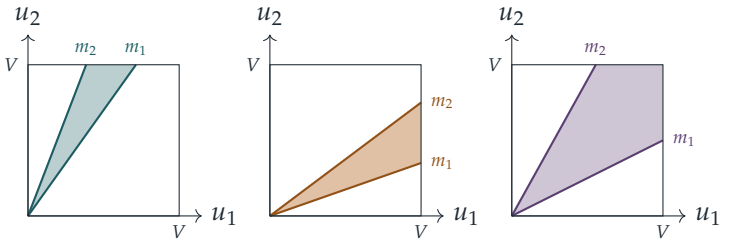
$$\frac{d_1}{u_1} < \frac{d_2 + l_2}{u_2} \quad \text{and} \quad \frac{d_2}{u_2} < \frac{d_1 + l_1}{u_1}.$$

Multiplying through by the positive quantity $u_1 u_2$ and rearranging, this is

$$\boxed{\frac{d_2}{d_1 + l_1}\, u_1 \;<\; u_2 \;<\; \frac{d_2 + l_2}{d_1}\, u_1.}$$

(If an inequality is an equality the vehicles only touch, front to back; this has probability zero, so it doesn't matter whether we count it.)

Write $m_1 = \dfrac{d_2}{d_1 + l_1}$ and $m_2 = \dfrac{d_2 + l_2}{d_1}$. Since $d_2 < d_2 + l_2$ and $d_1 + l_1 > d_1$, we have $m_1 < m_2$, so the collision region is the wedge in the positive quadrant between the lines $y = m_1 x$ and $y = m_2 x$. The pictures below show it, shaded, inside the square $0 < x, y < V$ that we need next.



(A) $d_1 + l_1 \le d_2$ (B) $d_2 + l_2 \le d_1$ (C) otherwise

The probability when the back of the train is nearer than the front of the lorry

Since u_1 and u_2 are independent and each uniform on $(0, V)$, the pair (u_1, u_2) has constant joint density $1/V^2$ on the square $(0, V)^2$. So the probability of a collision is

$$\frac{\text{area of the wedge inside the square}}{V^2}.$$

Ask yourself Which sides of the square do the two lines leave through?

The line $y = mx$ leaves the square through the top side if $m \ge 1$ (at $(V/m, V)$) and through the right-hand side if $m \le 1$ (at (V, mV)). In this case $d_1 + l_1 < d_2$, so $m_1 = d_2/(d_1 + l_1) > 1$, and therefore $m_2 > m_1 > 1$ as well: both lines leave through the top, as in picture (A).

The wedge inside the square is then the triangle with vertices $(0,0)$, $(V/m_2, V)$ and $(V/m_1, V)$. Its base along the top side has length $V/m_1 - V/m_2$ and its height is V, so

$$\begin{aligned}\mathrm{P}(\text{collision}) &= \frac{1}{V^2} \cdot \frac{1}{2} V \left(\frac{V}{m_1} - \frac{V}{m_2} \right) = \frac{1}{2} \left(\frac{d_1 + l_1}{d_2} - \frac{d_1}{d_2 + l_2} \right) \\ &= \frac{(d_1 + l_1)(d_2 + l_2) - d_1 d_2}{2 d_2 (d_2 + l_2)} = \frac{l_1 l_2 + l_2 d_1 + l_1 d_2}{2 d_2 (l_2 + d_2)},\end{aligned}$$

since the $d_1 d_2$ terms cancel. This is the required result.

The other two cases

Back of the lorry nearer than the front of the train, $d_2 + l_2 < d_1$. Now $m_2 = (d_2 + l_2)/d_1 < 1$, so $m_1 < m_2 < 1$ and both lines leave through the right-hand side, as

in picture (B). The wedge is the triangle with vertices $(0,0)$, (V, m_1V), (V, m_2V): base $(m_2 - m_1)V$ on the side $x = V$, height V. So

$$\mathrm{P(collision)} = \frac{1}{2}(m_2 - m_1) = \frac{1}{2}\left(\frac{d_2 + l_2}{d_1} - \frac{d_2}{d_1 + l_1}\right) = \frac{l_1 l_2 + l_1 d_2 + l_2 d_1}{2d_1(l_1 + d_1)}.$$

This is the first answer with the roles of train and lorry swapped ($l_1 \leftrightarrow l_2$, $d_1 \leftrightarrow d_2$), as it must be, since swapping the vehicles amounts to reflecting the picture in $y = x$.

Neither vehicle entirely ahead of the other, $d_2 \le d_1 + l_1$ and $d_1 \le d_2 + l_2$. Now $m_1 \le 1 \le m_2$, so the line $y = m_1 x$ leaves through the right-hand side and $y = m_2 x$ through the top, as in picture (C). It is easiest to subtract the two non-collision triangles from the square: the one below $y = m_1 x$ has area $\frac{1}{2}V \cdot m_1 V$, and the one to the left of $y = m_2 x$ (between it and the u_2-axis) has area $\frac{1}{2}V \cdot V/m_2$. Hence

$$\mathrm{P(collision)} = 1 - \frac{m_1}{2} - \frac{1}{2m_2} = 1 - \frac{d_2}{2(d_1 + l_1)} - \frac{d_1}{2(d_2 + l_2)}.$$

The answers agree where the cases meet: at $d_2 = d_1 + l_1$ (so $m_1 = 1$) the first and third formulae both give $\frac{1}{2} - \frac{d_1}{2(d_2 + l_2)}$, and similarly at $d_1 = d_2 + l_2$. Notice that none of the probabilities depend on V: scaling both speeds by the same factor doesn't change whether the vehicles collide.

Checks

- With $l_1 = 1$, $l_2 = 2$, $d_1 = 1$, $d_2 = 5$ (case A), the formula gives $\frac{9}{70} \approx 0.1286$; a simulation of 4×10^5 random pairs of speeds gave 0.1283. Simulations in cases B and C (for example $l_1 = l_2 = 1$, $d_1 = 1$, $d_2 = 1.5$, giving 0.425) agreed to within 0.002.
- Letting $l_1, l_2 \to 0$ in case A gives probability 0: two points meet at the crossing only if $u_2/u_1 = d_2/d_1$ exactly.
- In case C with $d_1 = d_2$ and very long vehicles, $m_1 \to 0$ and $m_2 \to \infty$, so the probability tends to 1, which is sensible.

Common slips

- Writing only one inequality, such as "the train arrives before the lorry has cleared the crossing"; overlap of two intervals needs *both* conditions.
- Using the formula for a triangle in case C, where the region is a quadrilateral; check where each line leaves the square first.

Beyond STEP

The method here, turning a probability for independent uniform variables into an area, is geometric probability, and it is the simplest instance of computing $\mathrm{P}((X, Y) \in A)$ as a double integral of a joint density. The same idea underlies classic problems such as Buffon's needle and the "meeting problem" of two people arriving at random times. Grimmett and Welsh, *Probability: An Introduction*, develops joint densities and their use in this way.

WORKED SOLUTION

STEP II 1987, Question 16

Probability and Statistics • Bayes' Theorem

STEP II 1987, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q16

The key idea

This is Bayes' theorem in its plainest form. Two reports of "yes" can arise in two ways: the claim is true and both friends are telling the truth, or it is false and both are lying. We weigh each route by its prior probability and the probability of that pair of reports, then ask what fraction of the total comes from the "true" route. The second part works the same way. The only new point is that the friends' answers say something about A alone, so we must sort the three possibilities by whether A is in the fridge.

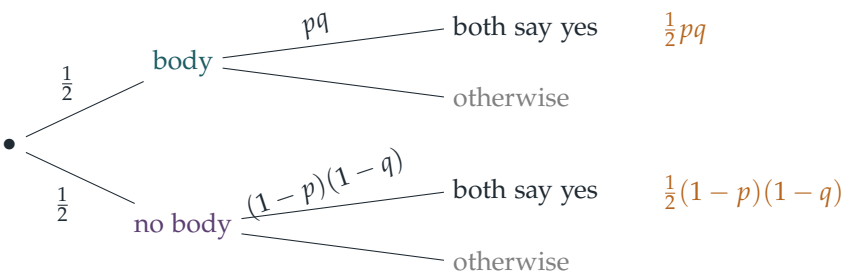
Solution

The first estimate

Let F be the event that there is a body in the fridge, and let Y be the event that both P and Q say there is. This morning $\Pr(F) = \Pr(F') = \frac{1}{2}$.

We model each friend as answering a yes/no question: telling the truth means giving the correct answer and lying means giving the wrong one. The friends' honesty does not depend on what is in the fridge, and P and Q act independently. So:

- if there is a body, both say "yes" exactly when both tell the truth, so $\Pr(Y \mid F) = pq$;
- if there is no body, both say "yes" exactly when both lie, so $\Pr(Y \mid F') = (1-p)(1-q)$.



By the definition of conditional probability (Bayes' theorem),

$$\Pr(F \mid Y) = \frac{\Pr(F)\Pr(Y \mid F)}{\Pr(F)\Pr(Y \mid F) + \Pr(F')\Pr(Y \mid F')} = \frac{\frac{1}{2}pq}{\frac{1}{2}pq + \frac{1}{2}(1-p)(1-q)}$$
$$= \frac{pq}{pq + 1 - p - q + pq} = \frac{pq}{1 - p - q + 2pq},$$

which is the required estimate. The factors of $\frac{1}{2}$ cancel because the prior was even.

Ask yourself Why isn't the answer simply pq, the chance that both are telling the truth?

Because we already know something: they agreed. The outcomes in which exactly one of them lies are now impossible, as those would give one "yes" and one "no". We have to rescale over what is left. Of the two surviving routes, "both truthful" has weight pq and "both lying" has weight $(1-p)(1-q)$. Dividing by their sum is exactly that rescaling. (The $\frac{1}{2}$ priors play a part too. With an uneven prior they would not cancel.)

The second estimate

Now there is certainly a body. The three equally likely possibilities, each with prior probability $\frac{1}{3}$, are

$$A \text{ only}, \qquad B \text{ only}, \qquad \text{both } A \text{ and } B.$$

Let Y_A be the event that both friends say A is in the fridge. Their answers depend only on whether A is there, by the same model as before:

- in the cases "A only" and "both", A is in the fridge, so $\Pr(Y_A) = pq$ in each;
- in the case "B only", A is not there, so $\Pr(Y_A) = (1-p)(1-q)$.

Hence

$$\Pr(\text{both} \mid Y_A) = \frac{\frac{1}{3}pq}{\frac{1}{3}pq + \frac{1}{3}(1-p)(1-q) + \frac{1}{3}pq}$$
$$= \frac{pq}{2pq + 1 - p - q + pq} = \frac{pq}{1 - p - q + 3pq}.$$

Notice that the reports tell us nothing directly about B. Their only effect is to change the chance that A is present. By the first part (with prior $\frac{2}{3}$ in place of $\frac{1}{2}$), that chance becomes $2pq/(1-p-q+3pq)$. Given that A is present, "A only" and "both" remain equally likely, so we halve it. This gives the same answer.

Checks

- If $p = q = \frac{1}{2}$ the friends are useless, and the answers return the priors: $\frac{1}{4}/\frac{1}{2} = \frac{1}{2}$ and $\frac{1}{4}/\frac{3}{4} = \frac{1}{3}$.

- If $p = q = 1$ the friends are infallible, and the answers become 1 (there is a body) and $\frac{1}{2}$ (A is there, so it's a toss-up whether B is too).
- With $p = q = \frac{3}{4}$ the answers are $\frac{9}{10}$ and $\frac{9}{19}$. Exact enumeration over every combination of state and truth-telling agrees for 200 random pairs (p, q). A simulation of 4×10^5 trials with $p = 0.7$, $q = 0.6$ gave 0.4375 for the second part, matching $\frac{0.42}{0.96} = 0.4375$.

Common slips

- Answering pq, and forgetting that "both lying" also produces two "yes" answers.
- In the second part, putting the "both" case alone in the numerator but leaving out the "A only" case from the denominator. Both cases are consistent with A being present.

Beyond STEP

This is Bayesian updating. A prior belief is multiplied by the *likelihood* of the evidence and then normalised. A tidy way to write it is in odds form: posterior odds equal prior odds times the likelihood ratio $\frac{pq}{(1-p)(1-q)}$. That makes it clear that independent witnesses simply multiply their ratios. The same calculation underlies medical testing, spam filters and the combining of evidence in court. Blitzstein and Hwang, *Introduction to Probability*, has an excellent chapter on conditional probability with many puzzles of exactly this flavour.

WORKED SOLUTION

STEP III 1987, Question 1

Pure • Divisibility of Factorials

STEP III 1987, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q1

The key idea

The factorial $(n-1)! = 1 \times 2 \times \cdots \times (n-1)$ contains every whole number smaller than n. If n can be built as a product of two *different* numbers smaller than n, both appear in the factorial and n divides it. So the question is really about which n *can't* be split up like that: primes clearly can't, and squares of primes need a closer look. Trying the first dozen values by hand points straight at the answer.

Solution

Experiment first

A quick table is worth making:

n	1	2	3	4	5	6	7	8
$(n-1)!$	1	1	2	6	24	120	720	5040
$n \mid (n-1)!$?	yes	no	no	no	no	yes	no	yes

The failures are $2, 3, 4, 5, 7$: the primes so far, together with 4. We now prove that this pattern continues.

Claim. n does not divide $(n-1)!$ exactly when n is prime or $n = 4$.

$n = 1$

Here $(n-1)! = 0! = 1$, and 1 divides 1. So $n = 1$ is *not* in the set.

$n = p$ **prime**

Each factor $1, 2, \ldots, p-1$ of $(p-1)!$ is smaller than p and positive, so none is divisible by p. Since p is prime, if p divided the product it would have to divide one of the factors (Euclid's lemma). Hence $p \nmid (p-1)!$, and every prime is in the set.

n composite

Let p be the smallest prime factor of n and write $n = pm$. Because n is composite, $m > 1$, and every prime factor of m is at least p, so $m \geq p$. There are two cases.

Case 1: $m > p$. Then $1 < p < m < n$, so p and m are two *different* numbers in the list $1, 2, \ldots, n-1$. Their product $pm = n$ therefore divides $(n-1)!$.

Case 2: $m = p$, **so** $n = p^2$.

> **?** **Ask yourself** Why doesn't Case 1's argument work here, and what can replace it?

The factor p appears only once in the list $1, \ldots, n-1$, so we can't use it twice. Instead use p and $2p$. If $p \geq 3$ then

$$2p < p \cdot p = n,$$

so p and $2p$ are two different numbers in the list $1, 2, \ldots, n-1$. Hence their product $2p^2 = 2n$ divides $(n-1)!$, and so does n.

If $p = 2$ this fails, because $2p = 4 = n$ is not in the list. Indeed $n = 4$ gives $3! = 6$, and $4 \nmid 6$. So 4 is in the set.

Conclusion

Every composite n other than 4 divides $(n-1)!$, and so does $n = 1$. Therefore the required set is

$$\{\, n : n \text{ is prime} \,\} \cup \{4\} = \{2, 3, 4, 5, 7, 11, 13, 17, \ldots\}.$$

Checks

The table above agrees with the claim for $n \leq 8$. Also try a prime square: $n = 9$ gives $8! = 40320 = 9 \times 4480$, as Case 2 predicts (using the factors 3 and 6). A computer check of every $n \leq 2000$, computing $(n-1)!$ mod n directly, finds no exceptions to the claim.

Common slips

- Arguing "write $n = ab$ with $a, b < n$, so $ab \mid (n-1)!$" without insisting that $a \neq b$. That argument wrongly "proves" that $4 \mid 3!$.
- Forgetting $n = 1$. Since $0! = 1$, it is not in the set.
- Saying a prime p fails to divide $(p-1)!$ "because p isn't one of the factors". The real reason is that p is prime and divides none of the factors; a composite number can divide a product without being one of the factors.

Beyond STEP

There is a sharper fact for primes, called Wilson's theorem: $(p-1)! \equiv -1 \pmod{p}$. Together with this question, it gives a (very slow) test for primality: $n > 1$ is prime exactly when $(n-1)! \equiv -1 \pmod{n}$. Wilson's theorem is proved by pairing each number with its inverse modulo p, which is a first taste of the group structure of the integers modulo p. Davenport's *The Higher Arithmetic* covers this beautifully.

WORKED SOLUTION

STEP III 1987, Question 2

Pure • Trigonometric Reduction Formulae

STEP III 1987, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q2

The key idea

The reduction formula comes from differentiating the "boundary term" $\cos^m x \cos nx$ and then using a compound-angle identity to turn $\sin x \cos nx$ into something involving $\sin(n-1)x$. Once we have it, both parts are about evaluating that boundary term at the limits: at 0 and π it either cancels or survives depending on parity, and at $\frac{\pi}{2}$ the factor $\cos^m x$ kills it.

(The printed formula has $-\cos^m \cos nx$; it should read $-\cos^m x \cos nx$.)

Solution

The reduction formula

Differentiate $\cos^m x \cos nx$ by the product rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\cos^m x \cos nx\right) = -m\cos^{m-1} x \, \sin x \cos nx \; - \; n\cos^m x \sin nx.$$

The first term is not yet in the form we want. The compound-angle formula

$$\sin(n-1)x = \sin nx \cos x - \cos nx \sin x \quad \text{gives} \quad \sin x \cos nx = \cos x \sin nx - \sin(n-1)x.$$

Substituting,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\cos^m x \cos nx\right) &= -m\cos^m x \sin nx + m\cos^{m-1} x \sin(n-1)x - n\cos^m x \sin nx \\ &= -(m+n)\cos^m x \sin nx + m\cos^{m-1} x \sin(n-1)x.\end{aligned}$$

Integrating both sides (this needs $m \geq 1$ and $n \geq 1$ so that $m-1$, $n-1$ are non-negative),

$$\cos^m x \cos nx = -(m+n)I_{m,n} + mI_{m-1,n-1},$$

that is,

$$(m+n)I_{m,n} = -\cos^m x \cos nx + mI_{m-1,n-1},$$

as required (with the usual arbitrary constant absorbed into the indefinite integrals).

Part (i)

Write $J_{m,n} = \int_0^{\pi} \cos^m x \sin nx \, \mathrm{d}x$. Applying the formula between 0 and π, and using $\cos^m \pi \cos n\pi = (-1)^m(-1)^n$, for $m, n \geq 1$:

$$(m+n)J_{m,n} = 1 - (-1)^{m+n} + mJ_{m-1,n-1}.$$

If m and n have the same parity then $m+n$ is even, the boundary term vanishes and

$$J_{m,n} = \frac{m}{m+n} J_{m-1,n-1}.$$

Moreover $m-1$ and $n-1$ again have the same parity, so we may keep going.

Ask yourself Where does this chain of reductions stop, and why is the value there zero?

Each step lowers both indices by 1, so after $\min(m,n)$ steps one index is 0 and the other, having the same parity, is even. So it is enough to show $J_{k,0} = 0$ and $J_{0,k} = 0$ for every even $k \geq 0$.

- $J_{k,0} = \int_0^{\pi} \cos^k x \sin 0 \, \mathrm{d}x = 0$, since the integrand is identically zero.
- For even $k \geq 2$, $J_{0,k} = \int_0^{\pi} \sin kx \, \mathrm{d}x = \left[-\frac{1}{k}\cos kx\right]_0^{\pi} = \frac{1}{k}\left(1-(-1)^k\right) = 0$.

Hence, by induction on $\min(m,n)$, $J_{m,n} = 0$ whenever m, n are both even or both odd.

Or otherwise: substituting $x = \pi - t$ sends $\cos^m x$ to $(-1)^m \cos^m t$ and $\sin nx$ to $-(-1)^n \sin nt$, so $J_{m,n} = (-1)^{m+n+1} J_{m,n}$, which forces $J_{m,n} = 0$ when $m+n$ is even. The integrand is antisymmetric about $x = \frac{\pi}{2}$.

Part (ii)

To bring in cosines, write $\sin^2 x = 1 - \cos^2 x$:

$$\int_0^{\pi/2} \sin^2 x \sin 3x \, \mathrm{d}x = K_{0,3} - K_{2,3}, \qquad K_{m,n} = \int_0^{\pi/2} \cos^m x \sin nx \, \mathrm{d}x.$$

For $m \geq 1$ the boundary term is $-\left[\cos^m x \cos nx\right]_0^{\pi/2} = 0 - (-1) = 1$, because $\cos \frac{\pi}{2} = 0$. So

$$(m+n)K_{m,n} = 1 + mK_{m-1,n-1} \qquad (m, n \geq 1).$$

Starting from $K_{0,1} = \int_0^{\pi/2} \sin x \, \mathrm{d}x = 1$:

$$3K_{1,2} = 1 + K_{0,1} = 2, \qquad K_{1,2} = \tfrac{2}{3},$$
$$5K_{2,3} = 1 + 2K_{1,2} = \tfrac{7}{3}, \qquad K_{2,3} = \tfrac{7}{15}.$$

Also $K_{0,3} = \left[-\frac{1}{3}\cos 3x\right]_0^{\pi/2} = \frac{1}{3}$, since $\cos \frac{3\pi}{2} = 0$. Therefore

$$\int_0^{\pi/2} \sin^2 x \sin 3x \, \mathrm{d}x = \frac{1}{3} - \frac{7}{15} = -\frac{2}{15}.$$

Checks

Part (ii) can be done directly with product-to-sum formulae: $\sin^2 x \sin 3x = \frac{1}{2}\sin 3x - \frac{1}{4}(\sin 5x + \sin x)$, which integrates over $[0, \frac{\pi}{2}]$ to $\frac{1}{6} - \frac{1}{4}\left(\frac{1}{5} + 1\right) = -\frac{2}{15}$. The negative sign is plausible: $\sin 3x < 0$ on $(\frac{\pi}{3}, \frac{\pi}{2})$, exactly where $\sin^2 x$ is largest. Simpson's rule gives -0.133333, and checks the reduction formula and part (i) numerically for many m, n.

Common slips

- Trying to make the substitution $x = \frac{\pi}{2} - t$ in (ii): it turns $\sin^2 x$ into $\cos^2 t$ but $\sin 3x$ into $-\cos 3t$, so the integral is not of the form $I_{m,n}$.
- Forgetting that the reduction formula only applies for $m, n \geq 1$, so the chain in (i) needs its own base cases with an index equal to 0.

Beyond STEP

Reduction formulae like this one are the start of a bigger story: integrals of products of trigonometric functions over a full period are the orthogonality relations behind Fourier series, where symmetry arguments like the one in (i) decide at a glance which coefficients vanish. Boyce and DiPrima, *Elementary Differential Equations*, has a readable introduction to Fourier series and their use in solving differential equations.

WORKED SOLUTION

STEP III 1987, Question 3

Pure • Arguments and Machin's Formula

STEP III 1987, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q3

This question has a full guided treatment in *Beyond STEP*, Section 5.1, *Machin's Formula*: https://beyondstep.keithmcnulty.org/ch05/machin.html

The key idea

Both parts rest on the polar form of a complex number. In (i), the point $(|x|, |y|)$ lies at distance $|z|$ from the origin, so $|x|\cos\alpha + |y|\sin\alpha$ is $|z|$ times a cosine. In (ii), arguments add when complex numbers multiply. So if a product of numbers like $5 - \mathrm{i}$ and $1 + \mathrm{i}$ turns out to be very nearly real, its argument gives an identity between inverse tangents. The only real care needed is the ambiguity of 2π in arguments.

Solution

Part (i)

Write $r = |z| = \sqrt{x^2 + y^2}$. If $r = 0$ then $x = y = 0$ and every term is 0, so suppose $r > 0$.

Right-hand inequality. Both sides are non-negative, so it is enough to compare squares:

$$(|x| + |y|)^2 = x^2 + y^2 + 2|x||y| \geq x^2 + y^2 = |z|^2,$$

since $|x||y| \geq 0$. Taking non-negative square roots gives $|z| \leq |x| + |y|$, with equality exactly when $x = 0$ or $y = 0$.

Left-hand inequality. The point $(|x|, |y|)$ lies in the closed first quadrant at distance r from the origin, so there is an angle $\varphi \in [0, \frac{\pi}{2}]$ with

$$|x| = r\cos\varphi, \qquad |y| = r\sin\varphi.$$

Then, by the compound-angle formula,

$$|x|\cos\alpha + |y|\sin\alpha = r(\cos\varphi\cos\alpha + \sin\varphi\sin\alpha) = r\cos(\alpha - \varphi) \leq r = |z|,$$

because $\cos(\alpha - \varphi) \leq 1$ for every real α. This proves both inequalities. Equality on the left holds when $\alpha = \varphi + 2k\pi$, that is, when $(\cos\alpha, \sin\alpha)$ points in the same direction as $(|x|, |y|)$.

(Another route to the left-hand inequality: by the Cauchy–Schwarz inequality, $|x|\cos\alpha + |y|\sin\alpha \leq \sqrt{x^2+y^2}\,\sqrt{\cos^2\alpha + \sin^2\alpha} = |z|$.)

Part (ii): the first formula

Expand step by step:

$$\begin{aligned}(5-\mathrm{i})^2 &= 25 - 10\mathrm{i} - 1 = 24 - 10\mathrm{i},\\ (5-\mathrm{i})^4 &= (24-10\mathrm{i})^2 = 576 - 480\mathrm{i} - 100 = 476 - 480\mathrm{i},\\ (5-\mathrm{i})^4(1+\mathrm{i}) &= 476 + 476\mathrm{i} - 480\mathrm{i} + 480 = 956 - 4\mathrm{i} = 4(239 - \mathrm{i}).\end{aligned}$$

Now compare arguments. Let $\theta = \tan^{-1}\frac{1}{5}$. Since $5 - \mathrm{i}$ is in the fourth quadrant, it has argument $-\theta$, and $1 + \mathrm{i}$ has argument $\frac{\pi}{4}$. Similarly $4(239 - \mathrm{i})$ has argument $-\tan^{-1}\frac{1}{239}$. Arguments add under multiplication, up to a multiple of 2π, so

$$\frac{\pi}{4} - 4\theta = -\tan^{-1}\frac{1}{239} + 2k\pi \quad \text{for some integer } k.$$

Ask yourself How do we know that $k = 0$?

We bound both sides. Since $\tan t > t$ for $0 < t < \frac{\pi}{2}$, we have $0 < \theta < \frac{1}{5}$, so $\frac{\pi}{4} - 4\theta$ lies between $\frac{\pi}{4} - \frac{4}{5} > -1$ and $\frac{\pi}{4} < 1$. Also $-\tan^{-1}\frac{1}{239}$ lies between -1 and 0. Two numbers in $(-1, 1)$ differ by less than $2 < 2\pi$, so $2k\pi = 0$ and $k = 0$. Rearranging,

$$\frac{\pi}{4} = 4\tan^{-1}\frac{1}{5} - \tan^{-1}\frac{1}{239},$$

as required.

Part (ii): the second formula

By the same reasoning, we want a product with argument

$$\frac{\pi}{4} - 3\tan^{-1}\frac{1}{4} - \tan^{-1}\frac{1}{20} - \tan^{-1}\frac{1}{1985},$$

namely $(4-\mathrm{i})^3(20-\mathrm{i})(1985-\mathrm{i})(1+\mathrm{i})$, and we hope that it is a positive real number. Leave the factor $1985 - \mathrm{i}$ until last:

$$\begin{aligned}(4-\mathrm{i})^2 &= 16 - 8\mathrm{i} - 1 = 15 - 8\mathrm{i},\\ (4-\mathrm{i})^3 &= (15-8\mathrm{i})(4-\mathrm{i}) = 60 - 15\mathrm{i} - 32\mathrm{i} - 8 = 52 - 47\mathrm{i},\\ (4-\mathrm{i})^3(20-\mathrm{i}) &= 1040 - 52\mathrm{i} - 940\mathrm{i} - 47 = 993 - 992\mathrm{i},\\ (4-\mathrm{i})^3(20-\mathrm{i})(1+\mathrm{i}) &= 993 + 993\mathrm{i} - 992\mathrm{i} + 992 = 1985 + \mathrm{i}.\end{aligned}$$

So the product so far is $1985 + \mathrm{i}$, and the last factor $1985 - \mathrm{i}$ is its conjugate: the full product is $1985^2 + 1$, which is real and positive. Equivalently, without the last factor, compare arguments:

$$\frac{\pi}{4} - 3\varphi - \psi = \tan^{-1}\frac{1}{1985} + 2k\pi, \qquad \varphi = \tan^{-1}\frac{1}{4},\ \psi = \tan^{-1}\frac{1}{20}.$$

Again $k = 0$: since $0 < \varphi < \frac{1}{4}$ and $0 < \psi < \frac{1}{20}$, the left-hand side lies between $\frac{\pi}{4} - \frac{4}{5} > -1$ and $\frac{\pi}{4} < 1$, while the right-hand side minus $2k\pi$ lies in $(0, 1)$. Rearranging,

$$\frac{\pi}{4} = 3\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{1}{20} + \tan^{-1}\frac{1}{1985}.$$

Checks

- The modulus is a good check on the expansions. $|5 - \mathrm{i}|^8\,|1 + \mathrm{i}|^2 = 26^4 \cdot 2 = 913\,952$, and $956^2 + 4^2 = 913\,952$ too. Likewise $17^3 \cdot 401 \cdot 2 = 3\,940\,226 = 1985^2 + 1$.
- Numerically, $4\tan^{-1}\frac{1}{5} \approx 0.789582$, and subtracting $\tan^{-1}\frac{1}{239} \approx 0.004184$ gives $0.785398 \approx \frac{\pi}{4}$. Also $3\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{1}{20} \approx 0.784894$, and adding $\tan^{-1}\frac{1}{1985} \approx 0.000504$ gives $\frac{\pi}{4}$ again.
- In (i), try $z = 3 - 4\mathrm{i}$: the bounds read $3\cos\alpha + 4\sin\alpha \le 5 \le 7$, and $3\cos\alpha + 4\sin\alpha = 5\cos(\alpha - \varphi)$ reaches 5 when $\tan\alpha = \frac{4}{3}$.

Common slips

- Writing $\arg(zw) = \arg z + \arg w$ as an exact equation and never checking the multiple of 2π. The bounds on the inverse tangents are what make the final equation true.
- Using $5 + \mathrm{i}$ in place of $5 - \mathrm{i}$: then $(5 + \mathrm{i})^4(1 + \mathrm{i}) = -4 + 956\mathrm{i}$, which is nearly imaginary rather than nearly real. The signs must be chosen so that the product is nearly real.
- In (i), forgetting that α is arbitrary, so $\cos\alpha$ or $\sin\alpha$ may be negative. The cosine form handles every α at once.

Beyond STEP

The first formula is Machin's formula (1706). Combined with the series $\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$, which converges fast for small x, it was used to compute π to many digits by hand. Finding such formulas is a question about factorising in the Gaussian integers $a + b\mathrm{i}$, where $1 + \mathrm{i}$, $5 \pm \mathrm{i}$ and $4 \pm \mathrm{i}$ behave like primes. This is a first taste of algebraic number theory. Davenport's *The Higher Arithmetic* discusses sums of two squares and the Gaussian integers, and Needham's *Visual Complex Analysis* shows the geometry of multiplying complex numbers.

WORKED SOLUTION

STEP III 1987, Question 4

Pure • Volumes of Revolution

STEP III 1987, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

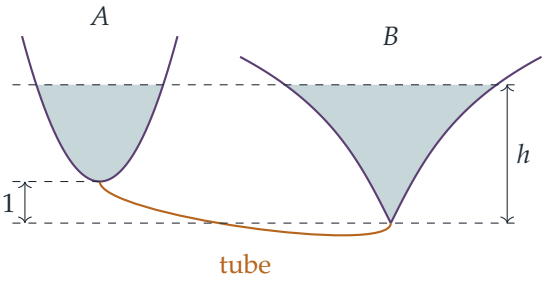
step.maths.org/questions/87-s3-q4

The key idea

The water just moves from one funnel to the other, so its volume is conserved. Each funnel is a solid of revolution about the y-axis, so the volume of water up to a depth d is $\int_0^d \pi x^2 \, \mathrm{d}y$, with x^2 written in terms of y. Once both volumes are known as functions of depth, the only remaining care is with the two different reference levels: the depths in A and B differ by one unit.

Solution

We treat the tube as having negligible volume, and we read the second "closed" in the question as "opened" (otherwise no water could move). Measure y upwards from the bottom of each funnel.



Volumes as functions of depth

For A, the curve is $y = x^2$, so the horizontal cross-section at height y is a disc of area $\pi x^2 = \pi y$. Water of depth d in A has volume

$$V_A(d) = \int_0^d \pi y \, \mathrm{d}y = \frac{\pi d^2}{2}.$$

For B, the curve $y = 2\sinh^{-1} x$ with $x > 0$ inverts to $x = \sinh\frac{y}{2}$. Using $\sinh^2\frac{y}{2} = \frac{1}{2}(\cosh y - 1)$, water of depth d in B has volume

$$V_B(d) = \int_0^d \pi \sinh^2\frac{y}{2}\,\mathrm{d}y = \frac{\pi}{2}\int_0^d (\cosh y - 1)\,\mathrm{d}y = \frac{\pi}{2}(\sinh d - d).$$

Conservation of water

At the start, A holds water of depth 4 and B is empty, so the total volume is

$$V_A(4) = \frac{16\pi}{2} = 8\pi.$$

At the end both surfaces are at height h above the bottom of B. That makes the depth in B equal to h. The bottom of A is one unit higher than the bottom of B, so the depth in A is $h - 1$.

Ask yourself Why do the two surfaces end up at the same level at all?

The two funnels and the tube form one connected body of water at rest. Its pressure depends only on the height, and both free surfaces are at atmospheric pressure, so they are at the same height. The question gives us this, but it explains why the two depths differ by exactly the difference in the heights of the bottoms.

The water was neither gained nor lost, so

$$\frac{\pi(h-1)^2}{2} + \frac{\pi}{2}(\sinh h - h) = 8\pi.$$

Multiplying by $2/\pi$ and expanding,

$$\begin{aligned}(h-1)^2 + \sinh h - h &= 16\\ h^2 - 2h + 1 + \sinh h - h &= 16\\ h^2 - 3h + \sinh h &= 15,\end{aligned}$$

which is the required equation.

The left-hand side of the middle line is a sensible model only for $h \geq 1$, where A actually contains water. On that range its derivative $2(h-1) + \cosh h - 1$ is positive, so the equation has exactly one relevant root. Numerically $h \approx 3.33$, so the water in A drops from depth 4 to about 2.33.

Checks

- Bisection gives $h \approx 3.3272$. Then $V_A(h-1) + V_B(h) \approx 25.1327$, which matches $8\pi \approx 25.1327$.
- Simpson's rule on $\int_0^d \pi\sinh^2(y/2)\,\mathrm{d}y$ agrees with $\frac{\pi}{2}(\sinh d - d)$ for several values of d (for example $d = 1.7$ gives 1.48540 both ways).
- Plausibility: $h - 1 < 4$, so the level in A has fallen, as it must when water flows out of it.

Common slips

- Using depth h in both funnels, which forgets that the bottom of A is one unit higher. That gives $h^2 - h + \sinh h = 16$ instead.
- Integrating $\pi y^2 \, \mathrm{d}x$ (rotation about the x-axis) rather than $\pi x^2 \, \mathrm{d}y$.
- Inverting $y = 2\sinh^{-1} x$ as $x = 2\sinh y$ or $x = \frac{1}{2}\sinh y$. The correct inverse is $x = \sinh \frac{y}{2}$.

Beyond STEP

The final equation mixes a polynomial with a hyperbolic function, so it has no closed-form solution. It has to be solved numerically, and the monotonicity argument above is exactly what guarantees that a method like bisection or Newton–Raphson finds the one physically meaningful root. First-year numerical analysis makes this precise with the intermediate value theorem and convergence rates. For the analysis behind it, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 1987, Question 5

Pure • Modular Arithmetic Ciphers

STEP III 1987, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q5

The key idea

The code is two separate shift ciphers, one for the even positions (key n_0) and one for the odd positions (key n_1), each with an extra shift of j. Strip off the known shift j first. What remains is each half of the message shifted by one unknown constant, and a shift modulo 50 is a bijection, so it keeps equal numbers equal and different numbers different. The three copies of 43 are the fingerprint that gives n_0 away. After that, a sum taken modulo 50 pins down n_1.

Solution

Only n_0 and n_1 modulo 50 matter, so we take $0 \leq n_0, n_1 \leq 49$. Every a_k lies between 0 and 49, so each a_k is fixed once we know its residue modulo 50.

Removing the known shift

For the even positions, set $c_j = b_{2j} - j$ for $j = 1, \ldots, 6$. For the odd positions, set $d_j = b_{2j+1} - j$ for $j = 0, \ldots, 5$. Then

$$a_{2j} \equiv c_j - n_0, \qquad a_{2j+1} \equiv d_j - n_1 \pmod{50}.$$

From the coded message:

j	1	2	3	4	5	6
b_{2j}	27	36	35	40	37	48
c_j	26	34	32	36	32	42

j	0	1	2	3	4	5
b_{2j+1}	35	2	15	8	40	24
d_j	35	1	13	5	36	19

Finding n_0

Subtracting a fixed n modulo 50 is one-to-one. So two positions of the same parity hold equal a's exactly when their c's (or d's) are equal.

Ask yourself Where can the three 43s go?

The six d_j are all different, so the odd positions hold six different numbers, and at most one of them is a 43. So at least two 43s sit in even positions. Among the c_j the only repeat is 32, which occurs twice ($j = 3, 5$) and no more. So the even positions hold exactly two equal numbers, and these must be the two 43s, because 43 is the only number that appears more than once in the message. Hence

$$32 - n_0 \equiv 43 \pmod{50} \quad \Longrightarrow \quad n_0 \equiv -11 \equiv 39 \pmod{50}.$$

Now $a_{2j} \equiv c_j - 39 \equiv c_j + 11$, which gives

$$(a_2, a_4, a_6, a_8, a_{10}, a_{12}) = (37,\ 45,\ 43,\ 47,\ 43,\ 3),$$

since $42 + 11 = 53 \equiv 3$. All six are in the message, as they must be.

Finding n_1

Removing these six numbers from the message leaves the odd-position numbers

$$S = \{1,\ 7,\ 23,\ 24,\ 39,\ 43\},$$

so $\{d_j - n_1\}$ must equal S modulo 50.

Ask yourself How can we get n_1 without trying all fifty values?

Add up both sides. If the two sets agree, their sums agree modulo 50:

$$\sum d_j - 6n_1 \equiv \sum_{s \in S} s \pmod{50}, \qquad \text{that is,} \qquad 109 - 6n_1 \equiv 137.$$

So $6n_1 \equiv -28 \equiv 22 \pmod{50}$. Dividing through by 2, which also divides the modulus, gives $3n_1 \equiv 11 \pmod{25}$. Since $3 \times 17 = 51 \equiv 1$, we get $n_1 \equiv 11 \times 17 = 187 \equiv 12 \pmod{25}$. So $n_1 = 12$ or $n_1 = 37$.

The sum condition is necessary but not sufficient, so we test both values. With $n_1 = 37$, $d_0 - 37 = -2 \equiv 48$, which is not in the message, so $n_1 = 37$ fails. With $n_1 = 12$,

$$(a_1, a_3, a_5, a_7, a_9, a_{11}) \equiv (23,\ -11,\ 1,\ -7,\ 24,\ 7) \equiv (23,\ 39,\ 1,\ 43,\ 24,\ 7),$$

which is exactly S. So $n_1 = 12$.

The message

Interleaving the two halves, the original message is

$$23,\ 37,\ 39,\ 45,\ 1,\ 43,\ 43,\ 47,\ 24,\ 43,\ 7,\ 3,$$

with $n_0 \equiv 39$ and $n_1 \equiv 12 \pmod{50}$. The argument shows this answer is the only one: n_0 was forced, and only one of the two candidates for n_1 works.

Checks

Re-encode the answer. For example, $b_{12} \equiv a_{12} + 39 + 6 = 48$ and $b_7 \equiv a_7 + 12 + 3 = 58 \equiv 8$, both correct. A computer that tries all 2500 pairs (n_0, n_1) finds that exactly one pair, $(39, 12)$, decodes to the given collection of numbers, and that it gives the message above.

Common slips

- Getting the indexing wrong. The odd positions start at b_1 with $j = 0$, but the even positions start at b_2 with $j = 1$.
- Dividing $6n_1 \equiv 22 \pmod{50}$ by 6 as if 6 were invertible. It isn't, since $\gcd(6, 50) = 2$, and the congruence has two solutions modulo 50.
- Finding a value of n_1 that works without showing that no other value does. The question asks you to explain your method, so uniqueness needs an argument.

Beyond STEP

This is a Caesar (shift) cipher, and the attack is a small case of frequency analysis: a shift preserves the pattern of repeated letters. In a first course, the step from $6n \equiv 22$ to two solutions is the theory of linear congruences $ax \equiv b \pmod{m}$, which are solvable exactly when $\gcd(a, m)$ divides b. Affine ciphers $x \mapsto ax + b$ and, later, RSA are built on the same ideas. Davenport's *The Higher Arithmetic* covers congruences beautifully.

WORKED SOLUTION

STEP III 1987, Question 6

Pure • Coupled Differential Equations

STEP III 1987, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q6

The key idea

A pair of coupled first-order equations is best turned into a single second-order equation for one unknown. The first equation lets us write y in terms of x and $\mathrm{d}x/\mathrm{d}t$, and substituting into the second leaves a forced oscillator equation for x. The whole character of the answer then depends on how the natural frequency of that oscillator compares with the frequency of the forcing $2\cos t$.

Solution

Write dots for derivatives with respect to t.

Eliminating y

From the first equation,

$$y = \tfrac{1}{5}\left(\dot{x} + 2x\right), \qquad \text{so} \qquad \dot{y} = \tfrac{1}{5}\left(\ddot{x} + 2\dot{x}\right).$$

Substituting both into the second equation and multiplying by 5,

$$\begin{aligned} \ddot{x} + 2\dot{x} + 5ax - 2\left(\dot{x} + 2x\right) &= 10\cos t, \\ \ddot{x} + (5a - 4)\,x &= 10\cos t. \end{aligned}$$

The $\dot{x}$ terms cancel, so there is no damping: this is a pure forced oscillator with natural frequency $\sqrt{5a - 4}$ (when $5a > 4$).

Translating the initial conditions

We are told $x(0) = 0$ and $\dot{y}(0) = 0$, but the equation for x needs $x(0)$ and $\dot{x}(0)$. Put $t = 0$ into the second original equation:

$$\dot{y}(0) + a\,x(0) - 2y(0) = 2 \quad \Longrightarrow \quad 0 + 0 - 2y(0) = 2 \quad \Longrightarrow \quad y(0) = -1.$$

Then the first equation at $t = 0$ gives $\dot{x}(0) = 5y(0) - 2x(0) = -5$. Note that neither step used the value of a, so

$$x(0) = 0, \qquad \dot{x}(0) = -5 \qquad \text{for every } a.$$

Ask yourself The condition is on $\mathrm{d}y/\mathrm{d}t$, not on y. How do we use it?

The differential equations themselves link the values of x, y and their derivatives at any single instant. At $t = 0$ the second equation turns the information about $\dot{y}(0)$ into information about $y(0)$, and the first equation then turns that into $\dot{x}(0)$, which is exactly what a second-order equation for x needs.

The case $a = 1$

Now the equation is

$$\ddot{x} + x = 10\cos t.$$

The complementary function is $A\cos t + B\sin t$. The forcing $\cos t$ is itself a solution of the homogeneous equation (this is resonance), so a trial $C\cos t$ fails and we try $x_p = t\,(C\cos t + D\sin t)$. Differentiating twice,

$$\ddot{x}_p + x_p = -2C\sin t + 2D\cos t,$$

so $C = 0$ and $D = 5$, giving $x_p = 5t\sin t$. Hence

$$x = A\cos t + B\sin t + 5t\sin t.$$

The condition $x(0) = 0$ gives $A = 0$. Then $\dot{x} = B\cos t + 5\sin t + 5t\cos t$, so $\dot{x}(0) = B = -5$. Therefore

$$\boxed{x = 5(t-1)\sin t}$$

For y, use $y = \frac{1}{5}(\dot{x} + 2x)$ with $\dot{x} = 5\sin t + 5(t-1)\cos t$:

$y = \sin t + (t-1)\cos t + 2(t-1)\sin t$, that is $\boxed{y = (t-1)\cos t + (2t-1)\sin t}$

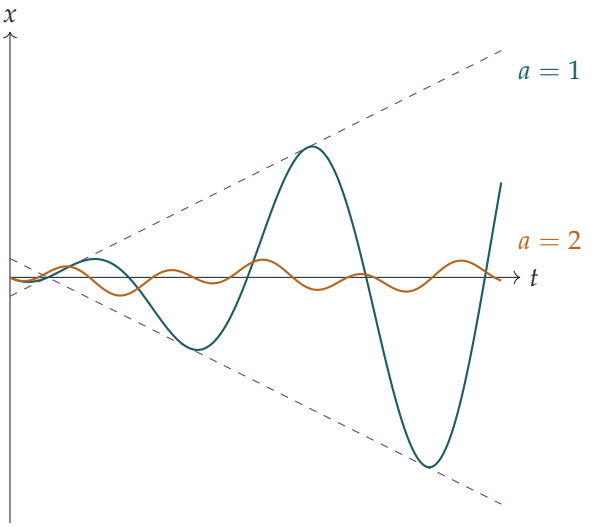
As a quick check, $y(0) = -1$, and $\dot{y} = \cos t - (t-1)\sin t + 2\sin t + (2t-1)\cos t$ gives $\dot{y}(0) = 1 - 1 = 0$, as required.

The case $a > 1$

Now $5a - 4 > 1$, so write $\omega = \sqrt{5a-4}$ with $\omega > 1$. The equation $\ddot{x} + \omega^2 x = 10\cos t$ is no longer at resonance, because the forcing frequency 1 differs from the natural frequency ω. So:

- The complementary function is $A\cos\omega t + B\sin\omega t$, oscillating with period $2\pi/\omega$ (shorter than 2π), rather than with period 2π.
- The particular integral is simply a multiple of $\cos t$, with no factor of t, so there is no term like $t\sin t$.

- Hence x (and likewise $y = \frac{1}{5}(\dot{x} + 2x)$) is a sum of sinusoids of frequencies 1 and ω. Both stay bounded for all t, instead of oscillating with an amplitude that grows linearly in t as when $a = 1$. The motion is periodic only when ω is rational; in general it is a superposition of two oscillations that "beat" against each other.



Checks

- Substituting $x = 5(t-1)\sin t$ and $y = (t-1)\cos t + (2t-1)\sin t$ back into both original equations works identically. A step-by-step numerical solution (fourth-order Runge–Kutta from $x = 0$, $y = -1$) agrees with the formulas to within 10^{-10} on $0 \le t \le 10$; for example $x(10) = 45\sin 10 \approx -24.481$.
- For $a > 1$ the full solution turns out to be $x = \frac{2}{a-1}(\cos t - \cos \omega t) - \frac{5}{\omega}\sin \omega t$. As $a \to 1$ we have $\omega - 1 \approx \frac{5}{2}(a-1)$ and $\cos t - \cos \omega t \approx (\omega - 1)\, t \sin t$, which recovers $5t\sin t - 5\sin t$. The resonant answer is the limit of the non-resonant one.

Common slips

- Using $y(0) = 0$ instead of deducing $y(0) = -1$ from the condition on $\dot{y}(0)$.
- Trying $C\cos t + D\sin t$ as the particular integral when $a = 1$: it is part of the complementary function, so it yields $0 = 10\cos t$.
- Solving separately for y with new arbitrary constants: once x is known, y is fixed by $y = \frac{1}{5}(\dot{x} + 2x)$.

Beyond STEP

A linear system like this is really $\dot{\mathbf{v}} = M\mathbf{v} + \mathbf{f}(t)$ with $\mathbf{v} = (x, y)$ and a constant matrix M; the eigenvalues of M, here $\pm\sqrt{4 - 5a}$, decide whether the free motion grows, decays or oscillates. Resonance happens when the forcing frequency matches an eigenvalue $\pm$i, which is exactly $a = 1$. Boyce and DiPrima, *Elementary Differential Equations*, treats both systems via matrices and forced oscillations with resonance and beats.

WORKED SOLUTION

STEP III 1987, Question 7

Pure • Series for Arctangent

STEP III 1987, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q7

The key idea

The series for $\tan^{-1} t$ comes from integrating $1/(1+x^2)$, and $1/(1+x^2)$ is the sum of a geometric series with ratio $-x^2$. Stopping the geometric series after finitely many terms leaves an *exact* remainder, and integrating that remainder gives the error term. On $[0,1]$ the factor $1/(1+x^2)$ lies between $\frac{1}{2}$ and 1, which traps the error between two easy integrals.

Solution

The identity

For any real x, the finite geometric series with first term 1 and ratio $-x^2$ gives

$$\sum_{r=0}^{n}(-x^2)^r = \frac{1-(-x^2)^{n+1}}{1+x^2},$$

which is valid because the ratio $-x^2$ is never equal to 1. Rearranging,

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots + (-1)^n x^{2n} + (-1)^{n+1}\frac{x^{2n+2}}{1+x^2}.$$

This is an exact identity, not an approximation. Integrate both sides from 0 to t. Since $\int_0^t \frac{\mathrm{d}x}{1+x^2} = \tan^{-1} t$ and $\int_0^t x^{2r}\,\mathrm{d}x = \frac{t^{2r+1}}{2r+1}$,

$$\tan^{-1} t = t - \frac{t^3}{3} + \frac{t^5}{5} - \cdots + \frac{(-1)^n t^{2n+1}}{2n+1} + (-1)^{n+1}\int_0^t \frac{x^{2n+2}}{1+x^2}\,\mathrm{d}x,$$

as required. This holds for every real t.

The bounds

Write $S_n(t) = \sum_{r=0}^{n} \frac{(-1)^r t^{2r+1}}{2r+1}$. By the identity,

$$\left|\tan^{-1} t - S_n(t)\right| = \int_0^t \frac{x^{2n+2}}{1+x^2}\,\mathrm{d}x,$$

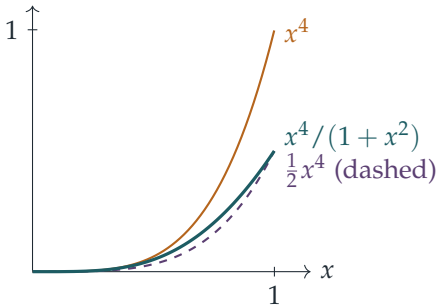
where we may drop the modulus on the right because, for $t \geq 0$, the integrand is non-negative and the integral runs forwards.

Ask yourself How can we estimate an integral we can't evaluate in closed form?

Bound the awkward part of the integrand by constants. For $0 \leq x \leq 1$ we have $1 \leq 1 + x^2 \leq 2$, so

$$\tfrac{1}{2}x^{2n+2} \leq \frac{x^{2n+2}}{1+x^2} \leq x^{2n+2}.$$

The picture shows this for $n = 1$: the integrand (teal) is squeezed between x^4 and $\frac{1}{2}x^4$.



Integrating these inequalities from 0 to t, which preserves them because $t \geq 0$, gives

$$\frac{t^{2n+3}}{2(2n+3)} \leq \left|\tan^{-1} t - S_n(t)\right| \leq \frac{t^{2n+3}}{2n+3} \qquad (0 \leq t \leq 1).$$

That is the required result.

The series for π

Put $t = 1$, where $\tan^{-1} 1 = \frac{\pi}{4}$ and $S_n(1) = \sum_{r=0}^{n} \frac{(-1)^r}{2r+1}$. The upper bound gives

$$\left|\pi - 4\sum_{r=0}^{n} \frac{(-1)^r}{2r+1}\right| \leq \frac{4}{2n+3} \to 0 \quad \text{as } n \to \infty,$$

so $4\sum_{r=0}^{n} \frac{(-1)^r}{2r+1} \to \pi$.

Ask yourself Which of the two bounds tells us the approximation is *bad*?

The lower bound does: it says the error can't be small. Multiplying the left-hand inequality at $t = 1$ by 4,

$$\left|\pi - 4\sum_{r=0}^{n}\frac{(-1)^r}{2r+1}\right| \geq \frac{4}{2(2n+3)} = \frac{2}{2n+3}.$$

If $n \leq 98$ then $2n+3 \leq 199 < 200$, so $\frac{2}{2n+3} > \frac{2}{200} = 10^{-2}$. Hence the error is at least 10^{-2} (in fact strictly more) for every $n \leq 98$. Since the error is below $4/(2n+3)$, a hundred terms buy only about two decimal places, so this series is a very slow way to compute π.

Checks

- $n = 0$, $t = 1$: the bounds say $\frac{1}{6} \leq \left|\frac{\pi}{4} - 1\right| \leq \frac{1}{3}$. Indeed $1 - \frac{\pi}{4} \approx 0.215$.
- Numerically, with $n = 98$ the error is $|\pi - 4S_{98}(1)| \approx 0.01010$, just above 10^{-2}. With $n = 99$ it is about 0.0099998, just below. So 98 is exactly the cut-off. We also checked the identity by quadrature for random t and n, and the two-sided bound for thousands of random $t \in [0, 1]$.

Common slips

- Expanding $1/(1+x^2)$ as an *infinite* series and integrating term by term. That needs $|x| < 1$ and a justification for swapping sum and integral. The finite version with an exact remainder avoids both.
- Using the upper bound to argue that the error is large. Only the lower bound can do that.

Beyond STEP

This is Taylor's theorem with an integral form of the remainder, worked out by hand for one function. The series at $t = 1$ is the Leibniz (or Madhava–Gregory) series. Its error of roughly $1/(2n)$ is typical of alternating series whose terms decrease slowly. Faster formulae, such as Machin's $\frac{\pi}{4} = 4\tan^{-1}\frac{1}{5} - \tan^{-1}\frac{1}{239}$, apply the same series at small t, where t^{2n+3} shrinks rapidly. For remainders and convergence done rigorously, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 1987, Question 8

Pure • Parallelogram Area Inequalities

STEP III 1987, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q8

The key idea

The diagonals of a parallelogram bisect each other, so they cut it into four triangles whose areas we can write down at once using $\frac{1}{2}ab\sin C$. That gives a formula for the area in terms of the diagonals and the angle between them, and both opening results drop out of it. The set A is where two strips cross, so it is a parallelogram. If we work out its area, the second result tells us its longer diagonal must be long, and the two ends of that diagonal are the points we want.

Solution

Area in terms of the diagonals

Let the diagonals have lengths p and q and meet at an angle θ, where $0 < \theta < \pi$. The diagonals of a parallelogram bisect each other, so they split it into four triangles, each with two sides $\frac{1}{2}p$ and $\frac{1}{2}q$ meeting at the centre. Two of these triangles have angle θ there and the other two have angle $\pi - \theta$. Since $\sin(\pi - \theta) = \sin\theta$, each triangle has area $\frac{1}{2} \cdot \frac{p}{2} \cdot \frac{q}{2}\sin\theta$, so

$$\text{area} = 4 \cdot \tfrac{1}{8}pq\sin\theta = \tfrac{1}{2}pq\sin\theta.$$

If p and q are fixed, this is largest when $\sin\theta$ is largest, which is when $\theta = \frac{\pi}{2}$. So the area is maximal exactly when the diagonals are perpendicular, and the maximum area is $\frac{1}{2}pq$.

The bound by the longer diagonal

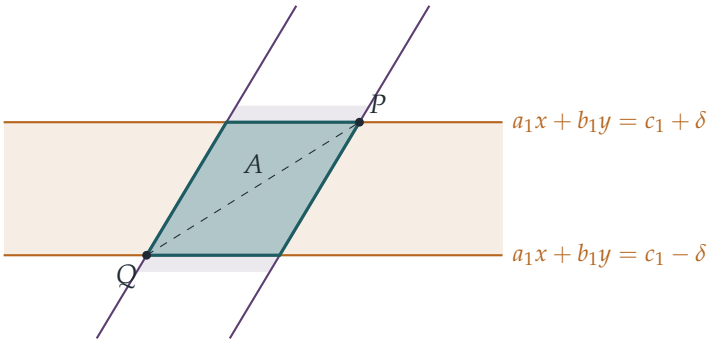
Suppose $p \geq q$, so that p is the longer diagonal (or one of two equal ones). Using $\sin\theta \leq 1$ and then $q \leq p$,

$$\text{area} = \tfrac{1}{2}pq\sin\theta \leq \tfrac{1}{2}pq \leq \tfrac{1}{2}p^2.$$

So the area is at most half the square of the longer diagonal. Equality needs $\sin\theta = 1$ and $q = p$, so it holds only for a square.

The set A is a parallelogram

Assume $\delta > 0$. (If $\delta = 0$ the bound is 0, and any single point of A will do twice.) The inequality $|a_1x + b_1y - c_1| \leq \delta$ says that $c_1 - \delta \leq a_1x + b_1y \leq c_1 + \delta$. So it describes the closed strip between the two parallel lines $a_1x + b_1y = c_1 \pm \delta$. The second inequality gives a second strip in the same way. The condition $a_1b_2 \neq a_2b_1$ says that the normal vectors $\mathbf{n}_1 = (a_1, b_1)$ and $\mathbf{n}_2 = (a_2, b_2)$ are not parallel. So the two strips cross, and neither $\mathbf{n}_1$ nor $\mathbf{n}_2$ is zero. Their intersection A is a parallelogram, bounded by two pairs of parallel lines.



The figure shows the first strip (amber) and part of the second strip (plum). Their intersection A (teal) is a parallelogram, and the dashed line is its longer diagonal.

The area of A

The distance between the parallel lines $a_1x + b_1y = c_1 \pm \delta$ is

$$h_1 = \frac{2\delta}{\sqrt{a_1^2 + b_1^2}},$$

using the formula for the distance from a point to a line. Similarly, the second strip has width $h_2 = 2\delta/\sqrt{a_2^2 + b_2^2}$. Let φ be the angle between the two families of lines, with $0 < \varphi < \pi$. The sides of A that lie along the second strip's edges cross the first strip at angle φ, so each has length $h_1 / \sin\varphi$. The perpendicular distance between these two sides is h_2, so

$$\text{area of } A = \frac{h_1 h_2}{\sin\varphi}.$$

Ask yourself How do we find $\sin\varphi$ from the coefficients?

The angle between two lines equals the angle between their normals, or its supplement, and both have the same sine. For the vectors $\mathbf{n}_1 = (a_1, b_1)$ and $\mathbf{n}_2 = (a_2, b_2)$, the area of the parallelogram they span is $|\mathbf{n}_1||\mathbf{n}_2| \sin\varphi$. That area is also $|a_1b_2 - a_2b_1|$, the size of

the 2×2 determinant. Hence

$$\sin\varphi = \frac{|a_1b_2 - a_2b_1|}{\sqrt{a_1^2+b_1^2}\sqrt{a_2^2+b_2^2}}.$$

Substituting, the square roots cancel:

$$\text{area of } A = \frac{2\delta}{\sqrt{a_1^2+b_1^2}} \cdot \frac{2\delta}{\sqrt{a_2^2+b_2^2}} \cdot \frac{\sqrt{a_1^2+b_1^2}\sqrt{a_2^2+b_2^2}}{|a_1b_2 - a_2b_1|} = \frac{4\delta^2}{|a_1b_2 - a_2b_1|}.$$

The two points

Let PQ be the longer diagonal of A, and write $P = (x_1, y_1)$ and $Q = (x_2, y_2)$. These are vertices of A, so they lie in A, because the strips include their boundary lines. By the bound from the first part,

$$\frac{4\delta^2}{|a_1b_2 - a_2b_1|} = \text{area of } A \leq \tfrac{1}{2}PQ^2,$$

so

$$(x_1 - x_2)^2 + (y_1 - y_2)^2 = PQ^2 \geq \frac{8\delta^2}{|a_1b_2 - a_2b_1|},$$

as required.

Checks

- Take $a_1 = b_2 = 1$ and $a_2 = b_1 = 0$, so that A is the square $|x - c_1| \leq \delta$, $|y - c_2| \leq \delta$. Its diagonal squared is $(2\delta)^2 + (2\delta)^2 = 8\delta^2$, which is exactly the bound. That fits, because a square is the equality case of the first part.
- Scaling: if we double a_1, b_1 and c_1, the first strip halves in width, the area of A halves, and $|a_1b_2 - a_2b_1|$ doubles. The formula behaves correctly.
- By computer we tested 10^5 random cases. We found the vertices of A by solving the boundary equations, and the shoelace area matched $4\delta^2/|a_1b_2 - a_2b_1|$ every time. The smallest ratio of (longer diagonal)^2 to the bound was 1.0007, which is never below 1.

Common slips

- Taking the strip widths to be 2δ. The width of $|a_1x + b_1y - c_1| \leq \delta$ is $2\delta/\sqrt{a_1^2+b_1^2}$.
- Forgetting to say why the chosen points lie in A. They do because they are vertices, and A includes its boundary.
- Writing the area of A as h_1h_2, which is only right when the strips are perpendicular.

Beyond STEP

The area calculation is a special case of a fundamental fact: a linear map with matrix M multiplies all areas by $|\det M|$. Here the map $(x, y) \mapsto (a_1x + b_1y,\ a_2x + b_2y)$ sends A to a square of area $4\delta^2$, so A has area $4\delta^2/|\det M|$. This is the Jacobian factor you meet when changing variables in double integrals. Strang's *Introduction to Linear Algebra* develops determinants as areas and volumes in this way.

WORKED SOLUTION

STEP III 1987, Question 9

Pure • Direct Products of Groups

STEP III 1987, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q9

The key idea

Everything in $G \times H$ happens one coordinate at a time, so each group axiom and each property of $G \times H$ comes down to the same property in G and in H separately. For the true/false parts, a "true" needs a general proof, while a "false" needs just one solid obstruction. Isomorphic groups agree in everything that can be described using only the multiplication, such as their size and the orders of their elements. So we compare those.

Solution

Throughout, e_G and e_H are the identities of G and H, and we write g^{-1} and h^{-1} for inverses in G and H.

$G \times H$ is a group

We check the four axioms.

Closure. If $g_1, g_2 \in G$ and $h_1, h_2 \in H$, then $g_1 * g_2 \in G$ and $h_1 \circ h_2 \in H$, because G and H are closed. So $(g_1, h_1)(g_2, h_2) = (g_1 * g_2,\ h_1 \circ h_2) \in G \times H$.

Associativity. For any three elements,

$$\begin{aligned}\big((g_1,h_1)(g_2,h_2)\big)(g_3,h_3) &= \big((g_1 * g_2) * g_3,\ (h_1 \circ h_2) \circ h_3\big)\\ &= \big(g_1 * (g_2 * g_3),\ h_1 \circ (h_2 \circ h_3)\big)\\ &= (g_1,h_1)\big((g_2,h_2)(g_3,h_3)\big),\end{aligned}$$

where the middle step uses associativity in G and in H.

Identity. The pair (e_G, e_H) satisfies

$$(e_G, e_H)(g,h) = (e_G * g,\ e_H \circ h) = (g,h),$$

and similarly $(g,h)(e_G, e_H) = (g,h)$. So it is an identity.

Inverses. Given (g,h), the pair (g^{-1},h^{-1}) lies in $G\times H$, and

$$(g,h)(g^{-1},h^{-1}) = (g*g^{-1},\ h\circ h^{-1}) = (e_G,e_H),$$

and the product the other way round is also (e_G,e_H). So every element has an inverse. All four axioms hold, so $G\times H$ is a group under this multiplication.

Part (i): true

If G and H are abelian, then so is $G\times H$. For any two elements,

$$(g_1,h_1)(g_2,h_2) = (g_1*g_2,\ h_1\circ h_2) = (g_2*g_1,\ h_2\circ h_1) = (g_2,h_2)(g_1,h_1),$$

using commutativity in G and in H.

If $G\times H$ is abelian, then so are G and H. Take any $g_1,g_2\in G$. Since $G\times H$ is abelian,

$$(g_1*g_2,\ e_H) = (g_1,e_H)(g_2,e_H) = (g_2,e_H)(g_1,e_H) = (g_2*g_1,\ e_H).$$

Ordered pairs are equal only when their first coordinates are equal, so $g_1*g_2 = g_2*g_1$, and G is abelian. Using pairs (e_G,h_1) and (e_G,h_2) in the same way shows that H is abelian.

> **Ask yourself** Why do we need the identity e_H in the second direction?

We need some second coordinate to make elements of $G\times H$ from g_1 and g_2. Any fixed $h\in H$ would do, since only the first coordinates matter in the comparison. A group always contains an identity, so e_H is a safe choice that we know exists.

Part (ii): true

Let $\widetilde{G} = \{(g,e_H) : g\in G\}$, and define $\phi\colon G\to\widetilde{G}$ by $\phi(g) = (g,e_H)$.

$\widetilde{G}$ is a subgroup of $G\times H$. It contains the identity (e_G,e_H). It is closed under products, since $(g_1,e_H)(g_2,e_H) = (g_1*g_2,e_H)$. It is closed under inverses, since $(g,e_H)^{-1} = (g^{-1},e_H)$.

ϕ is an isomorphism. It is onto $\widetilde{G}$ by definition. It is one-to-one, because $(g_1,e_H) = (g_2,e_H)$ forces $g_1 = g_2$. It preserves the operation, because

$$\phi(g_1*g_2) = (g_1*g_2,\ e_H) = (g_1,e_H)(g_2,e_H) = \phi(g_1)\phi(g_2).$$

So $G\times H$ contains the subgroup $\widetilde{G}$, which is isomorphic to G.

Part (iii): false

Write both groups additively: $\mathbb{Z}_n = \{0,1,\ldots,n-1\}$ under addition mod n. In $\mathbb{Z}_2\times\mathbb{Z}_2$, every element $x=(a,b)$ satisfies

$$x+x = (2a,2b) = (0,0),$$

so every element has order 1 or 2. In $\mathbb{Z}_4$, by contrast, $1+1=2\neq 0$, so 1 has order 4.

Suppose there were an isomorphism $\phi\colon \mathbb{Z}_4 \to \mathbb{Z}_2 \times \mathbb{Z}_2$. Then

$$\phi(2) = \phi(1+1) = \phi(1) + \phi(1) = (0,0) = \phi(0),$$

because $\phi(1)+\phi(1)=(0,0)$ by the calculation above. The last equality holds because an isomorphism sends the identity to the identity. This gives $\phi(2)=\phi(0)$ with $2\neq 0$, so ϕ is not one-to-one, which is a contradiction. Hence $\mathbb{Z}_2 \times \mathbb{Z}_2$ is *not* isomorphic to $\mathbb{Z}_4$. (In short, $\mathbb{Z}_4$ is cyclic and $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not.)

Part (iv): false

An isomorphism is a bijection, so isomorphic groups have the same number of elements. Now $|S_n| = n!$, because there are $n!$ arrangements of n objects, and $|G \times H| = |G|\,|H|$, since each pair is a free choice of g and of h. So

$$|S_2 \times S_3| = 2! \times 3! = 12, \qquad |S_6| = 6! = 720.$$

These sizes differ, so $S_2 \times S_3$ is *not* isomorphic to S_6.

Checks

- By brute force on small examples (such as $\mathbb{Z}_2 \times \mathbb{Z}_3$, $\mathbb{Z}_2 \times S_3$, $S_3 \times S_3$ and $S_2 \times S_3$), the product satisfies all four axioms with identity (e_G, e_H). It is abelian exactly when both factors are, and $g \mapsto (g, e_H)$ is an injective homomorphism.
- The element orders in $\mathbb{Z}_2 \times \mathbb{Z}_2$ are $1,2,2,2$, and in $\mathbb{Z}_4$ they are $1,2,4,4$, so the two groups really are different.
- Listing $S_2 \times S_3$ gives 12 elements, against 720 in S_6.

Common slips

- Proving only one direction of the "if and only if" in (i). The converse is the part that needs the trick of putting e_H in the second coordinate.
- In (ii), saying "G is a subgroup of $G \times H$". Strictly, G is not even a subset of $G \times H$. The subgroup is $G \times \{e_H\}$, which is *isomorphic* to G.
- In (iii), noting that both groups have order 4 and stopping there. Equal size is necessary for an isomorphism but not sufficient, so we need a property that tells the groups apart.

Beyond STEP

This is the *direct product* construction, the simplest way of building new groups from old. A major result of a first course is that every finite abelian group is a direct product

of cyclic groups. A related fact is that $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$ exactly when $\gcd(m, n) = 1$, which is the group-theoretic face of the Chinese remainder theorem. Part (iii) is the smallest case where this fails. For more, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP III 1987, Question 10

Pure • Bernoulli Polynomials

STEP III 1987, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q10

The key idea

The defining rule fixes P_n only through its derivative, so the natural thing to induct on is the *derivative* of the difference $P_n(x+1) - P_n(x) - nx^{n-1}$: the inductive hypothesis kills it, leaving a constant, and the integral condition pins that constant down. After that, the sum is a telescoping sum, and for the cubes we only need P_4 up to its constant term, which cancels anyway.

Solution

The difference identity

For $n \geq 1$ let

$$Q_n(x) = P_n(x+1) - P_n(x) - nx^{n-1}.$$

We prove by induction that $Q_n(x) = 0$ for all x.

Base case $n = 1$. Since $P_1' = 1 \cdot P_0 = 1$, we have $P_1(x) = x + c$ for some constant c. Then $P_1(x+1) - P_1(x) = 1 = 1 \cdot x^0$, so $Q_1 \equiv 0$. (In fact $\int_0^1 (x+c)\,\mathrm{d}x = 0$ gives $c = -\frac{1}{2}$, but we don't need it.)

Inductive step. Suppose $n \geq 2$ and $Q_{n-1} \equiv 0$. Differentiating, and using $P_n' = nP_{n-1}$,

$$\begin{aligned} Q_n'(x) &= nP_{n-1}(x+1) - nP_{n-1}(x) - n(n-1)x^{n-2} \\ &= n\left[P_{n-1}(x+1) - P_{n-1}(x) - (n-1)x^{n-2}\right] \\ &= nQ_{n-1}(x) = 0. \end{aligned}$$

So Q_n is a constant, and it equals its value at $x = 0$. Because $n \geq 2$, the term nx^{n-1} vanishes at $x = 0$, so

$$Q_n(0) = P_n(1) - P_n(0).$$

Ask yourself How do we know $P_n(1) = P_n(0)$? The integral condition is about P_n, not its values.

Use the integral condition one level down. By the fundamental theorem of calculus,

$$P_n(1) - P_n(0) = \int_0^1 P_n'(x)\,\mathrm{d}x = n\int_0^1 P_{n-1}(x)\,\mathrm{d}x = 0,$$

since $n-1 \geq 1$, so the condition $\int_0^1 P_{n-1}\,\mathrm{d}x = 0$ applies. (This is exactly why the case $n=1$ had to be done separately: $\int_0^1 P_0\,\mathrm{d}x = 1$.) Hence $Q_n \equiv 0$, which completes the induction:

$$P_n(x+1) - P_n(x) = nx^{n-1} \qquad \text{for all } n \geq 1.$$

The sum of powers

Put $x = m$ in the identity and add for $m = 0, 1, \ldots, k$. The right-hand side telescopes:

$$\begin{aligned} n\sum_{m=0}^{k} m^{n-1} &= \sum_{m=0}^{k}\big[P_n(m+1) - P_n(m)\big] \\ &= \big[P_n(1) - P_n(0)\big] + \big[P_n(2) - P_n(1)\big] + \cdots + \big[P_n(k+1) - P_n(k)\big] \\ &= P_n(k+1) - P_n(0), \end{aligned}$$

as required. (For $n = 1$ the $m = 0$ term is 0^0, which must be read as 1; both sides are then $k+1$.)

The sum of cubes

Take $n = 4$, so we need P_4. Working up from $P_0 = 1$, integrating each time and choosing the constant so that the integral over $[0,1]$ is zero:

$$\begin{aligned} P_1(x) &= x - \tfrac{1}{2}, \\ P_2(x) &= x^2 - x + \tfrac{1}{6} && \text{since } \textstyle\int_0^1 (x^2 - x)\,\mathrm{d}x = -\tfrac{1}{6}, \\ P_3(x) &= x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x && \text{since } \textstyle\int_0^1 (x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x)\,\mathrm{d}x = \tfrac{1}{4} - \tfrac{1}{2} + \tfrac{1}{4} = 0, \\ P_4(x) &= x^4 - 2x^3 + x^2 + c \end{aligned}$$

for some constant c. We don't need c: it cancels in $P_4(k+1) - P_4(0)$. Writing $y = k+1$,

$$P_4(y) - P_4(0) = y^4 - 2y^3 + y^2 = y^2(y-1)^2 = (k+1)^2k^2.$$

So the sum formula with $n = 4$ gives

$$\sum_{m=0}^{k} m^3 = \frac{k^2(k+1)^2}{4} = \left(\frac{k(k+1)}{2}\right)^2.$$

With $k = 1000$, $\frac{1}{2}k(k+1) = \frac{1}{2}\cdot 1000 \cdot 1001 = 500500$, so

$$\sum_{m=0}^{1000} m^3 = (500500)^2,$$

which is the required result.

Checks

- Small k: $k=2$ gives $0+1+8=9=\left(\frac{2\cdot3}{2}\right)^2$, and $k=3$ gives $36=6^2$.
- The same method with $n=2$ gives $2\sum m = P_2(k+1)-P_2(0)=(k+1)^2-(k+1)=k(k+1)$, the familiar triangular-number formula. Notice that the cube sum is the square of this, which is a nice thing to spot.
- By computer, building $P_1,\dots,P_{12}$ with exact fractions confirms the difference identity at random rational points and the sum formula for $k\le 39$. Summing the cubes directly gives $250\,500\,250\,000 = 500500^2$. (The constant we skipped turns out to be $c=-\frac{1}{30}$.)

Common slips

- Trying to fix the constant in Q_n by integrating Q_n over $[0,1]$. That brings in $\int_1^2 P_n$, which we know nothing about. Evaluating at $x=0$ and using the condition on P_{n-1} is much cleaner.
- Forgetting that the step $P_n(1)=P_n(0)$ needs $n\ge 2$, so the base case must be $n=1$ and checked by hand.
- Spending time finding the constant term of P_4, which cancels.

Beyond STEP

The Bernoulli polynomials have the generating function $\dfrac{te^{xt}}{e^t-1}=\sum_{n\ge0}P_n(x)\dfrac{t^n}{n!}$, and their values at 0 are the Bernoulli numbers $1,-\frac{1}{2},\frac{1}{6},0,-\frac{1}{30},\dots$. They are central to the Euler–Maclaurin formula, which compares a sum with an integral and its error terms, and they appear in Euler's evaluation of $\sum 1/n^{2k}$. For more on these ideas, try Graham, Knuth and Patashnik, *Concrete Mathematics*, which treats sums of powers and Bernoulli numbers in depth.

WORKED SOLUTION

STEP III 1987, Question 11

Mechanics • Relative Velocity Optimisation

STEP III 1987, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

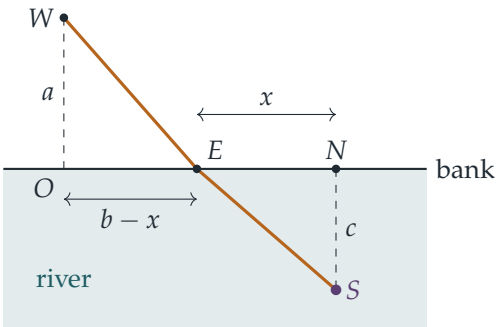
step.maths.org/questions/87-s3-q11

The key idea

Write the total time as a function of the single variable x and set its derivative to zero. In still water this is just the refraction problem from optics, and the answer is Snell's law. With a current the only new work is the swimming time: the current and her swimming velocity have the same size, so their sum is the diagonal of a rhombus, which makes the swimming time a surprisingly simple function of x.

Solution

Model the woman as a particle moving at constant speed u on land and at constant speed v relative to the water, each along a straight line. Take the bank as the X-axis with downstream as the positive direction, the field as $Y > 0$, and put the woman at $W = (0, a)$. The stump is then $S = (b, -c)$, the nearest point of the bank to it is $N = (b, 0)$, and she enters the river at $E = (b - x, 0)$, which is x upstream of N.



Her running distance is $WE = \sqrt{a^2 + (b-x)^2}$ and her swimming distance is $ES = \sqrt{x^2 + c^2}$.

Still water

The total time is

$$T(x) = \frac{\sqrt{a^2 + (b-x)^2}}{u} + \frac{\sqrt{x^2 + c^2}}{v},$$

so

$$T'(x) = -\frac{b-x}{u\sqrt{a^2 + (b-x)^2}} + \frac{x}{v\sqrt{x^2 + c^2}}.$$

The function $s \mapsto s/\sqrt{k^2 + s^2}$ is increasing, so both terms of T' increase with x. Hence T' is strictly increasing and T has exactly one stationary point, which is its minimum. Since $T'(0) < 0 < T'(b)$, this point lies in $0 < x < b$. The required equation is

$$\boxed{\frac{b-x}{u\sqrt{a^2 + (b-x)^2}} = \frac{x}{v\sqrt{x^2 + c^2}}}$$

or equivalently $v(b-x)\sqrt{x^2 + c^2} = ux\sqrt{a^2 + (b-x)^2}$. In terms of the angles α and β that WE and ES make with the normal to the bank, this reads $\dfrac{\sin\alpha}{u} = \dfrac{\sin\beta}{v}$, which is Snell's law of refraction.

Flowing river

Now the water moves downstream at speed v, and she swims at speed v relative to the water. To travel along the straight line ES she must choose a heading, at angle θ below the downstream direction, such that her velocity relative to the ground points along ES.

> **Ask yourself** How fast does she actually move along ES, and why must she enter upstream of the stump?

Her ground velocity is the current plus her swimming velocity:

$$v\begin{pmatrix}1\\0\end{pmatrix} + v\begin{pmatrix}\cos\theta\\-\sin\theta\end{pmatrix} = 2v\cos\tfrac{\theta}{2}\begin{pmatrix}\cos\frac{\theta}{2}\\-\sin\frac{\theta}{2}\end{pmatrix},$$

using $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$. Geometrically, the two velocities have equal length, so they form a rhombus and their sum bisects the angle between them. Her ground path therefore makes angle $\phi = \theta/2$ below the downstream direction. Because $0 \le \theta \le \pi$, we have $0 \le \phi \le \frac{\pi}{2}$, so she can only drift downstream: the stump must be downstream of E, which means $x > 0$. For the path to run along ES we need

$$\tan\phi = \frac{c}{x}, \qquad \text{so} \qquad \cos\phi = \frac{x}{\sqrt{x^2 + c^2}},$$

and her speed along ES is $2v\cos\phi = \dfrac{2vx}{\sqrt{x^2 + c^2}}$. The swimming time is distance over speed:

$$\frac{\sqrt{x^2 + c^2}}{2vx/\sqrt{x^2 + c^2}} = \frac{x^2 + c^2}{2vx}.$$

So the total time is

$$T(x) = \frac{\sqrt{a^2+(b-x)^2}}{u} + \frac{x^2+c^2}{2vx} = \frac{\sqrt{a^2+(b-x)^2}}{u} + \frac{1}{2v}\left(x + \frac{c^2}{x}\right), \qquad x > 0,$$

and

$$T'(x) = -\frac{b-x}{u\sqrt{a^2+(b-x)^2}} + \frac{1}{2v}\left(1 - \frac{c^2}{x^2}\right).$$

Both terms again increase strictly with x for $x > 0$. Also $T'(x) \to -\infty$ as $x \to 0^+$, and $T'(x) > 0$ for large x, so there is exactly one stationary point and it gives the minimum time. Setting $T'(x) = 0$ and multiplying by $2vx^2u\sqrt{a^2+(b-x)^2}$ (which is positive) gives

$$2vx^2(b-x) = u(x^2-c^2)\left[a^2+(b-x)^2\right]^{\frac{1}{2}},$$

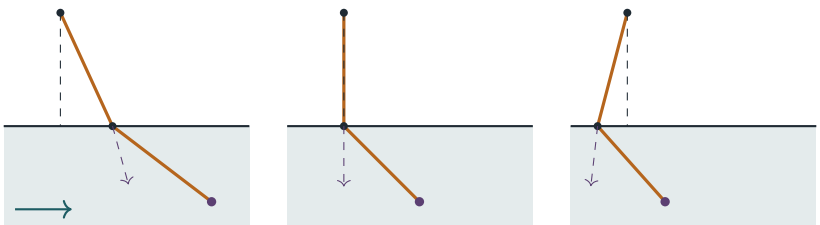
as required.

The three sketches

The equation tells us where x lies without solving it. The factor $[a^2+(b-x)^2]^{1/2}$ is positive, so $b-x$ and $x-c$ must have the same sign (or both be zero).

- $b > c$: then $c < x < b$. If $x \le c$ we would have $b - x > 0 \ge x - c$; if $x \ge b$ we would have $b - x \le 0 < x - c$; neither is possible. She enters at $b - x > 0$, downstream of O, the point of the bank nearest her. Since $x > c$, her swimming path makes an angle $\phi < 45°$ with the bank, and her heading $\theta = 2\phi < 90°$ is angled downstream.
- $b = c$: then $x = c = b$, by the same argument. She runs straight to O, heads directly across the river ($\theta = 90°$), and the current carries her along a path at $45°$ to the bank.
- $b < c$: then $b < x < c$, so $b - x < 0$. She runs *upstream* to enter the river, swims along a path steeper than $45°$, and her heading $\theta > 90°$ points partly upstream.

The sketches below are drawn to scale for $a = 3$, $c = 2$, $u = 2$, $v = 1$, with $b = 4, 2, 1$. Numerically, $x \approx 2.62$, $x = 2$ and $x \approx 1.79$. The current flows to the right; the dashed arrow at E shows the direction she heads relative to the water.



$b > c$ $\qquad$ $b = c$ $\qquad$ $b < c$

Checks

- If $b = c$, then $x = c$ makes both sides of the given equation zero, as found above.

- Straight swimming across ($\theta = 90°$): she drifts v downstream for each v across, so the path is at $45°$. The formula gives speed $2v\cos 45° = \sqrt{2}\,v$ along a path of length $\sqrt{2}\,c$, so the time is c/v, as it obviously must be.
- The swimming time $(x^2 + c^2)/(2vx)$ is smallest at $x = c$, by AM–GM. This makes sense: if running cost nothing she would enter the river a distance c upstream of N and head straight across.
- Numerically, minimising T directly for 2000 random choices of a, b, c, u, v always gave a minimiser satisfying both equations, and lying between b and c in the second case. Stepping the swim through the current confirmed the swimming time. For example, $a = 3$, $b = 4$, $c = 2$, $u = 2$, $v = 1$ gives $x \approx 2.621$.

Common slips

- Adding the current to her speed as if she swam straight at the stump. She must aim off so that the *resultant* velocity points at the stump.
- Forgetting that only $x > 0$ is possible with the current: with equal speeds she can't make any headway upstream.
- Squaring the equation for x too early. That loses the sign information that places x between b and c.

Beyond STEP

The first answer is Snell's law, and it comes from Fermat's principle: light takes the path of least time. Choosing a whole path to minimise a time is the subject of the calculus of variations. A close cousin is Zermelo's navigation problem: steer a boat through a current that varies from place to place so as to arrive as soon as possible. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers relative velocity and moving frames carefully.

WORKED SOLUTION

STEP III 1987, Question 12

Mechanics • Variable Mass Rotation

STEP III 1987, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q12

The key idea

This is a variable-mass problem, so $F = ma$ (or its rotational cousin) can't be used blindly. The safe method is to go back to first principles: compare the angular momentum about the pivot at time t with that at time $t + \delta t$, when a little burnt fuel has left the system. The pivot is smooth and gravity is vertical, so nothing exerts a moment about the vertical axis through the pivot, and that angular momentum is conserved.

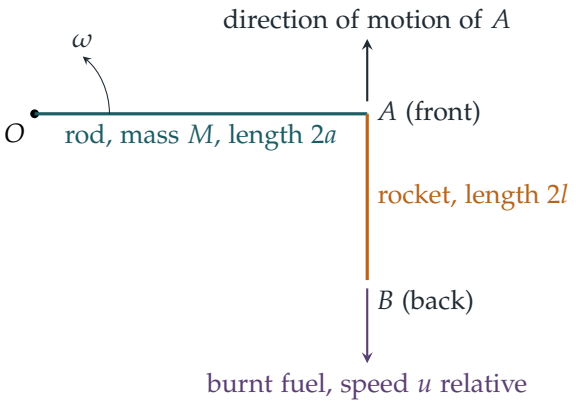
Solution

Setting up

Model the rod and the rocket as uniform rigid rods, rotating in a horizontal plane about the vertical axis through the pivot O. Let the rod be OA, with $OA = 2a$. Use unit vectors $\mathbf{e}_r$ along OA and $\mathbf{e}_\theta$ perpendicular to it in the direction of rotation, and let $\mathbf{k}$ be the upward vertical, so $\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}$.

The rocket's front end is at A and it points forwards, in the direction of motion of A. So the rocket lies along $\mathbf{e}_\theta$, extending *backwards* from A to its back end B, where

$$\overrightarrow{OB} = 2a\,\mathbf{e}_r - 2l\,\mathbf{e}_\theta.$$



Moment of inertia

The rod about its end has moment of inertia $\frac{1}{3}M(2a)^2 = \frac{4}{3}Ma^2$.

The rocket's centre is at $2a\,\mathbf{e}_r - l\,\mathbf{e}_\theta$, at distance $\sqrt{4a^2+l^2}$ from O. Its moment of inertia about its centre is $\frac{1}{3}ml^2$, so by the parallel axis theorem it contributes $\frac{1}{3}ml^2 + m(4a^2+l^2)$. The total is

$$I = \tfrac{4}{3}Ma^2 + m\left(4a^2 + \tfrac{4}{3}l^2\right).$$

How I changes

Since $m = 2\alpha l$ and $\dfrac{\mathrm{d}m}{\mathrm{d}t} = -\alpha\beta$, we have $\dfrac{\mathrm{d}l}{\mathrm{d}t} = -\frac{1}{2}\beta$. Differentiating I with the product rule,

$$\begin{aligned}\frac{\mathrm{d}I}{\mathrm{d}t} &= \frac{\mathrm{d}m}{\mathrm{d}t}\left(4a^2 + \tfrac{4}{3}l^2\right) + m\cdot\tfrac{8}{3}l\,\frac{\mathrm{d}l}{\mathrm{d}t}\\ &= -\alpha\beta\left(4a^2 + \tfrac{4}{3}l^2\right) + 2\alpha l\cdot\tfrac{8}{3}l\cdot\left(-\tfrac{1}{2}\beta\right) = -\alpha\beta\,(4a^2+4l^2).\end{aligned}$$

Notice that $4a^2 + 4l^2 = OB^2$: the moment of inertia drops at exactly the rate you'd get by removing mass from the back end B. That makes sense, because a uniform rocket that stays uniform, with its front fixed at A, can only shrink from the back.

Angular momentum of the ejected fuel

In the short time from t to $t+\delta t$ a mass $\delta\mu = \alpha\beta\,\delta t$ of burnt fuel leaves from B. Before it burnt it moved with the rocket, at velocity $\mathbf{v}_B = \omega\mathbf{k}\times\overrightarrow{OB}$. Afterwards its velocity is $\mathbf{v}_B - u\,\mathbf{e}_\theta$, since it moves directly backwards relative to the rocket. Its angular momentum about O is then

$$\begin{aligned}\delta\mu\,\overrightarrow{OB}\times(\mathbf{v}_B - u\,\mathbf{e}_\theta) &= \delta\mu\left[\omega\,OB^2\,\mathbf{k} - (2a\,\mathbf{e}_r - 2l\,\mathbf{e}_\theta)\times u\,\mathbf{e}_\theta\right]\\ &= \delta\mu\left[\omega(4a^2+4l^2) - 2au\right]\mathbf{k},\end{aligned}$$

using $\mathbf{e}_\theta\times\mathbf{e}_\theta = \mathbf{0}$ and $\mathbf{r}\times(\omega\mathbf{k}\times\mathbf{r}) = \omega r^2\mathbf{k}$ for $\mathbf{r}$ perpendicular to $\mathbf{k}$.

Ask yourself Why does only the $2a\,\mathbf{e}_r$ part of $\overrightarrow{OB}$ matter for the exhaust term?

The exhaust's relative velocity is along $\mathbf{e}_\theta$, the same line as the rocket. So the component of $\overrightarrow{OB}$ along the rocket has no moment. The perpendicular distance from O to the line of the exhaust (and of the thrust) is just $OA = 2a$, wherever along the rocket the fuel leaves.

Conservation of angular momentum

The pivot is smooth, so its reaction has no moment about the vertical axis through O, and gravity is vertical, so it has none either. The angular momentum about that axis of the whole system (firework plus the fuel burnt in δt) is therefore the same at t and $t + \delta t$:

$$I(t+\delta t)\,\omega(t+\delta t) + \delta\mu\left[\omega(4a^2+4l^2) - 2au\right] = I(t)\,\omega(t),$$

up to terms of order $(\delta t)^2$. Dividing by δt and letting $\delta t \to 0$, with $\delta\mu/\delta t = \alpha\beta$,

$$\frac{\mathrm{d}}{\mathrm{d}t}(I\omega) + \alpha\beta\,\omega(4a^2+4l^2) - 2\alpha\beta au = 0.$$

Expanding $\dfrac{\mathrm{d}}{\mathrm{d}t}(I\omega) = I\dfrac{\mathrm{d}\omega}{\mathrm{d}t} + \omega\dfrac{\mathrm{d}I}{\mathrm{d}t}$ and using $\dfrac{\mathrm{d}I}{\mathrm{d}t} = -\alpha\beta(4a^2+4l^2)$, the two ω terms cancel exactly, leaving

$$I\,\frac{\mathrm{d}\omega}{\mathrm{d}t} = 2\alpha\beta au.$$

This is just "moment of inertia $\times$ angular acceleration $=$ moment of the thrust": the thrust is $\alpha\beta u$, acting along the rocket at perpendicular distance $2a$ from O. Finally,

$$\frac{\mathrm{d}\omega}{\mathrm{d}t} = \frac{2\alpha\beta au}{\frac{4}{3}Ma^2 + 2\alpha l\left(4a^2 + \frac{4}{3}l^2\right)} = \frac{6\alpha\beta au}{4Ma^2 + 8\alpha l(3a^2+l^2)}$$
$$= \frac{3\alpha\beta au}{2\left[Ma^2 + 2\alpha l(3a^2+l^2)\right]},$$

as required. This holds while there is fuel left, that is, while $l > 0$.

Checks

- With no rocket ($l \to 0$) the result becomes $\frac{4}{3}Ma^2\,\dot{\omega} = 2a \cdot \alpha\beta u$: thrust times lever arm, as expected.
- Dimensions: $\alpha\beta u$ is a force (α is mass per length and β is a speed), so the numerator is force $\times$ length, and the denominator is a moment of inertia.
- Numerically, with $M = 1.3$, $a = 0.7$, $\alpha = 0.9$, $\beta = 0.4$, $u = 2.5$, $l(0) = 1.1$ and $\omega(0) = 0.3$, we stepped the motion by ejecting 2×10^5 small chunks from the back end and conserving angular momentum each time. At $t = 4.95$ this gave $\omega \approx 2.39817$, and integrating the formula gave 2.39816.

Common slips

- Writing $\dfrac{\mathrm{d}}{\mathrm{d}t}(I\omega) =$ torque for the firework alone. The firework isn't a closed system, and this gives the wrong answer unless you also account for the angular momentum carried off by the fuel.
- Giving the exhaust an absolute velocity of u backwards, rather than u relative to the moving rocket.
- Forgetting that l changes, so that the rocket's moment of inertia depends on time through both m and l.

Beyond STEP

The cancellation here is a general fact. For a body losing mass with the local rigid-body velocity plus a relative exhaust velocity, the equation of motion is the ordinary one for the instantaneous body, with an extra "thrust" force from the relative velocity. That is the rotational version of the rocket equation. Kleppner and Kolenkow's *An Introduction to Mechanics* treats variable-mass systems and angular momentum carefully, at just this level.

WORKED SOLUTION

STEP III 1987, Question 13

Mechanics • Rotating Rod with Friction

STEP III 1987, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q13

The key idea

While the particle grips the rod, rod and particle turn together as one rigid body about the hinge, so conservation of energy with the combined moment of inertia gives $\dot{\theta}^2$ at once. To find when slipping starts, we then look at the particle on its own: it moves in a circle about the hinge, so Newton's second law in the radial and transverse directions tells us exactly how much friction and normal reaction the rod must supply. Slipping begins when the friction needed reaches $\frac{1}{2}$ of the normal reaction.

Solution

The equation of motion

Model the rod as a uniform rigid rod turning about a smooth hinge O, and the particle as a point mass at distance a from O. Until the particle slides, it stays at distance a from O and turns with the rod.

The moment of inertia of the system about O is

$$I = \tfrac{4}{3}(3m)a^2 + ma^2 = 4ma^2 + ma^2 = 5ma^2,$$

using $\frac{4}{3}Ml^2$ for a uniform rod of length $2l$ about one end.

The hinge is smooth, and the friction and normal reaction between particle and rod are internal to the system, so they do no net work. When the rod has turned through θ, the rod's centre and the particle (both at distance a from O) have each fallen $a\sin\theta$. Starting from rest, conservation of energy gives

$$\tfrac{1}{2}(5ma^2)\dot{\theta}^2 = 3mg\,a\sin\theta + mg\,a\sin\theta = 4mga\sin\theta,$$

that is,

$$5a\dot{\theta}^2 = 8g\sin\theta,$$

as required. Differentiating with respect to t gives $10a\dot{\theta}\ddot{\theta} = 8g\cos\theta\,\dot{\theta}$, so (for $\dot{\theta} \neq 0$,

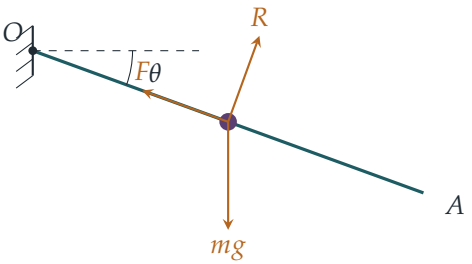
which holds once motion has begun)

$$a\ddot{\theta} = \tfrac{4}{5}g\cos\theta.$$

(The same comes from taking moments about O: $5ma^2\ddot{\theta} = 4mga\cos\theta$.)

When the particle slides

Now consider the particle alone. The forces on it are its weight mg, a normal reaction R perpendicular to the rod, and friction F along the rod. Since the particle moves on a circle of radius a about O, its acceleration has a component $a\dot{\theta}^2$ along the rod *towards* O and a component $a\ddot{\theta}$ perpendicular to the rod, in the direction of increasing θ.



We take F acting along the rod towards O, which is the direction the particle will turn out to need. Resolving along the rod, positive towards O: the weight has component $mg\sin\theta$ *away* from O, so

$$F - mg\sin\theta = ma\dot{\theta}^2 = \tfrac{8}{5}mg\sin\theta \quad\Longrightarrow\quad F = \tfrac{13}{5}mg\sin\theta.$$

Resolving perpendicular to the rod, positive in the direction of increasing θ (downwards from the rod): the weight has component $mg\cos\theta$, so

$$mg\cos\theta - R = ma\ddot{\theta} = \tfrac{4}{5}mg\cos\theta \quad\Longrightarrow\quad R = \tfrac{1}{5}mg\cos\theta.$$

Ask yourself Why does the particle need friction towards the hinge, when gravity seems to pull it down the slope?

It is not only gravity: to keep moving in a circle about O the particle needs a net force $ma\dot{\theta}^2$ towards O. Gravity's component along the rod points the other way, away from O, so friction must provide both the centripetal force and the force to cancel gravity's component. That is why F comes out positive towards O and grows like $\sin\theta$, while R is small because the rod is "falling away" beneath the particle (at $\theta = 0$ the particle has acceleration $\frac{4}{5}g$ downwards, so the rod carries only $\frac{1}{5}$ of its weight).

Both F and R are positive for $0 < \theta < \frac{\pi}{2}$, so the particle stays in contact with the rod. It stays at rest relative to the rod as long as $F \le \frac{1}{2}R$, that is,

$$\tfrac{13}{5}mg\sin\theta \le \tfrac{1}{2}\cdot\tfrac{1}{5}mg\cos\theta \iff \tan\theta \le \tfrac{1}{26}.$$

At $\theta = 0$ we have $F = 0 < \frac{1}{2}R$, and $F/R = 13\tan\theta$ increases with θ, so the particle first begins to slide (outwards, away from the hinge) when

$$\tan\theta = \tfrac{1}{26},$$

as required.

Checks

- At $\theta = 0$ the moment equation gives angular acceleration $\frac{4g}{5a}$; so the particle accelerates at $\frac{4}{5}g$, less than g, and $R = \frac{1}{5}mg > 0$ is consistent.
- Numerically stepping the rigid motion (with $g = 9.81$, $a = 0.7$) reproduced $5a\dot{\theta}^2 = 8g\sin\theta$ to 10^{-12}, and the friction and reaction found directly from the particle's computed acceleration matched $\frac{13}{5}mg\sin\theta$ and $\frac{1}{5}mg\cos\theta$. The ratio F/R first reached $\frac{1}{2}$ at $\theta \approx 0.03845$ rad, which is $\arctan\frac{1}{26} \approx 2.20°$.

Common slips

- Forgetting the centripetal term and resolving as though the particle were in equilibrium on a slope; this gives the wrong direction for friction as well as the wrong angle.
- Using $\frac{1}{3}(3m)(2a)^2$ correctly but then forgetting to add the particle's ma^2 to the moment of inertia.
- Taking $R = mg\cos\theta$, as on a fixed incline: the rod is accelerating, so the reaction is much smaller.

Beyond STEP

This question is a first taste of working in a rotating frame: viewed from the rod, the particle feels a "centrifugal" force $ma\dot{\theta}^2$ outwards and an "Euler" force $-ma\ddot{\theta}$, and these fictitious forces are exactly the terms we found. Undergraduate mechanics develops this systematically, along with Coriolis forces once the particle starts to slide. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation and non-inertial frames very clearly.

WORKED SOLUTION

STEP III 1987, Question 14

Mechanics • Gravitational Collapse

STEP III 1987, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q14

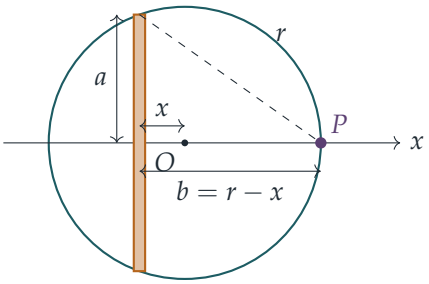
The key idea

A solid sphere is a stack of thin discs, so the force from the whole sphere is an integral of the given disc formula along the axis through the particle. The square root looks unpleasant until we notice that, for a particle on the surface, the expression under it factorises. After that, the cloud question is Newton's second law with the chain rule $\frac{dv}{dt} = v\frac{dv}{dr}$, and the substitution $r = R\cos^2\alpha$ is designed to tidy up the square root that appears when we integrate a second time.

Solution

The force at the surface of a sphere

Put the centre of the sphere at the origin and the particle P at $x = r$ on the x-axis. Slice the sphere into discs perpendicular to the axis. The slice at position x (where $-r \le x \le r$) has thickness δx, radius a with $a^2 = r^2 - x^2$, and lies at distance $b = r - x \ge 0$ from P, on its axis.



The key simplification is in the square root:

$$a^2 + b^2 = (r^2 - x^2) + (r - x)^2 = 2r^2 - 2rx = 2r(r - x),$$

so

$$\frac{b}{(a^2+b^2)^{1/2}} = \frac{r-x}{\sqrt{2r(r-x)}} = \sqrt{\frac{r-x}{2r}}.$$

Each disc pulls P along the axis towards the centre, so the forces all act in the same direction and simply add. Letting $\delta x \to 0$, the total force is

$$\begin{aligned} F &= 2\pi mk\rho \int_{-r}^{r} \left(1 - \sqrt{\frac{r-x}{2r}}\right) \mathrm{d}x \\ &= 2\pi mk\rho \left(2r - \frac{1}{\sqrt{2r}} \int_0^{2r} u^{1/2}\,\mathrm{d}u\right) \qquad (u = r - x) \\ &= 2\pi mk\rho \left(2r - \frac{1}{\sqrt{2r}} \cdot \tfrac{2}{3}(2r)^{3/2}\right) = 2\pi mk\rho \left(2r - \tfrac{4}{3}r\right) = \tfrac{4}{3}\pi mk\rho\, r. \end{aligned}$$

The mass of the sphere is $M = \frac{4}{3}\pi r^3\rho$, so $\frac{4}{3}\pi\rho = M/r^3$ and

$$F = \frac{kmM}{r^2},$$

directed towards the centre, as required.

The equation of motion of the cloud

Model the particles of the cloud as a continuous uniform sphere of mass M, and consider a particle of mass m on its surface at distance r from the centre. By the result above, the only force on it is kmM/r^2 towards the centre. Taking r as the coordinate (positive outwards), Newton's second law gives

$$m\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = -\frac{kmM}{r^2}.$$

Now $\frac{\mathrm{d}r}{\mathrm{d}t} = -v$, so $\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = -\frac{\mathrm{d}v}{\mathrm{d}t}$, and by the chain rule

$$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}r}\frac{\mathrm{d}r}{\mathrm{d}t} = -v\frac{\mathrm{d}v}{\mathrm{d}r}.$$

Hence $\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = v\frac{\mathrm{d}v}{\mathrm{d}r}$, and cancelling m,

$$v\frac{\mathrm{d}v}{\mathrm{d}r} = -\frac{kM}{r^2}.$$

Ask yourself The disc result was for a fixed sphere. Why may we use it for a cloud that is collapsing?

The force at any instant depends only on where the mass is at that instant, so we need the cloud to remain a uniform sphere with the same mass M inside the surface. It does: a particle that starts at rest at radius λR (with $0 < \lambda \le 1$) has mass $\lambda^3 M$ inside it, and $\lambda r(t)$ satisfies $\ddot{s} = -k\lambda^3 M/s^2$ whenever $r(t)$ satisfies $\ddot{r} = -kM/r^2$. So every particle stays at the same fraction of the surface radius: the cloud shrinks by a uniform scale factor, stays uniform, and no particle overtakes the surface particle. (Mass outside a particle would only matter for interior particles, and none lies outside the surface.)

The collapse from rest

Separate the variables and integrate from $r = R$, where $v = 0$:

$$\tfrac{1}{2}v^2 = \int_R^r -\frac{kM}{s^2}\,\mathrm{d}s = kM\left(\frac{1}{r} - \frac{1}{R}\right) = \frac{kM(R-r)}{rR}.$$

The acceleration is always inwards and the cloud starts at rest, so r decreases and $v \geq 0$ throughout.

Now substitute $r = R\cos^2\alpha$, with α increasing from 0 at $t = 0$ towards $\frac{\pi}{2}$ as r decreases from R towards 0. Then $R - r = R\sin^2\alpha$, so

$$v^2 = \frac{2kM\,R\sin^2\alpha}{R^2\cos^2\alpha} = \frac{2kM}{R}\tan^2\alpha, \qquad v = \left(\frac{2kM}{R}\right)^{1/2}\tan\alpha,$$

taking the positive root since $v \geq 0$ and $\tan\alpha \geq 0$ for $0 \leq \alpha < \frac{\pi}{2}$.

? **Ask yourself** How do we get from v as a function of α to t as a function of α?

Use $v = -\frac{\mathrm{d}r}{\mathrm{d}t}$ and $\frac{\mathrm{d}r}{\mathrm{d}\alpha} = -2R\cos\alpha\sin\alpha$:

$$\frac{\mathrm{d}t}{\mathrm{d}\alpha} = -\frac{1}{v}\frac{\mathrm{d}r}{\mathrm{d}\alpha} = \frac{2R\sin\alpha\cos\alpha}{(2kM/R)^{1/2}\tan\alpha} = \left(\frac{R^3}{2kM}\right)^{1/2} 2\cos^2\alpha = \left(\frac{R^3}{2kM}\right)^{1/2}(1+\cos 2\alpha).$$

Integrating, with $t = 0$ when $\alpha = 0$,

$$t = \left(\frac{R^3}{2kM}\right)^{1/2}\left(\alpha + \tfrac{1}{2}\sin 2\alpha\right),$$

which is the time at which $r = R\cos^2\alpha$, as required.

Checks

- Small times: for small α, $t \approx 2\alpha(R^3/2kM)^{1/2}$ and $R - r \approx R\alpha^2$, so $R - r \approx \frac{1}{2}(kM/R^2)\,t^2$: constant acceleration kM/R^2 from rest, exactly as expected at the start.
- Letting $\alpha \to \frac{\pi}{2}$ gives the total collapse time $\frac{\pi}{2}(R^3/2kM)^{1/2}$, finite even though $v \to \infty$.
- Numerically, summing 200 000 thin discs reproduces kmM/r^2 to a relative error of about 2×10^{-9}. Stepping $\ddot{r} = -kM/r^2$ with $k = 1.7$, $M = 2.3$, $R = 1.9$ to the time given by $\alpha = 0.8$ gives $r = 0.922260$, matching $R\cos^2 0.8 = 0.922260$.

Common slips

- Measuring b from the centre of the sphere instead of from the particle: $b = r - x$, not x.

- Forgetting that v is the *inward* speed, which flips the sign in $\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = v\frac{\mathrm{d}v}{\mathrm{d}r}$.
- Taking the negative square root for v, which makes the time come out negative.

Beyond STEP

The first part is a special case of Newton's shell theorem: outside a spherically symmetric body, gravity acts as if all the mass were at the centre. University courses prove it neatly with Gauss's law for gravity, $\nabla \cdot \mathbf{g} = -4\pi G\rho$. The collapse time you found is the astrophysicists' *free-fall time*, used to estimate how quickly gas clouds collapse to form stars, and the parametrisation by α is the same one that describes radial Kepler orbits. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats central forces and the shell theorem in this spirit.

WORKED SOLUTION

STEP III 1987, Question 15

Probability and Statistics • Expected Cost Optimisation

STEP III 1987, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q15

The key idea

Rather than minimising E_n directly, look at what one *extra* test changes. Comparing the strategies "stop after n" and "stop after $n+1$" only matters for patients who have had n positive results so far. For those patients, the extra test costs £10, and it saves the £100 treatment for a Steppe-syndrome patient whose test comes back negative. So the sign of $E_{n+1} - E_n$ decides everything, and that sign changes at most once.

Solution

Let F be the event that the patient has Fenland fever, so $\mathrm{P}(F) = p$, and let $S = F'$ be Steppe syndrome. We assume the test results for a Steppe patient are independent, each positive with probability q, where $0 < q < 1$. Treatment costs £100 whoever receives it.

Part (i)

Finding E_n. If the patient has F, every test is positive. So all n tests are given and then the treatment, at a cost of $10n + 100$.

If the patient has S, test number k (for $1 \leq k \leq n$) is given only when the first $k-1$ tests were all positive, which happens with probability q^{k-1}. So the expected number of tests is

$$\sum_{k=1}^{n} q^{k-1} = \frac{1-q^n}{1-q},$$

and the patient is treated only if all n tests are positive, with probability q^n. By linearity of expectation,

$$E_n = p\,(10n+100) + (1-p)\left(\frac{10(1-q^n)}{1-q} + 100q^n\right).$$

This also gives $E_0 = 100$, as it should, since then everyone is treated.

The difference. The F part increases by 10. In the S part, $\dfrac{1-q^{n+1}}{1-q} - \dfrac{1-q^n}{1-q} = q^n$, so

$$\begin{aligned} E_{n+1} - E_n &= 10p + (1-p)\left(10q^n + 100q^{n+1} - 100q^n\right) \\ &= 10p + 10(1-p)q^n(1 + 10q - 10) \\ &= 10p - 10(1-p)q^n(9 - 10q), \end{aligned}$$

as required.

Reading the result: with probability p, test $n+1$ is wasted on a fever patient. With probability $(1-p)q^n$, it is given to a Steppe patient and costs £10, but with probability $1-q$ it saves £100. The net expected saving is $100(1-q) - 10 = 10(9 - 10q)$.

The numbers. Write $D_n = E_{n+1} - E_n$. With $p = 10^{-4}$ and $q = 10^{-2}$ we have $9 - 10q = 8.9$ and $1 - p = 0.9999$, so

$$D_n = 10^{-3} - 89 \times 0.9999 \times 10^{-2n}.$$

Ask yourself Why is it enough to find where D_n changes sign?

Here $9 - 10q > 0$ and q^n strictly decreases as n grows, so D_n strictly increases with n. Hence the sequence E_n goes down while $D_n < 0$ and up once $D_n > 0$, and the minimum is at the first n with $D_n > 0$. Now

$$\begin{aligned} D_2 &= 10^{-3} - 0.0088991 < 0, \\ D_3 &= 10^{-3} - 0.000088991 = 0.000911\ldots > 0. \end{aligned}$$

So $E_0 > E_1 > E_2 > E_3$ and $E_3 < E_4 < E_5 < \cdots$, and the doctor should choose $n = 3$. (The expected cost is then about £10.11, against £100 for treating at once.)

Part (ii)

Take $q_0 = \frac{9}{10}$, and suppose $q > \frac{9}{10}$. Then $9 - 10q < 0$, so

$$D_n = 10p + 10(1-p)q^n(10q - 9).$$

Both terms are non-negative for every $p \in [0,1]$. They can't both be zero: if $p > 0$, the first term is positive, and if $p = 0$, the second is $10q^n(10q - 9) > 0$. So $D_n > 0$ for every $n \geq 0$, which means

$$E_0 < E_1 < E_2 < \cdots.$$

Whatever the value of p, the doctor should choose $n = 0$ and treat at once.

This value of q_0 can't be lowered. If $q < \frac{9}{10}$, then $D_0 = 10p - 10(1-p)(9 - 10q)$ is negative for small enough p, so testing at least once beats $n = 0$. In words, when $q > \frac{9}{10}$ a Steppe patient's test is negative less than one time in ten, so a £10 test saves less than £10 of treatment on average, even before counting the fever patients.

Checks

- $n = 1$ directly: $E_1 = 10 + 100\big(p + (1-p)q\big)$. With $p = 0$ this gives $E_1 - E_0 = 10 - 100(1-q)$, matching the formula's $-10q^0(9 - 10q)$.
- With $p = 10^{-4}$ and $q = 10^{-2}$ the formula gives $E_0 = 100$, $E_1 \approx 11.010$, $E_2 \approx 10.1210$, $E_3 \approx 10.11309$ and $E_4 \approx 10.11400$. The minimum is at $n = 3$, as found.
- Exact enumeration with fractions confirmed the formula for E_n and the difference identity for hundreds of random (p, q) and $n \leq 8$, and a simulation with $p = 0.3$, $q = 0.5$, $n = 2$ gave a mean cost of 63.98, against the exact 64.

Common slips

- Charging a Steppe patient for all n tests. The process stops at the first negative result, so test k is given only with probability q^{k-1}.
- Finding the first n with $D_n > 0$ without saying why D_n is monotonic. Without that, the sequence E_n could in principle fall again later.
- In part (ii), forgetting the case $p = 0$. That case is why the condition is $q > q_0$ rather than $q \geq q_0$: at $q = \frac{9}{10}$ and $p = 0$, every E_n equals 100.

Beyond STEP

This is a small example of *optimal stopping* and of decision theory: a sequence of actions with costs, a random state, and a rule for when to stop. Comparing neighbouring strategies ("marginal analysis") and showing the differences change sign only once is the standard way to handle such problems. The ideas lead on to Bayesian decision theory and dynamic programming. Grimmett and Welsh, *Probability: An Introduction*, covers the conditional-expectation tools that underpin them.

WORKED SOLUTION

STEP III 1987, Question 16

Probability and Statistics • Order Statistics and Poisson

STEP III 1987, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q16

The key idea

In part (i), the event "$A \leq k$ and $B \geq 1-k$" is awkward to handle directly, but its complement is a union of two simple events, "all the X_i exceed k" and "all the X_i are below $1-k$". Inclusion–exclusion then does the work. In part (ii), the number of winners is binomial with a huge n and a tiny p, which is exactly where the Poisson distribution lives. So we maximise the Poisson probability of one or two winners, and then make sure the approximation can't push the answer over 60%.

Solution

Part (i)

Let E be the event $A > k$ and F the event $B < 1-k$. Then $A > k$ means every X_i exceeds k, and $B < 1-k$ means every X_i is less than $1-k$. By independence,

$$\mathrm{P}(E) = (1-k)^n, \qquad \mathrm{P}(F) = (1-k)^n, \qquad \mathrm{P}(E \cap F) = (1-2k)^n,$$

where the last one holds because $E \cap F$ says every X_i lies in $(k, 1-k)$, an interval of length $1-2k > 0$.

The event we want is the complement of $E \cup F$, so by inclusion–exclusion

$$p_n = 1 - \mathrm{P}(E \cup F) = 1 - 2(1-k)^n + (1-2k)^n.$$

Since $0 < 1-2k < 1-k < 1$, both powers tend to 0, and so

$$p_n \to 1 \quad \text{as } n \to \infty.$$

Comment. However small we make k, with enough observations we become almost certain that some observation lies within k of 0 and another lies within k of 1. In other words, the sample minimum converges to 0 and the sample maximum to 1 (in probability), so the sample range $B - A$ tends to 1, the length of the whole interval. This is why, for example, A and B are natural estimators of the endpoints of an unknown uniform distribution.

Part (ii)

Let $N = 4\,000\,000$ and let X be the number of winners. Readers win independently, each with probability p, so $X \sim \mathrm{B}(N, p)$. Write $q = 1 - p$. You keep your job exactly when $X = 1$ or $X = 2$, which has probability

$$K(p) = Npq^{N-1} + \binom{N}{2} p^2 q^{N-2}.$$

The Poisson picture. With N huge and p small, X is approximately Poisson with mean $\lambda = Np$, so

$$K(p) \approx h(\lambda), \qquad \text{where } h(\lambda) = \mathrm{e}^{-\lambda}\Big(\lambda + \frac{\lambda^2}{2}\Big).$$

Differentiating,

$$h'(\lambda) = \mathrm{e}^{-\lambda}\Big(1 + \lambda - \lambda - \frac{\lambda^2}{2}\Big) = \mathrm{e}^{-\lambda}\Big(1 - \frac{\lambda^2}{2}\Big),$$

which is positive for $0 < \lambda < \sqrt{2}$ and negative for $\lambda > \sqrt{2}$. So h has its greatest value at $\lambda = \sqrt{2}$:

$$h(\sqrt{2}) = \mathrm{e}^{-\sqrt{2}}\big(\sqrt{2} + 1\big) \approx 0.2431 \times 2.4142 \approx 0.5869.$$

Your best choice is $p = \sqrt{2}/N \approx 3.5 \times 10^{-7}$, and even then your chance of survival is only about 58.7%. That is the reasoning the question asks for.

> **Ask yourself** The Poisson distribution is only an approximation. Could the true binomial probability creep above 0.6?

No. We can bound $K(p)$ directly. Since $q \le 1$ and $\binom{N}{2} \le \frac{1}{2}N^2$,

$$K(p) = q^{N-2}\Big(Npq + \binom{N}{2}p^2\Big) \le q^{N-2}\Big(\lambda + \frac{\lambda^2}{2}\Big).$$

Now $1 - p \le \mathrm{e}^{-p}$ for all p (the line $1 - p$ is the tangent to the convex curve e^{-p} at $p = 0$), so $q^{N-2} \le \mathrm{e}^{-(N-2)p} = \mathrm{e}^{-\mu}$, where $\mu = (N-2)p$. Put $c = N/(N-2)$, so that $\lambda = c\mu$ with $c > 1$. Then

$$K(p) \le \mathrm{e}^{-\mu}\Big(c\mu + \frac{c^2\mu^2}{2}\Big) \le c^2\,\mathrm{e}^{-\mu}\Big(\mu + \frac{\mu^2}{2}\Big) = c^2 h(\mu) \le c^2 h(\sqrt{2}).$$

Here $c^2 = \big(1 - \frac{2}{N}\big)^{-2} < 1.000002$, so for every choice of p,

$$K(p) < 1.000002 \times 0.58694 < 0.587 < 0.6.$$

So however you choose p, you have less than a 60% chance of keeping your job, as required.

Checks

- In part (i), $n = 1$ gives $p_1 = 1 - 2(1-k) + (1-2k) = 0$. That's right: a single observation can't be both $\le k$ and $\ge 1-k$ when $k < \frac{1}{2}$. For $n = 2$ we get $p_2 = 2k^2$, the probability that one of the two points lands in each end strip, in either order.
- A simulation with $n = 10$, $k = 0.3$ gave 0.944, against the formula's 0.9436.
- Computing the exact binomial $K(p)$ over a fine grid of $\lambda = Np$ gives a maximum of 0.58694 at $\lambda \approx 1.414$, agreeing with the Poisson value to five decimal places.

Common slips

- Writing $p_n = \mathrm{P}(A \le k)\,\mathrm{P}(B \ge 1-k)$. The minimum and maximum of the same sample are *not* independent.
- Maximising the binomial probability with respect to p directly. It can be done, but the algebra is unpleasant; working with $\lambda = Np$ makes it a one-line calculus problem.
- Stopping at the Poisson approximation without noticing that 0.587 is close enough to 0.6 that some justification is worthwhile.

Beyond STEP

Part (i) is a first taste of *order statistics* and of convergence in probability: the minimum and maximum of a sample converge to the ends of the support. Part (ii) is the Poisson "law of rare events", and the bound we used is a simple instance of the general fact that the binomial and Poisson distributions are very close in total variation when p is small. Both ideas are developed in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

This book takes the most interesting STEP questions, from the specimen papers that preceded the first examination in 1987 to the papers of 2025, and uses each one as a doorway. Every question is solved carefully, with the questions a good student asks along the way, and then the book steps through the door: where the idea comes from, what it becomes at university, and what to read next.

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